

MPLS Layer 3 VPNs

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Agenda

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Agenda

- MPLS L3VPNs
- Traceroute in an MPLS VPN
- PE-CE Routing Protocols
  - Static
  - RIPv2
  - BGP
  - OSPF
  - EIGRP
  - Deployment Scenario with BGP
  - Deployment Scenario with OSPF
  - Deployment Scenario with EIGRP
- 6vPE



MPLS Layer 3 VPNs

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Driving Forces for MPLS L3-VPNs

- Low cost and scalable WAN to replace traditional Layer 2 Technologies (ATM/FR)
- Separate networks for security, policy and economic/political reasons - Segmentation
- Not a Core Business activity and expectation of cost savings
  - No need to invest in high cost equipment
  - Pay as you go model
  - Reduce the cost of maintaining high cost and skilled IT staff

Driving Forces for MPLS L3-VPNs

- Other factors include:
  - Limited Geographic reach (cost)
  - Reluctance to share the routing info
  - Limited flexibility
- Larger enterprises with disparate business entities and economies of scale approaching that of a Tier 2 or Tier 3 ISP
- Bandwidth, performance, or other requirements that cannot be met by public service provider networks
- Business mandates on separation, segmentation and Virtualization of Traffic

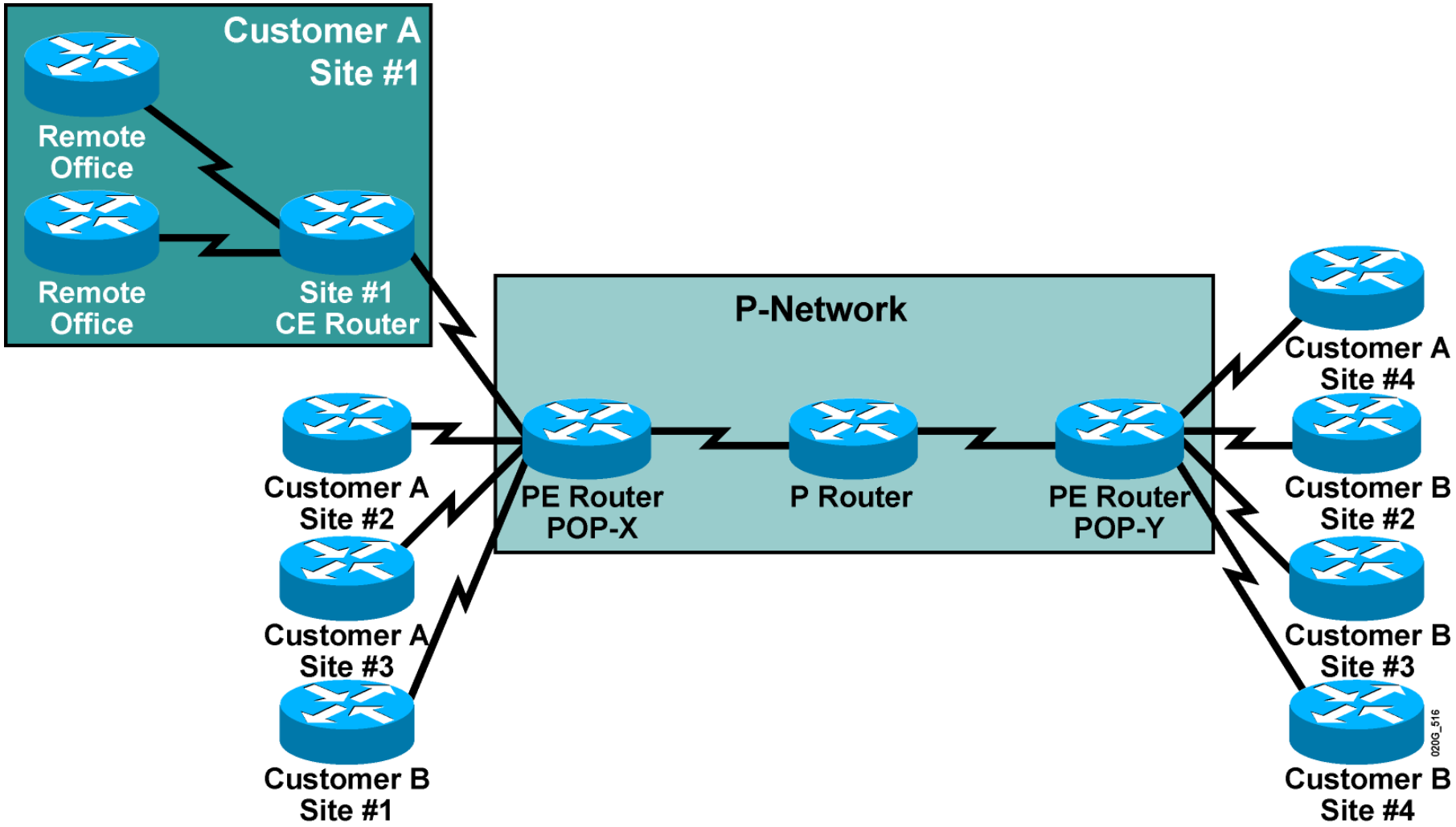
Terminology

- LSR: label switch router
- LSP: label switched path
  - The chain of labels that are swapped at each hop to get from one LSR to another
- VRF: VPN routing and forwarding
  - Mechanism in Cisco IOS® used to build per-customer RIB and FIB
- MP-BGP: multiprotocol BGP
- PE: provider edge router interfaces with CE routers
- P: provider (core) router, without knowledge of VPN
- VPNv4: address family used in BGP to carry MPLS-VPN routes
- RD: route distinguisher
  - Distinguish same network/mask prefix in different VRFs
- RT: route target
  - Extended community attribute used to control import and export policies of VPN routes
- LFIB: label forwarding information base
- FIB: forwarding information base

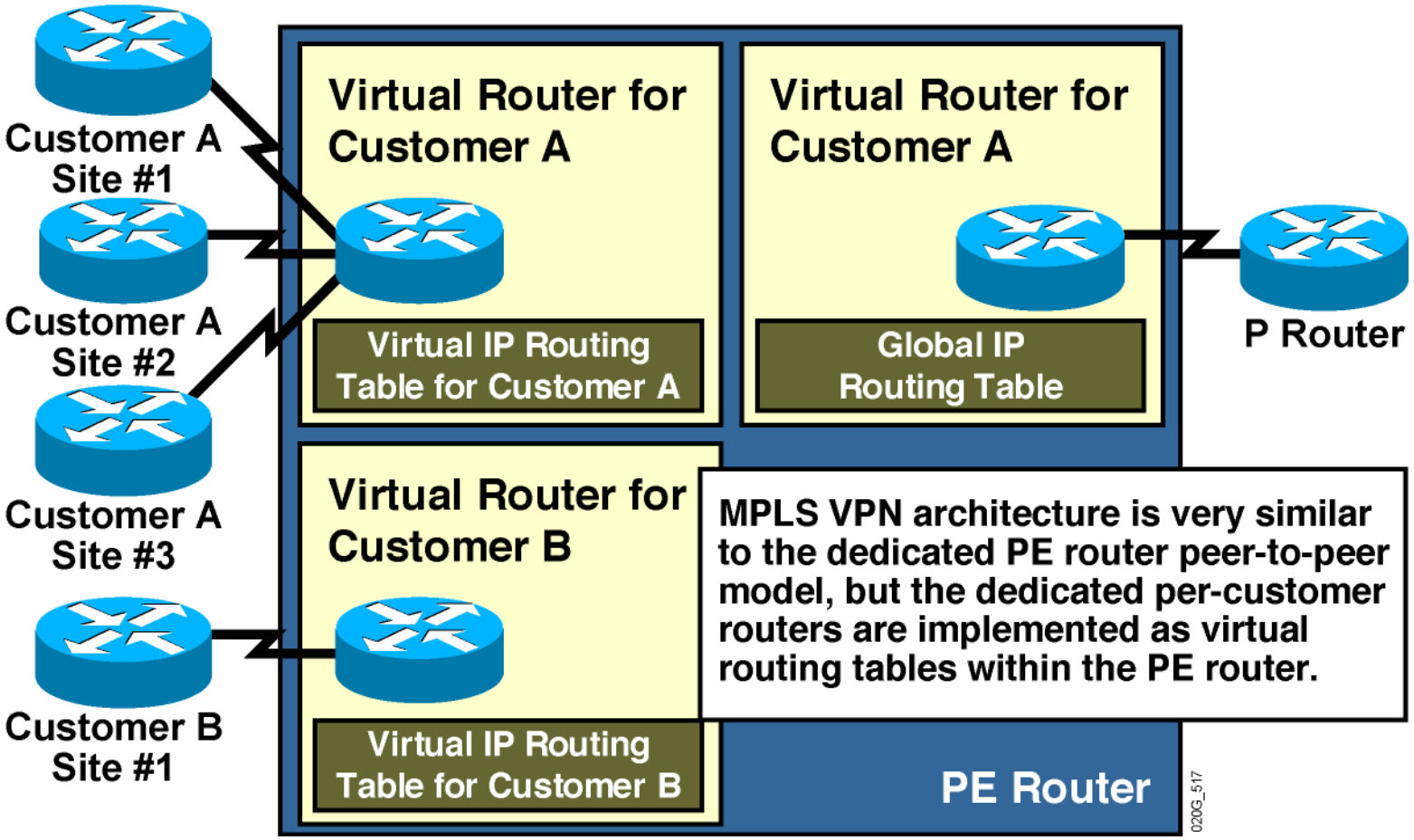
MPLS VPN Architecture

- An MPLS VPN combines the best features of an overlay VPN and a peer-to-peer VPN:
  - PE routers participate in customer routing, guaranteeing optimum routing between sites and easy provisioning.
  - PE routers carry a separate set of routes for each customer (similar to the dedicated PE router approach).
  - Customers can use overlapping addresses.

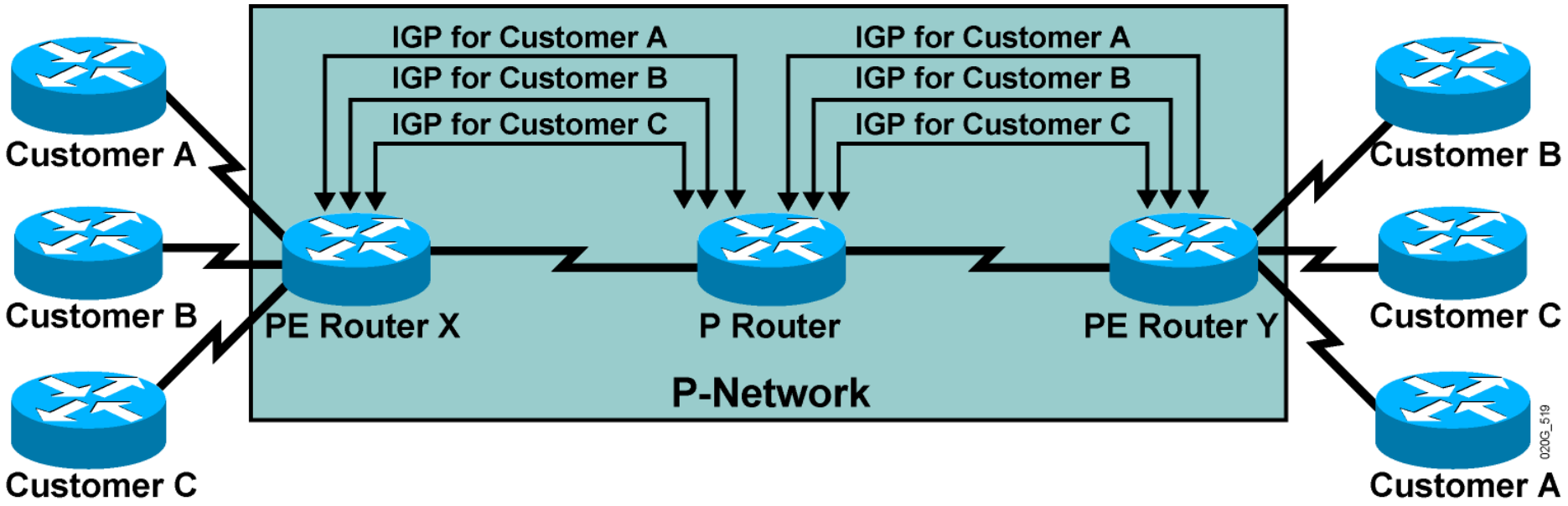
MPLS VPN Architecture (Cont.)
Terminology



PE Router Architecture



Propagation of Routing Information Across the P-Network



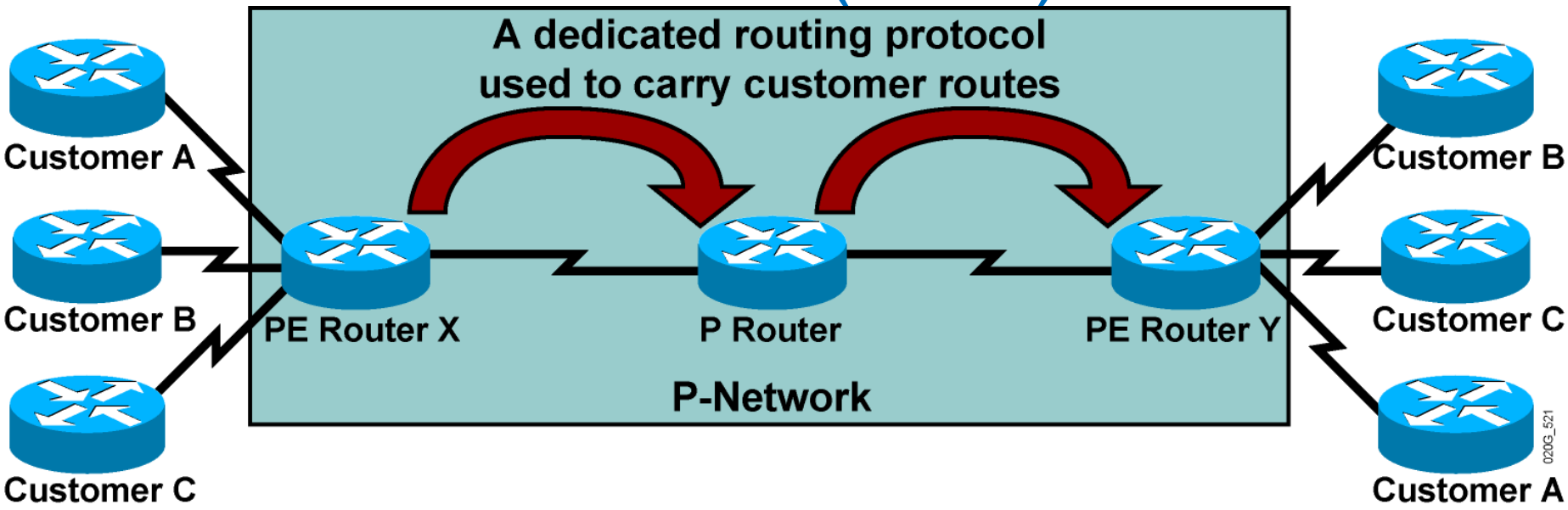
Question: How will PE routers exchange customer routing information?

Answer #1: Run a dedicated Interior Gateway Protocol (IGP) for each customer across the P-network.

This is the wrong answer for the following reasons:

- The solution does not scale.
- P routers carry all customer routes.

Propagation of Routing Information Across the P-Network (Cont.)



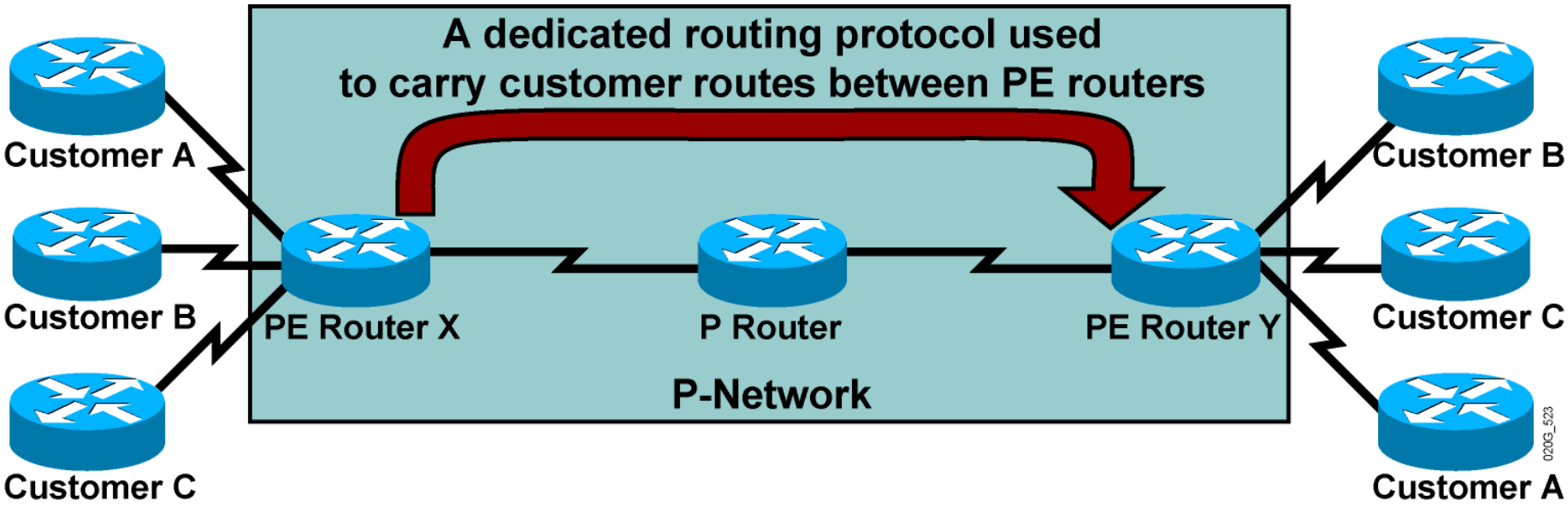
Question: How will PE routers exchange customer routing information?

Answer #2: Run a single routing protocol that will carry all customer routes inside the provider backbone.

Better answer, but still not good enough:

- P routers carry all customer routes.

Propagation of Routing Information Across the P-Network (Cont.)



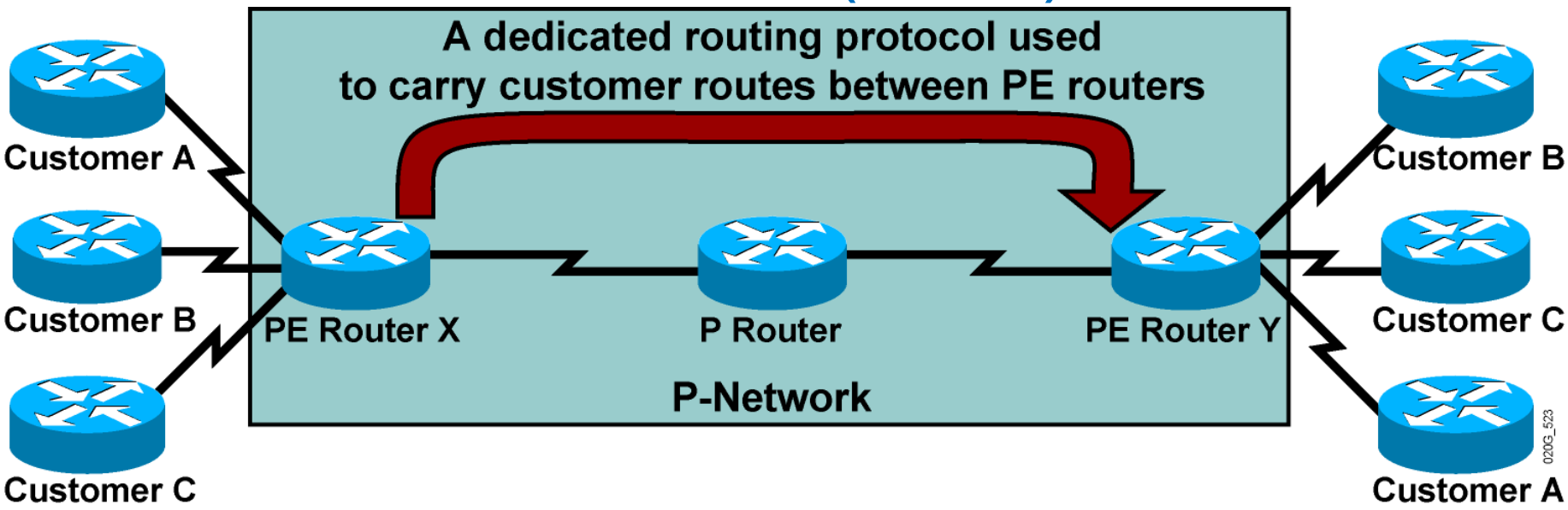
Question: How will PE routers exchange customer routing information?

Answer #3: Run a single routing protocol that will carry all customer routes between PE routers. Use MPLS labels to exchange packets between PE routers.

The best answer:

- P routers do not carry customer routes; the solution is scalable.

Propagation Routing Information Across the P-Network (Cont.)



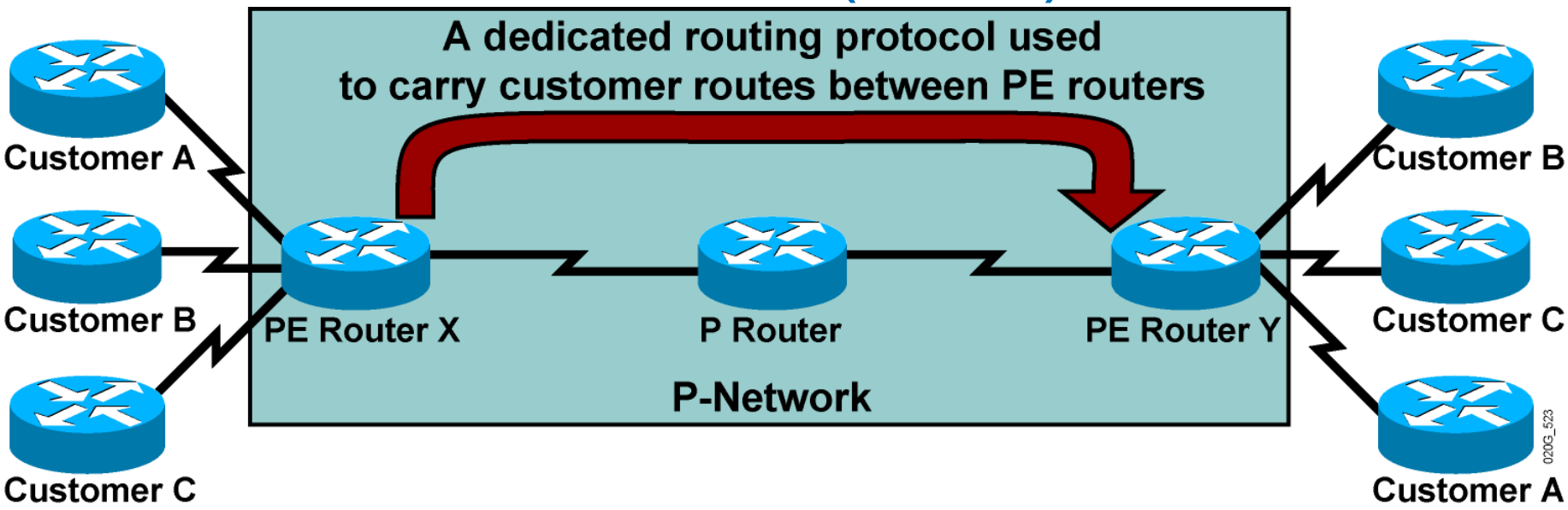
Question: Which protocol can be used to carry customer routes between PE routers?

Answer: The number of customer routes can be very large. BGP is the only routing protocol that can scale to a very large number of routes.

Conclusion:

BGP is used to exchange customer routes directly between PE routers.

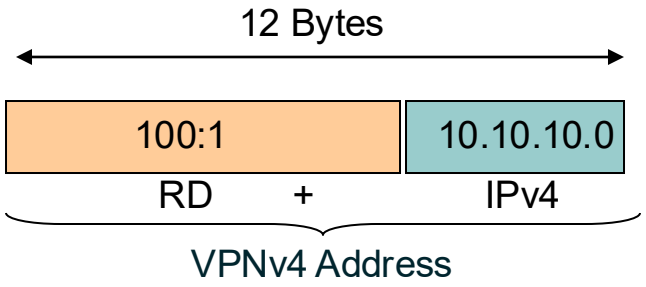
Propagation of Routing Information Across the P-Network (Cont.)



Question: How will information about the overlapping subnets of two customers be propagated via a single routing protocol?

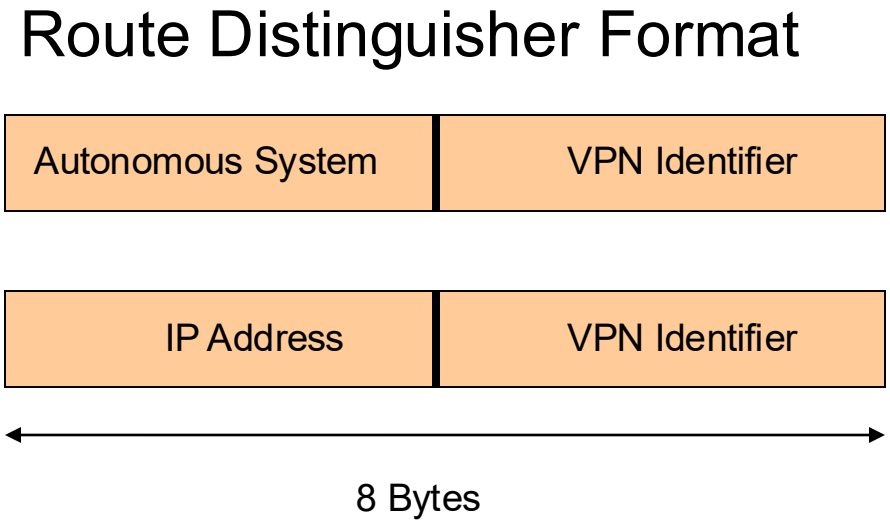
Answer: Extend the customer addresses to make them unique.

Route Distinguishers



- The 64-bit route distinguisher (RD) is prepended to an IPv4 address to make it globally unique.
- The resulting address is a VPNv4 address.
- VPNv4 addresses are exchanged between PE routers via BGP.
  - BGP that supports address families other than IPv4 addresses is called Multiprotocol BGP (MP-BGP).

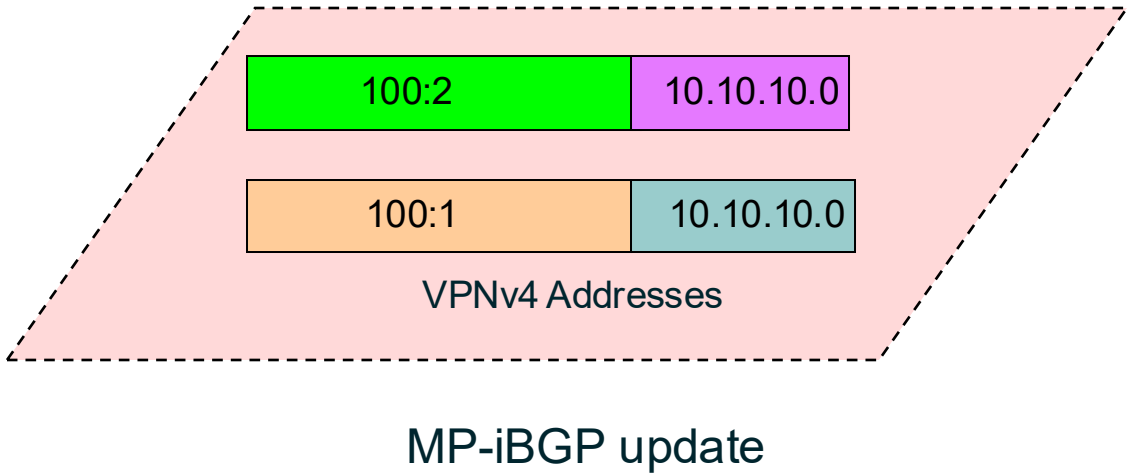
Route Distinguishers (Cont.)



Service Providers can use their BGP AS along with VPN customer identifier

Service Provider who do not have BGP AS, can use an IP address

Route Distinguishers (Cont.)

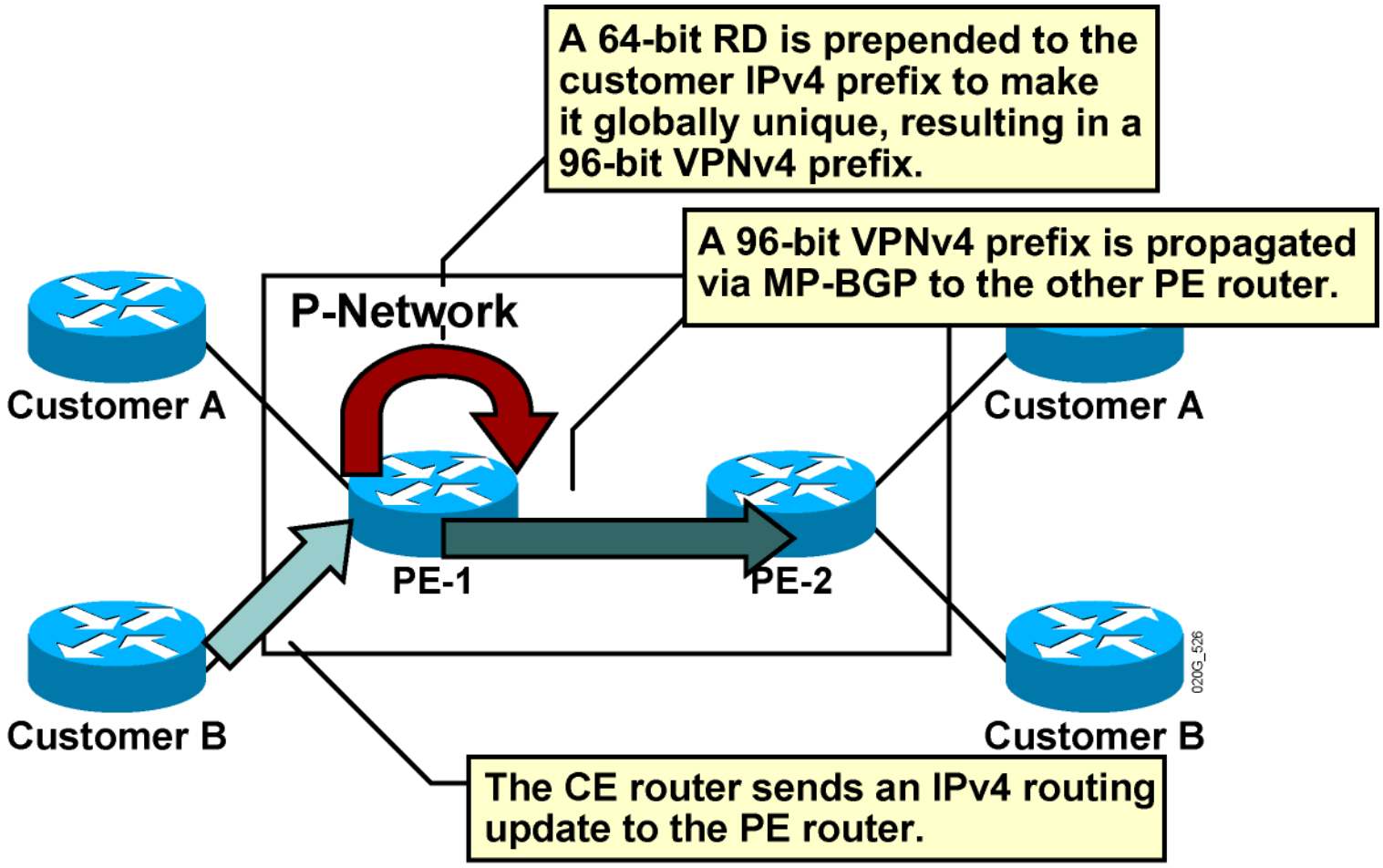


Customer A has RD of 100:1

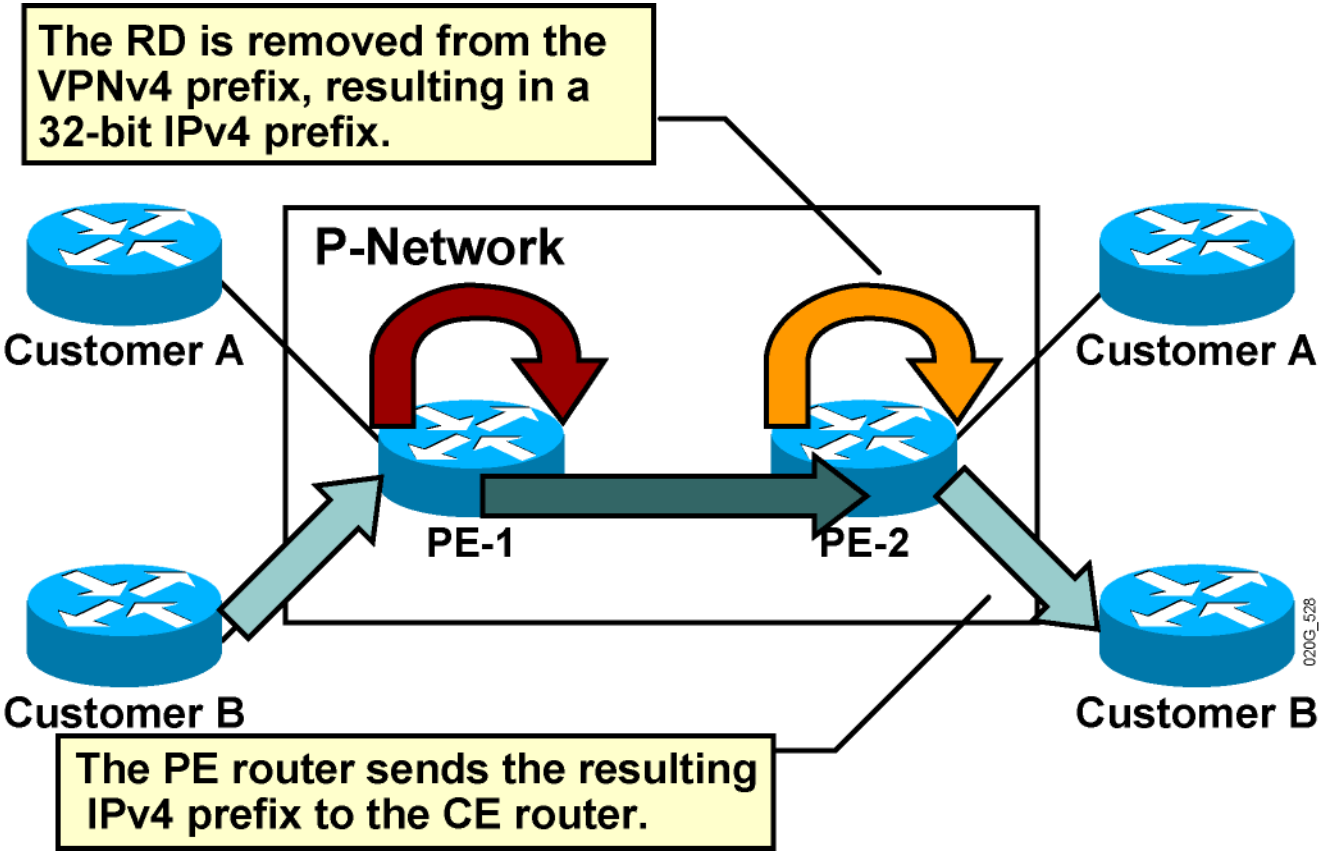
Customer B has RD of 100:2

Route Distinguisher keeps Customer A's update unique from Customer B in the MP-iBGP update, although they use the same IP address

Route Distinguishers (Cont.)



Route Distinguishers (Cont.)



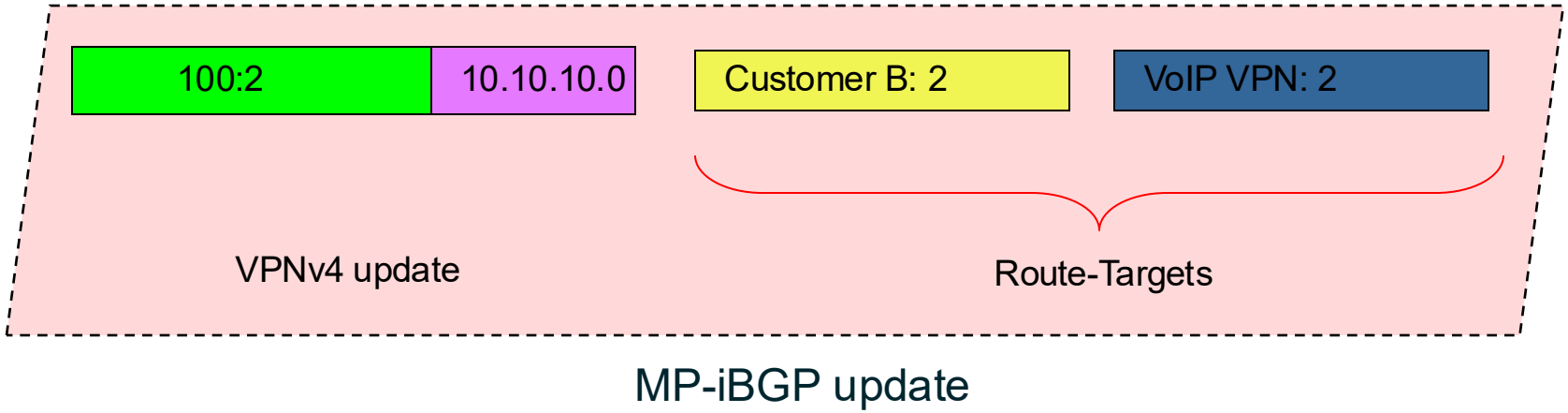
Route Distinguishers (Cont.) Usage in an MPLS VPN

- The RD has no special meaning.
- Used only to make potentially overlapping IPv4 addresses globally unique.
- The RD could serve as a VPN identifier, but this design could not support all topologies required by the customers.

Route Targets
Why Are They Needed?

- Some sites have to participate in more than one VPN.
- The RD cannot identify participation in more than one VPN.
- RTs were introduced in the MPLS VPN architecture to support complex VPN topologies.
  - A different method is needed in which a set of identifiers can be attached to a route.

Route Targets (Cont.)
What Are They?



- RTs are additional attributes attached to VPNv4 BGP routes to indicate VPN membership.
- Format is same as Route Distinguisher
- Extended BGP communities are used to encode these attributes.
  - Extended communities carry the meaning of the attribute together with its value.
- Any number of RTs can be attached to a single route.

Route Targets (Cont.)
How Do They Work?

- Export RTs:
  - Identifying VPN membership
  - Appended to the customer route when it is converted into a VPNv4 route
- Import RTs:
  - Associated with each virtual routing table
  - Select routes to be inserted into the virtual routing table

Virtual Private Networks Redefined

- With the introduction of complex VPN topologies, VPNs have had to be redefined:
  - A VPN is a collection of sites sharing common routing information.
  - A site can be part of different VPNs.
  - A VPN can be seen as a community of interest (closed user group, or CUG).
  - Complex VPN topologies are supported by multiple virtual routing tables on the PE routers.

Impact of Complex VPN Topologies on Virtual Routing Tables

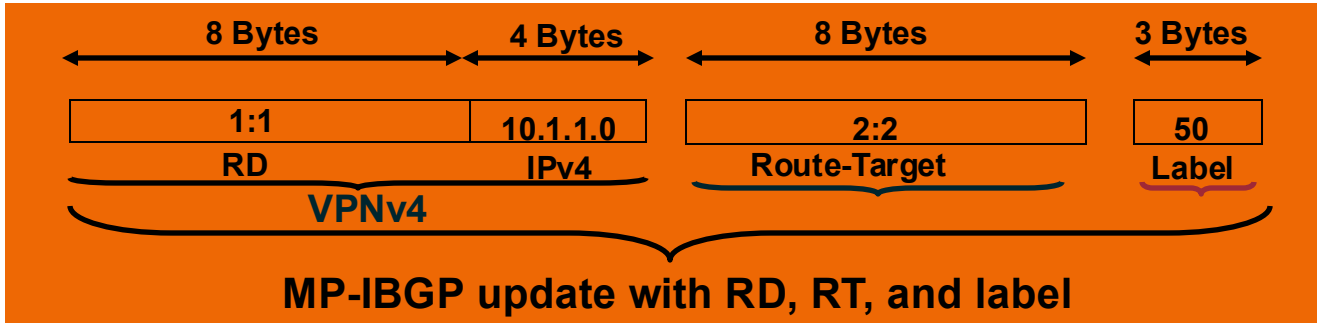
- A virtual routing table in a PE router can be used only for sites with identical connectivity requirements.
- Complex VPN topologies require more than one virtual routing table per VPN.
- As each virtual routing table requires a distinct RD value, the number of RDs in the MPLS VPN network increases.

Important points to note for RT and RD

- Route Distinguishers (RD) are only used to make ipv4 VPN addresses unique when advertising them over MP-iBGP, by making them vpnv4 prefixes
- We can have one RD per vrf
- Only one vrf can be assigned to an interface
- Route Targets (RT) are used for VPN membership, so that complex scenarios can be addressed
- VPN is the set of rules for customer connectivity and can be very complex
- A VPN may have several RTs

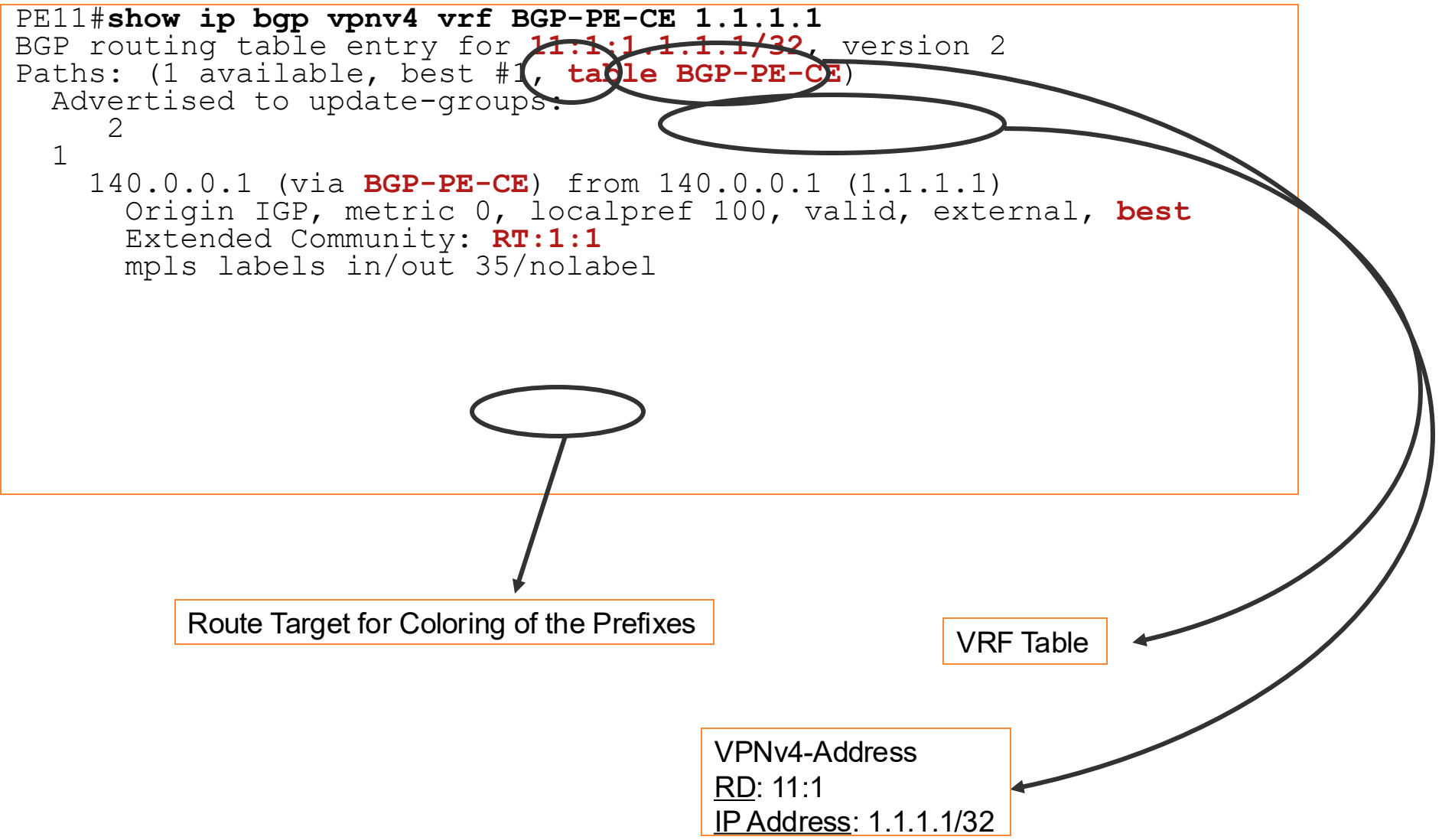
MP-BGP Extensions

- BGP has been extended in the form in Multi-protocol (MP)-BGP to carry routes from multiple “address families” in the form of VPNv4-AF
- VPNv4-AF is a 12-byte field of the form:
  - Route-Distinguisher [RD] 8-byte
  - IPv4-Address 4-byte
- Route-Target Attribute
- Label value



MP-BGP Extensions

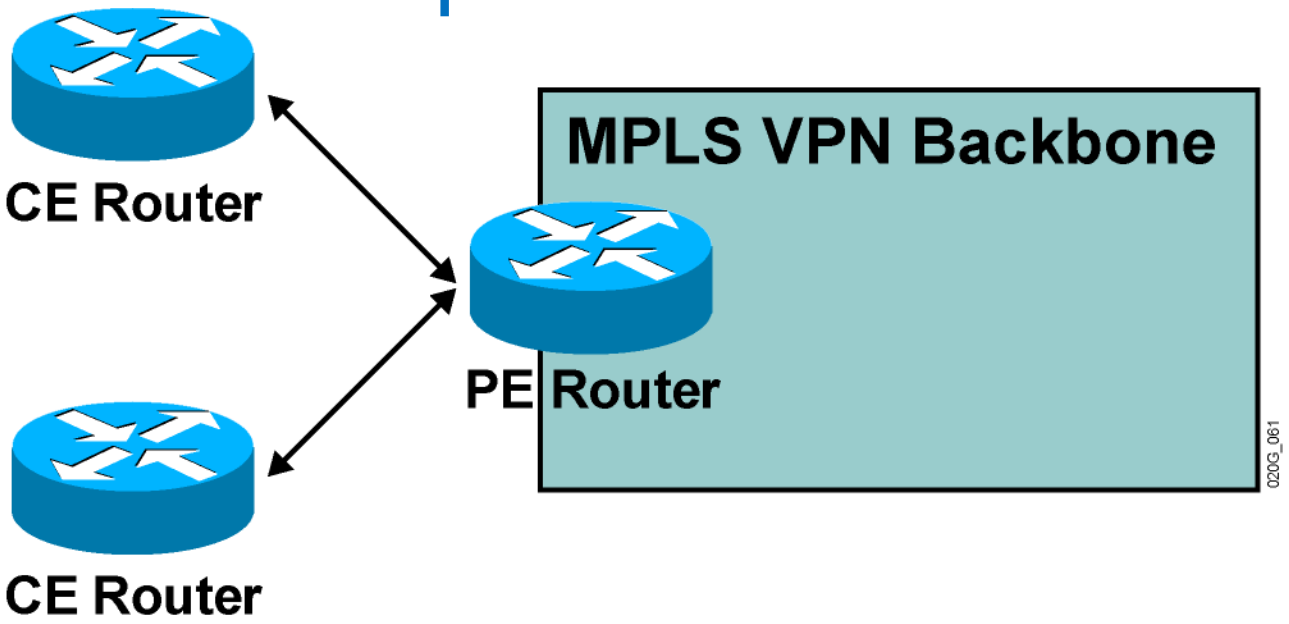
```
PE11#show ip bgp vpnv4 vrf BGP-PE-CE 1.1.1.1
BGP routing table entry for 11:1:1.1.1.1/32, version 2
Paths: (1 available, best #1, table BGP-PE-CE)
  Advertised to update-groups:
     2
  1
    140.0.0.1 (via BGP-PE-CE) from 140.0.0.1 (1.1.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Extended Community: RT:1:1
      mpls labels in/out 35/nolabel
```



MPLS VPN Routing Requirements

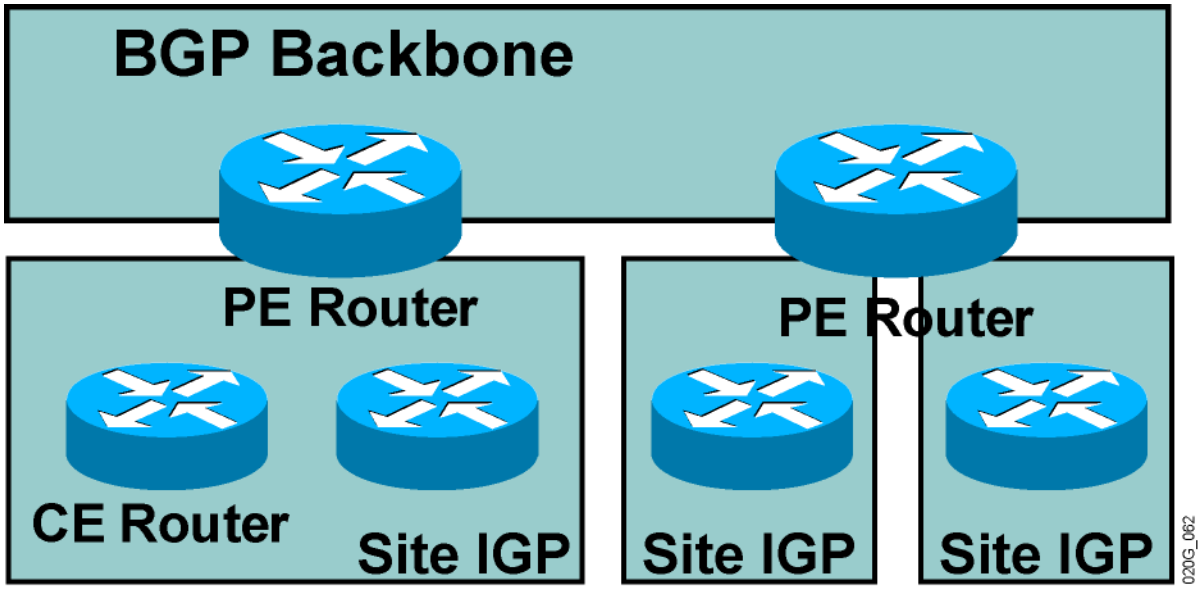
- CE routers have to run standard IP routing software.
- PE routers have to support MPLS VPN services and Internet routing.
- P routers have no VPN routes.

MPLS VPN Routing
CE Router Perspective



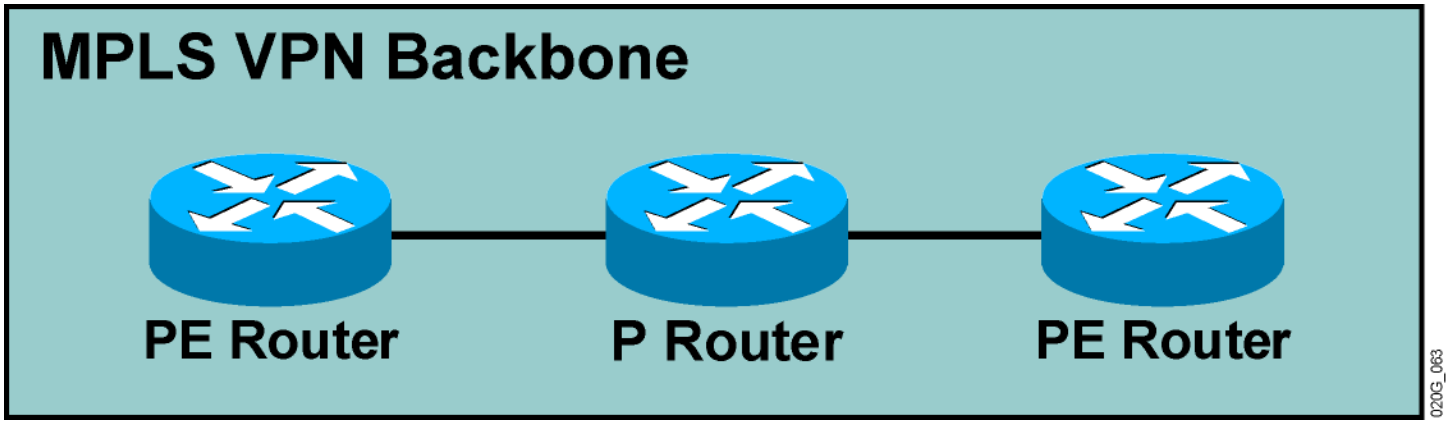
- The CE routers run standard IP routing software and exchange routing updates with the PE router.
  - PE-CE protocols can be EBGP, OSPF, RIPv2, EIGRP, and static routes.
- The PE router appears as another router in the C-network.

MPLS VPN Routing (cont.)
Overall Customer Perspective



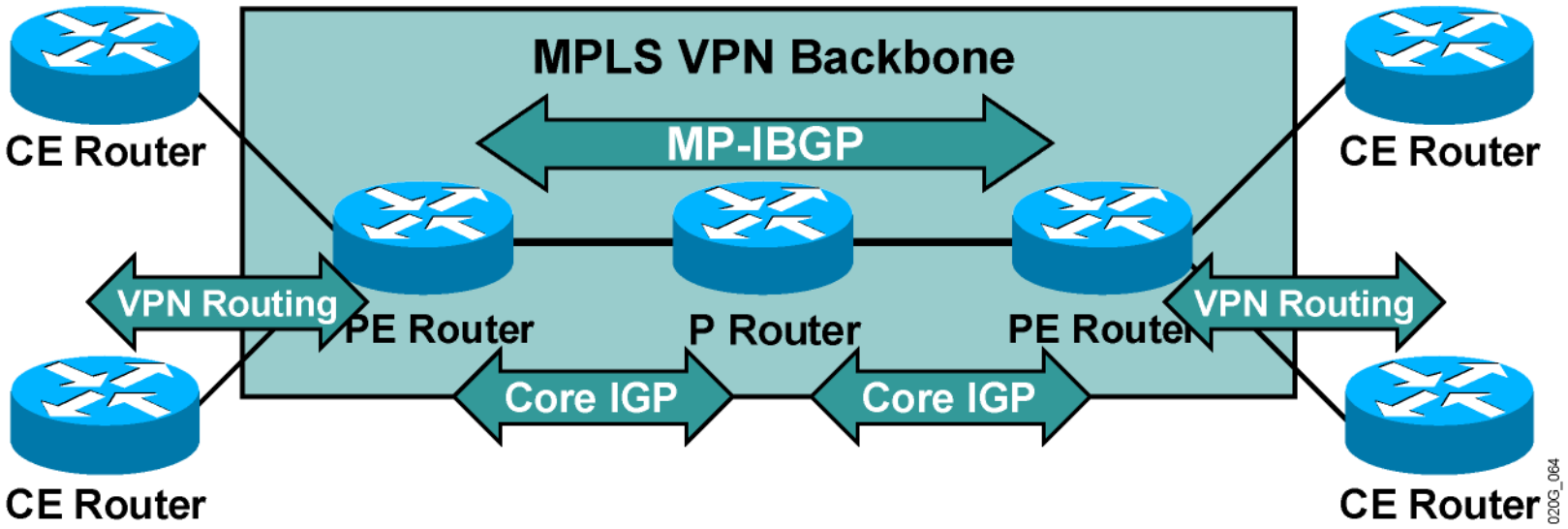
- To the customer, the PE routers appear as core routers connected via a BGP backbone.
- The usual BGP and IGP design rules apply.
- The P routers are hidden from the customer.

MPLS VPN Routing (Cont.)
P Router Perspective



- P routers do not participate in MPLS VPN routing and do not carry VPN routes.
- P routers run backbone IGP with the PE routers and exchange information about global subnets (core links and loopbacks).

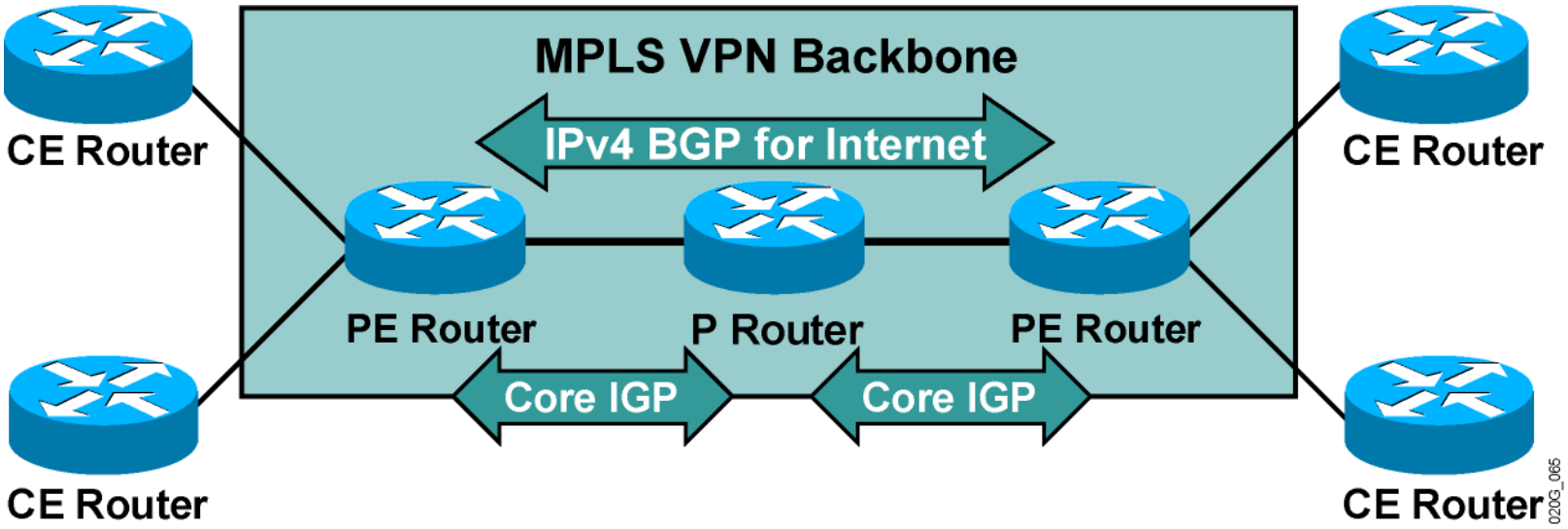
MPLS VPN Routing (Cont.)
PE Router Perspective



PE routers:

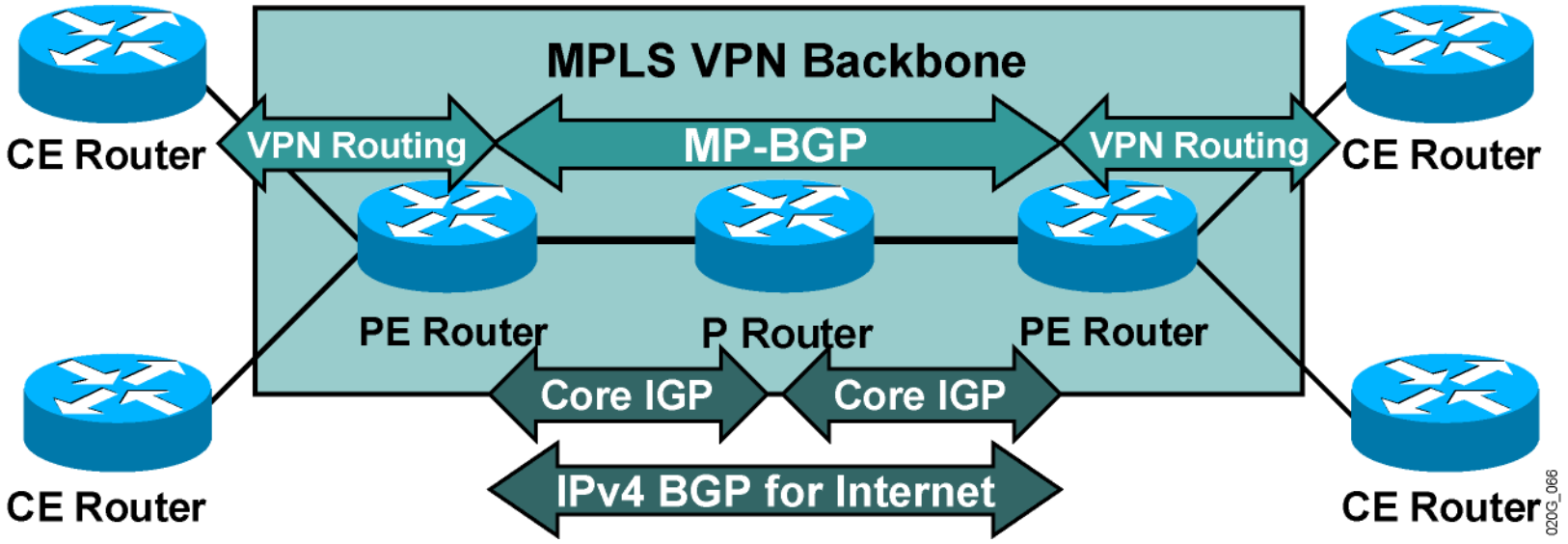
- Exchange VPN routes with CE routers via per-VPN routing protocols
- Exchange core routes with P routers and PE routers via core IGP
- Exchange VPNv4 routes with other PE routers via MP-IBGP sessions

Support for Existing Internet Routing



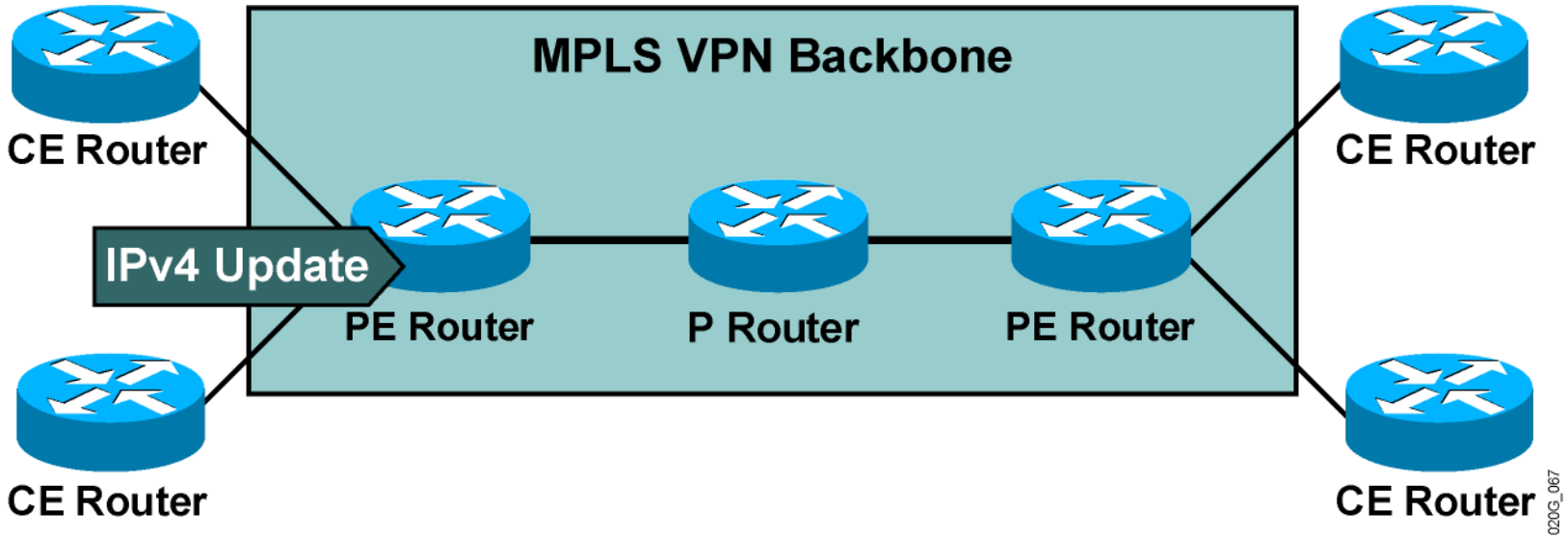
- PE routers can run standard IPv4 BGP in the global routing table:
  - PE routers exchange Internet routes with other PE routers.
  - CE routers do not participate in Internet routing.
  - P routers do not need to participate in Internet routing.

Routing Tables on PE Routers



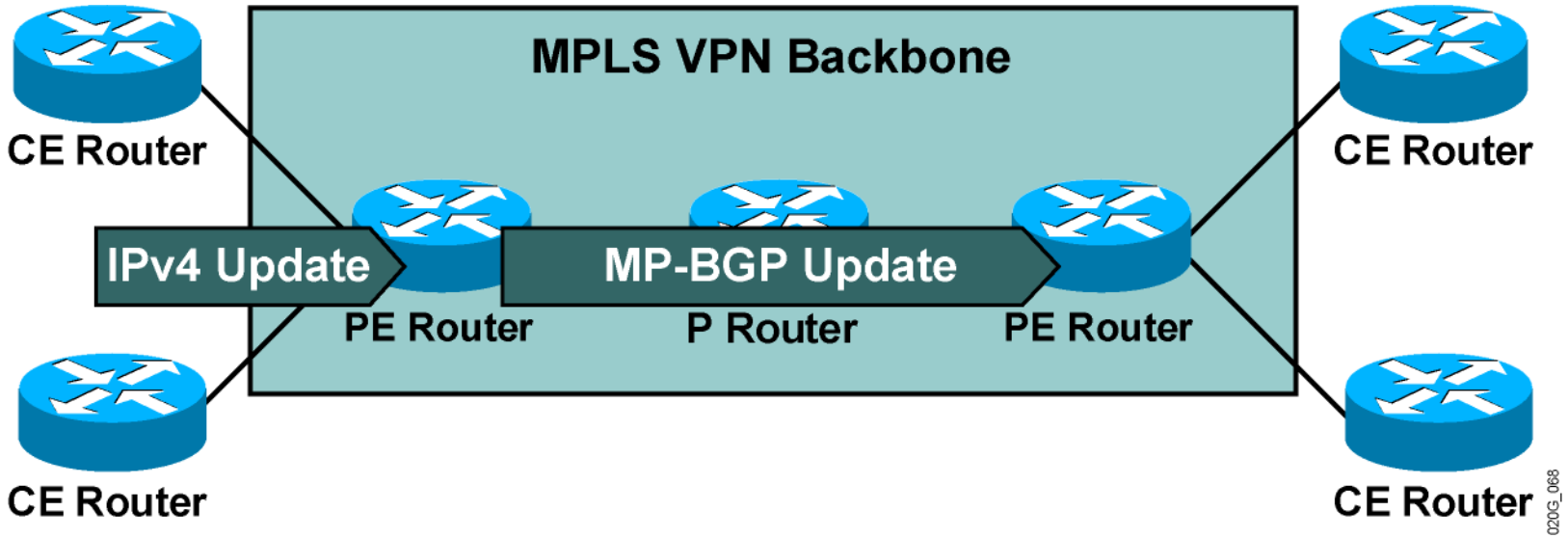
- PE routers contain a number of routing tables:
  - Global routing table, which contains core routes (filled with core IGP) and Internet routes (filled with IPv4 BGP)
  - VRF tables for sets of sites with identical routing requirements
  - VRFs filled with information from CE routers and MP-BGP information from other PE routers

End-to-End Routing Update Flow



- PE routers receive IPv4 routing updates from CE routers and install them in the appropriate VRF table.

End-to-End Routing Update Flow (Cont.)

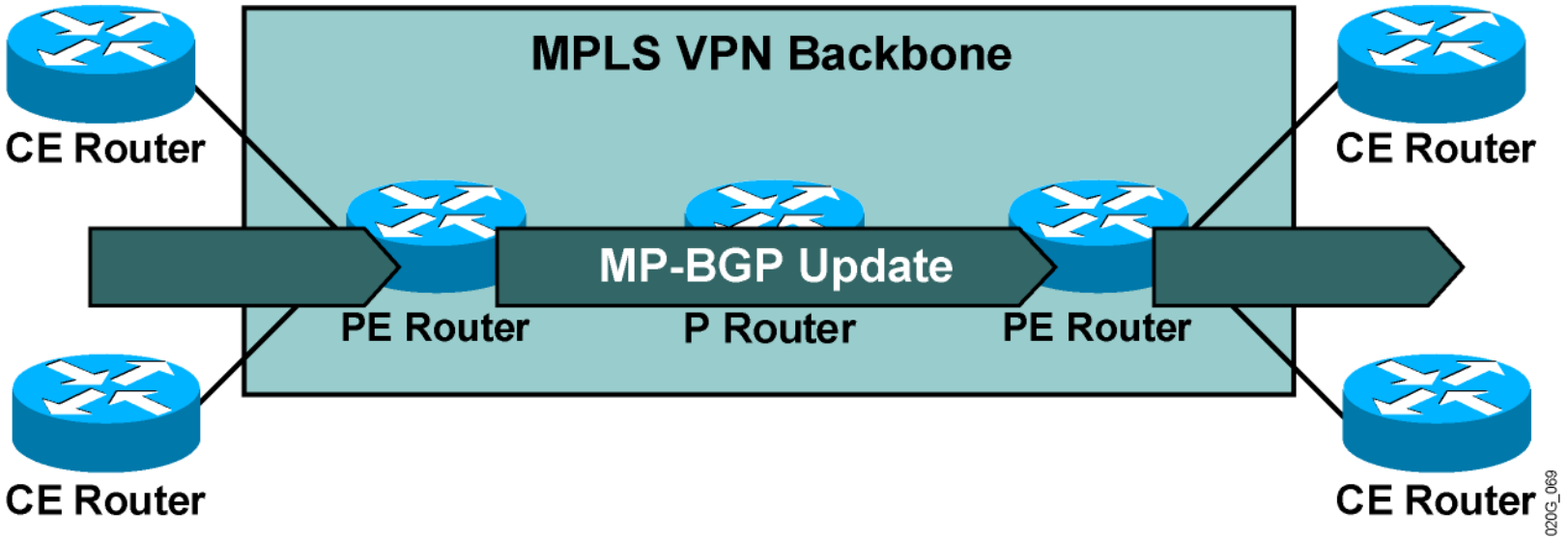


- PE routers export VPN routes from VRF tables into MP-BGP and propagate them as VPNv4 routes to other PE routers.

End-to-End Routing Update Flow (Cont.)
MP-BGP Update

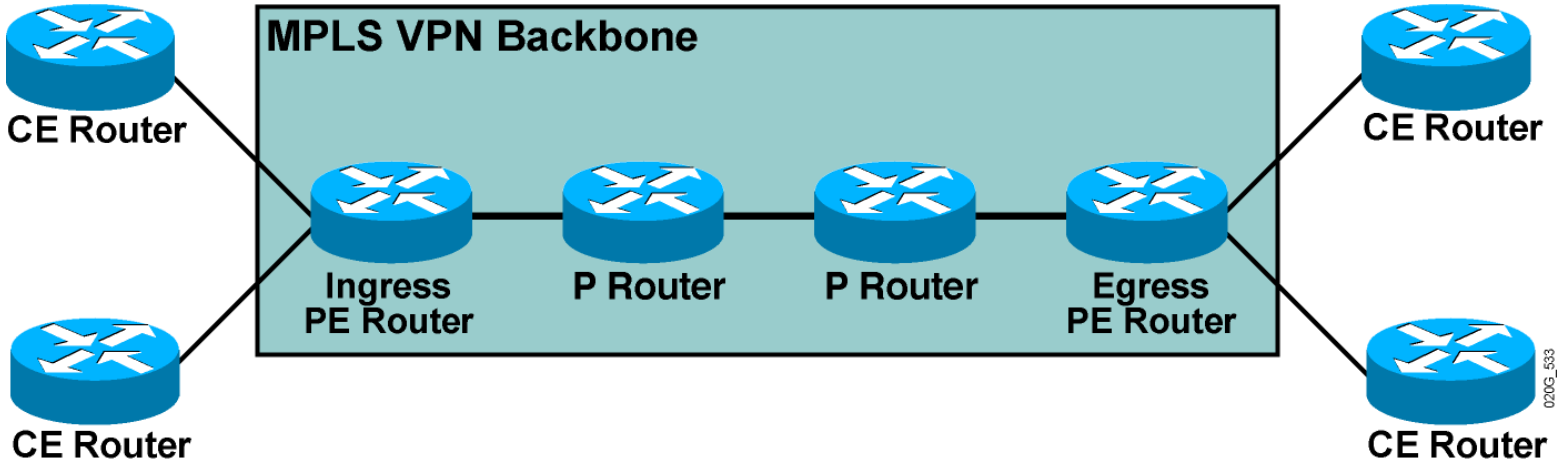
- An MP-BGP update contains the following:
  - VPNv4 address
  - Extended communities
    (route targets, optionally SOO)
  - Label used for VPN packet forwarding
  - Any other BGP attribute (for example, AS path, local preference, MED, standard community)

End-to-End Routing Update Flow (Cont.)



- Receiving PE router imports incoming VPNv4 routes into the appropriate VRF based on route targets attached to the routes.
- Routes installed in VRF are propagated to CE routers.

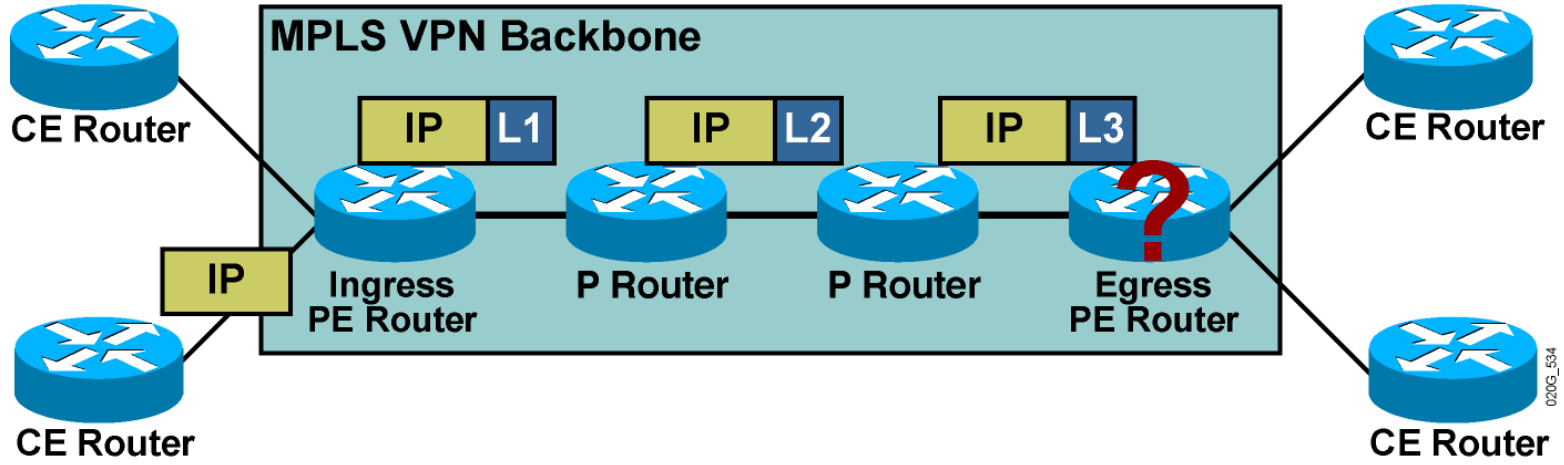
VPN Packet Forwarding Across an MPLS VPN Backbone



Question: How will the PE routers forward the VPN packets across the MPLS VPN backbone?

Answer #1: They will label the VPN packets with an LDP label for the egress PE router and forward the labeled packets across the MPLS backbone.

VPN Packet Forwarding Across an MPLS VPN Backbone



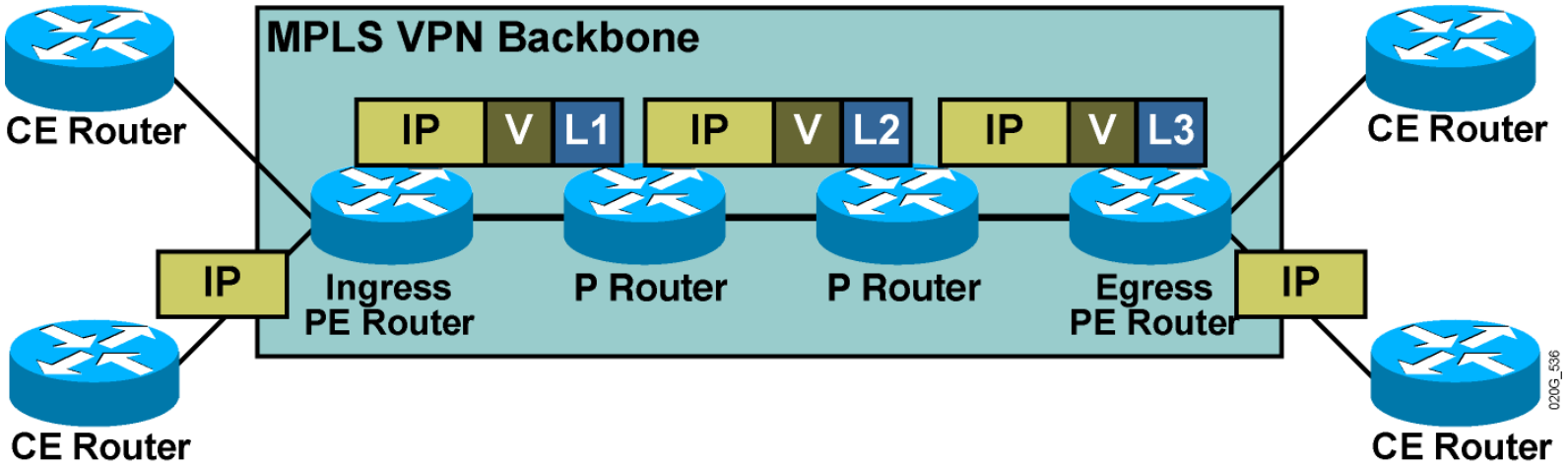
Question: How will the PE routers forward the VPN packets across the MPLS VPN backbone?

Answer #1: They will label the VPN packets with an LDP label for the egress PE router and forward the labeled packets across the MPLS backbone.

Results:

- The P routers perform the label switching, and the packet reaches the egress PE router.
- However, the egress PE router does not know which VRF to use for packet switching, so the packet is dropped.
- How about using a label stack?

VPN Packet Forwarding Across an MPLS VPN Backbone (Cont.)



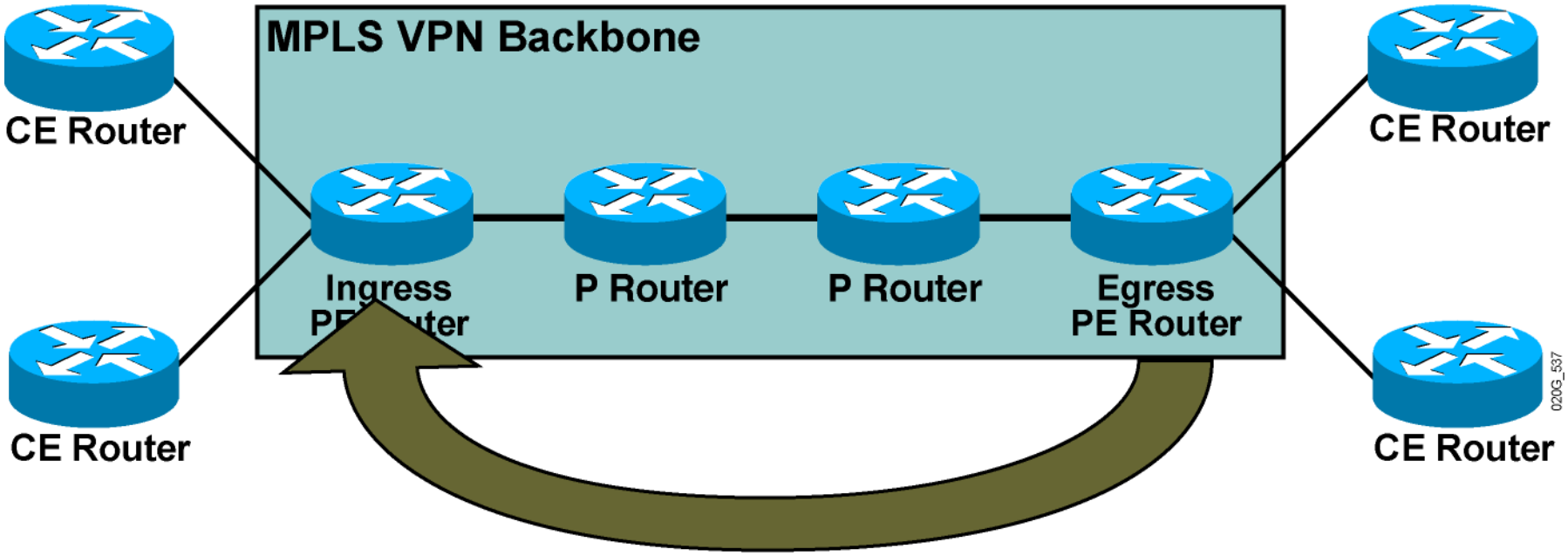
Question: How will the PE routers forward the VPN packets across the MPLS VPN backbone?

Answer #2: They will label the VPN packets with a label stack, using the LDP label for the egress PE router as the top label, and the VPN label assigned by the egress PE router as the second label in the stack.

Result:

- The P routers perform label switching, and the packet reaches the egress PE router.
- The egress PE router performs a lookup on the VPN label and forwards the packet toward the CE router.

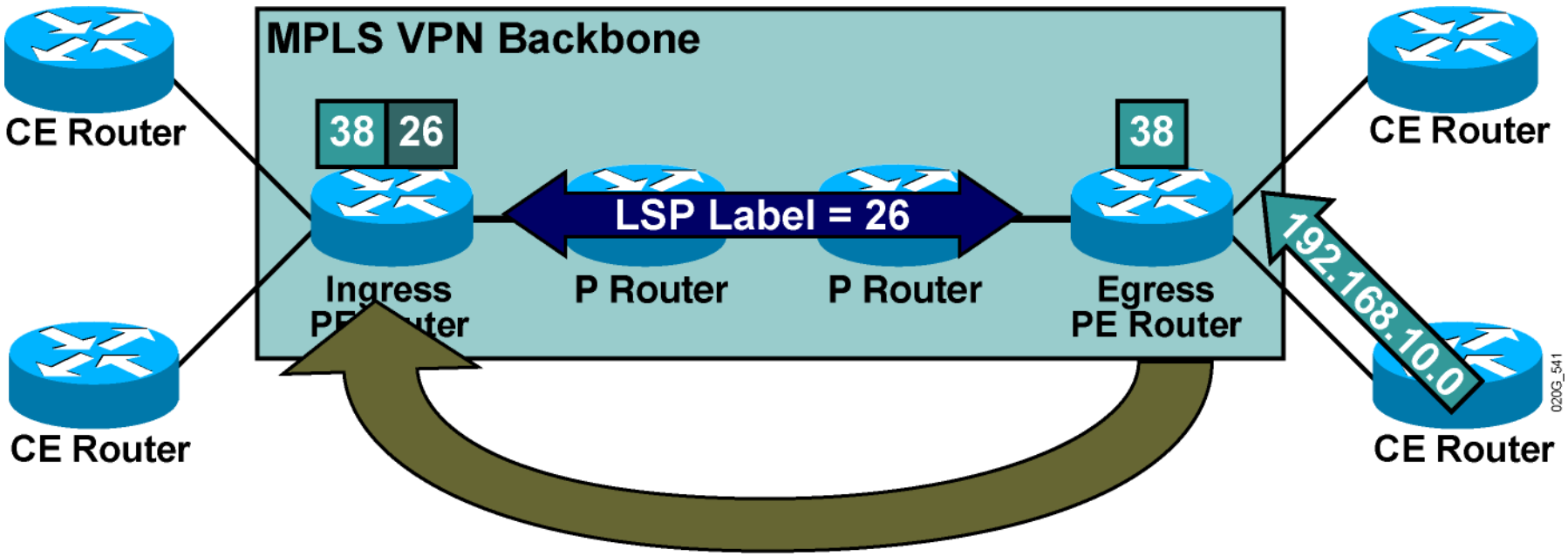
VPN Label Propagation



Question: How will the ingress PE router get the second label in the label stack from the egress PE router?

Answer: Labels are propagated in MP-BGP VPNv4 routing updates.

VPN Label Propagation (Cont.)



Step 1: A VPN label is assigned to every VPN route by the egress PE router.

Step 2: The VPN label is advertised to all other PE routers in an MP-BGP update.

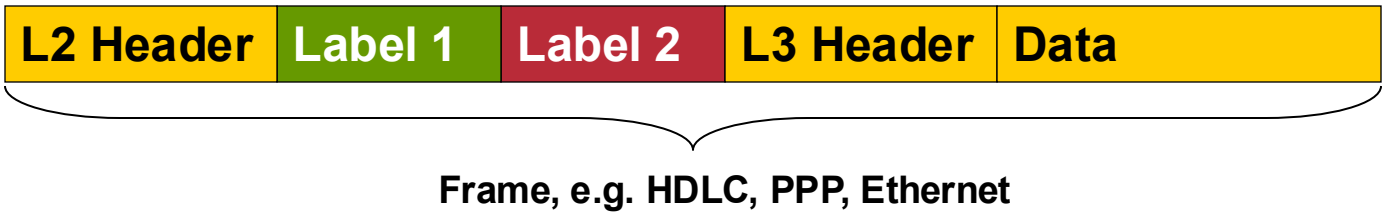
Step 3: A label stack is built in the VRF table.

MPLS VPNs and Label Propagation

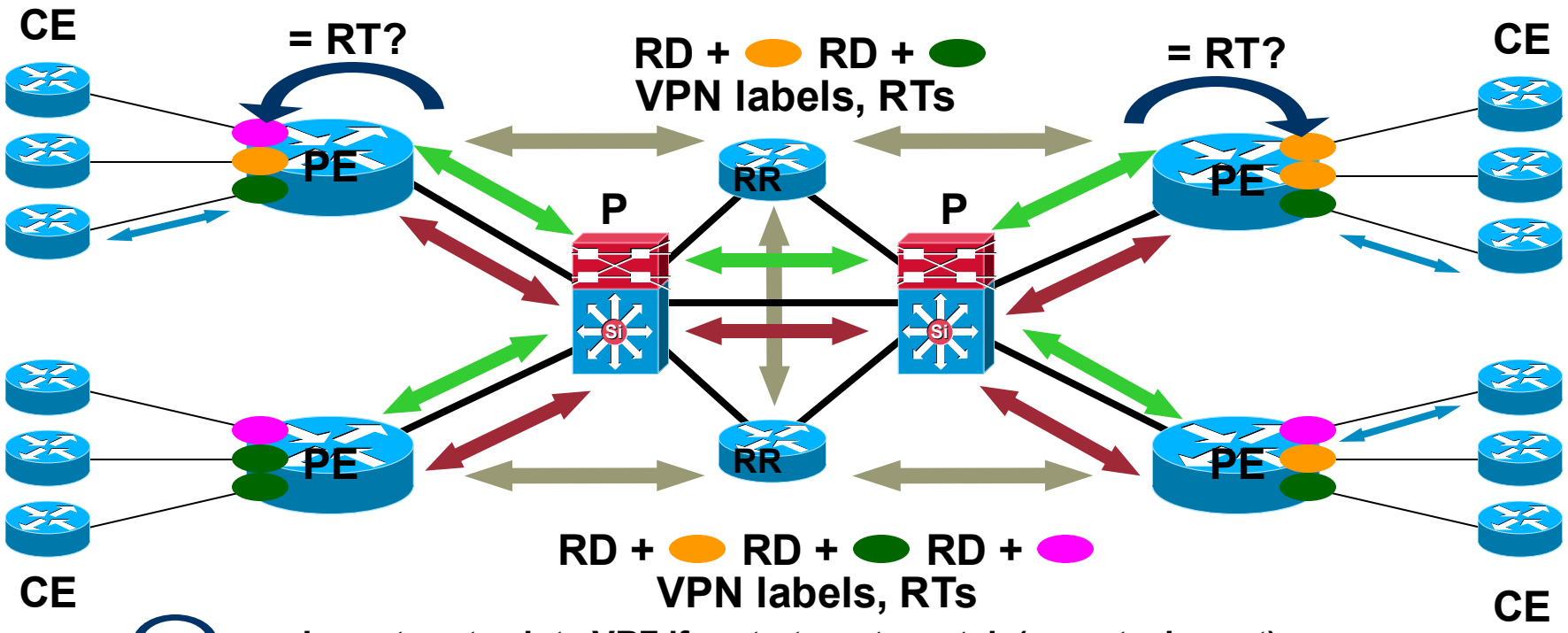
- The VPN label must be assigned by the BGP next hop.
- The BGP next hop should not be changed in the MP-IBGP update propagation.
- The PE router must be the BGP next hop.
- The label must be reoriginated if the next hop is changed.
  - A new label is assigned every time that the MP-BGP update crosses the AS boundary where the next hop is changed.

MPLS VPN Label Stack

- There are at least two labels when using MPLS-VPN
- The first label is distributed by TDP/LDP
  - Derived from an IGP route
  - Corresponds to a PE address (VPN egress point)
  - PE addresses are MP-BGP next-hops of VPN routes
- The second label is distributed by MP-BGP
  - Corresponds to the actual VPN route
  - Identifies the PE outgoing interface or routing table



MPLS VPN Operation



Import routes into VRF if route-targets match (export = import)

Customer routes placed into separate VRF tables at each PE

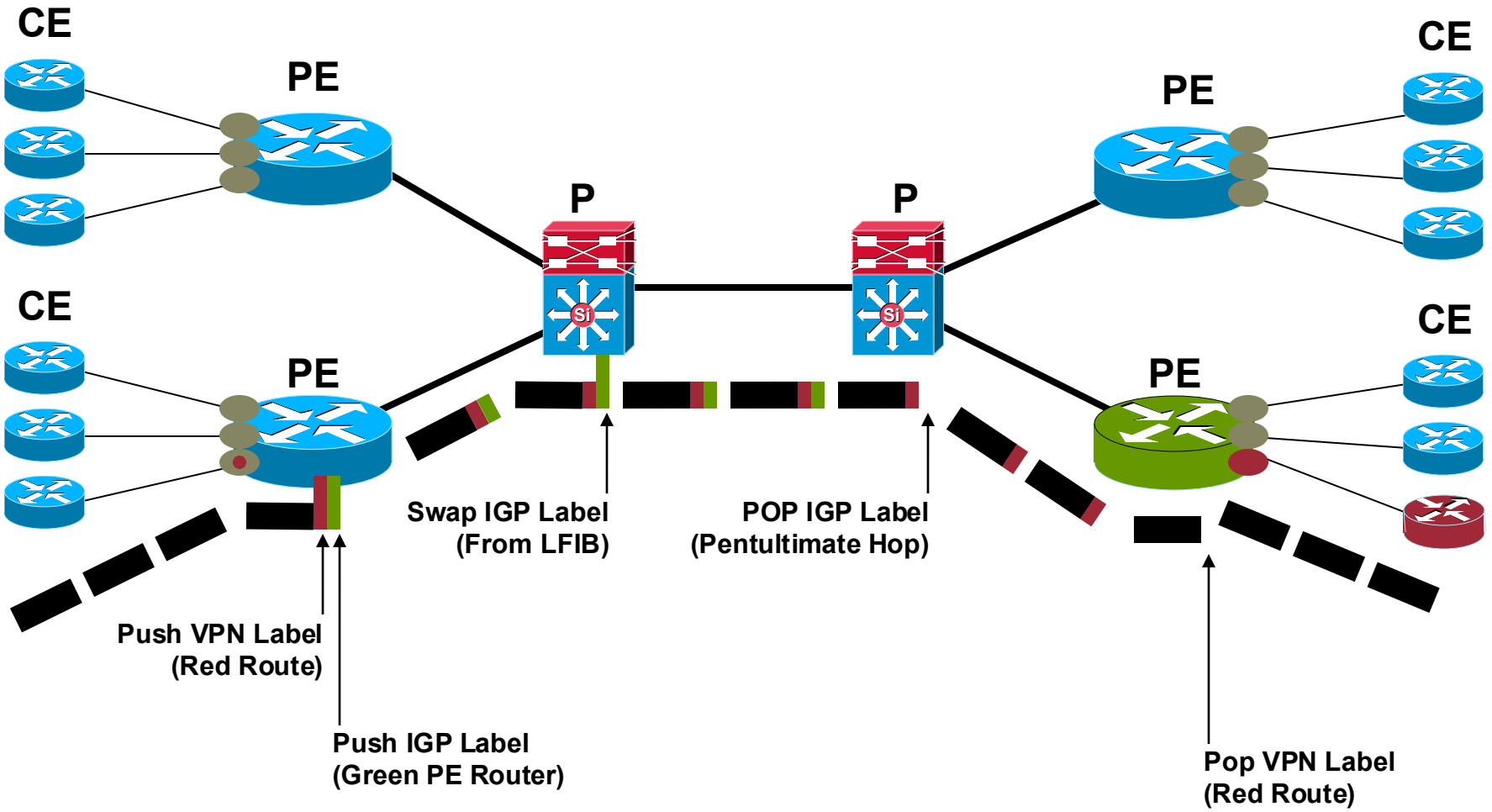
IGP (OSPF,ISIS) used to establish reachability to destination networks.

Label Distribution Protocol establishes mappings to IGP addresses

CE-PE dynamic routing (or static) populate the VRF routing tables

MP-BGP between PE router to distribute routes between VPNs

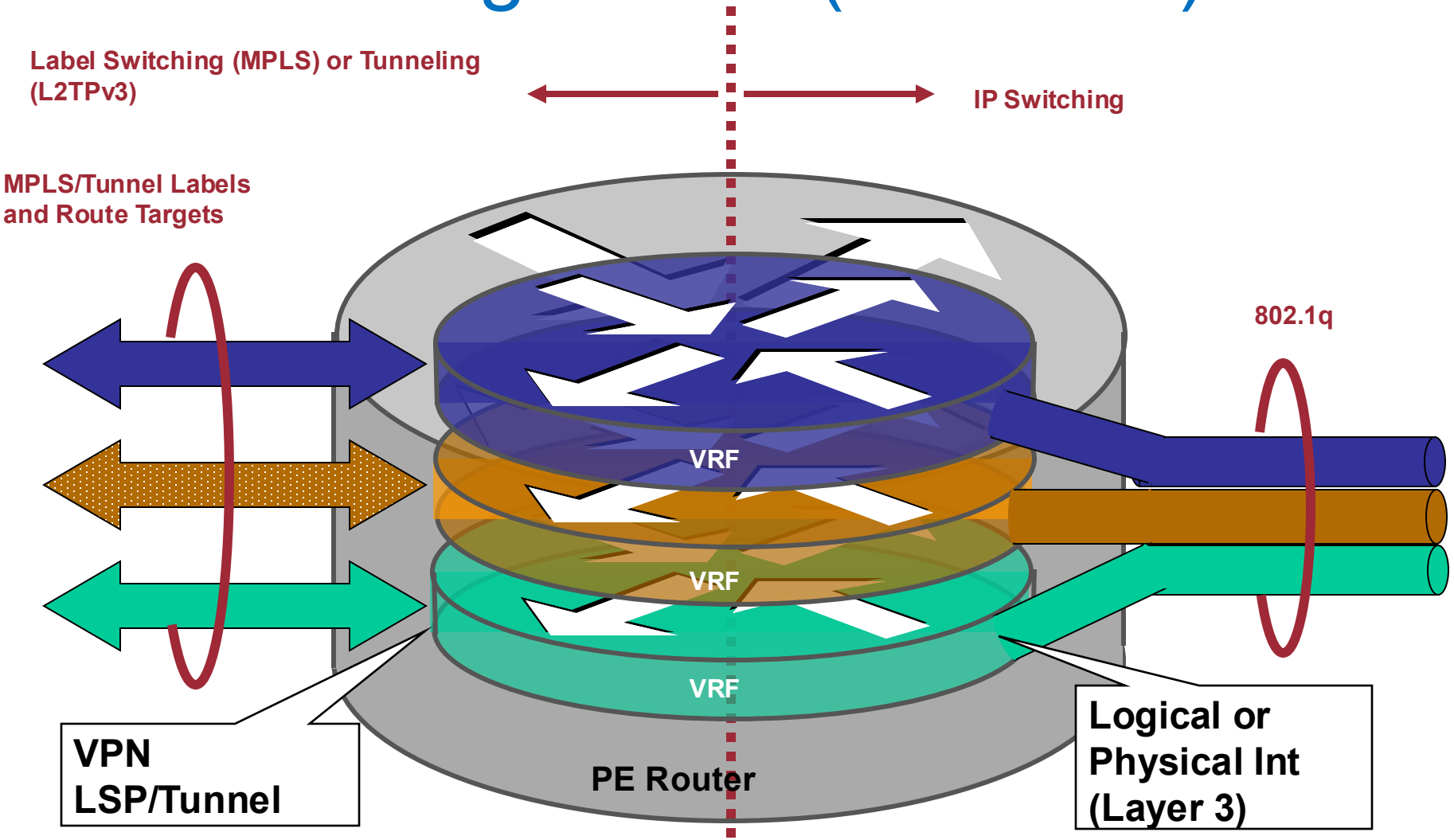
MPLS VPN Forwarding Example



Virtual Routing and Forwarding

- A VRF is the routing and forwarding instance for a set of sites with identical connectivity requirements.
- Data structures associated with a VRF:
  - IP routing table
  - Cisco Express Forwarding (CEF) forwarding table
  - Set of rules and routing protocol parameters (routing protocol contexts)
  - List of interfaces that use the VRF
- Other information associated with a VRF:
  - Route Distinguisher (RD)
  - Set of import and export route targets

Provider Edge VRFs (PE VRFs)



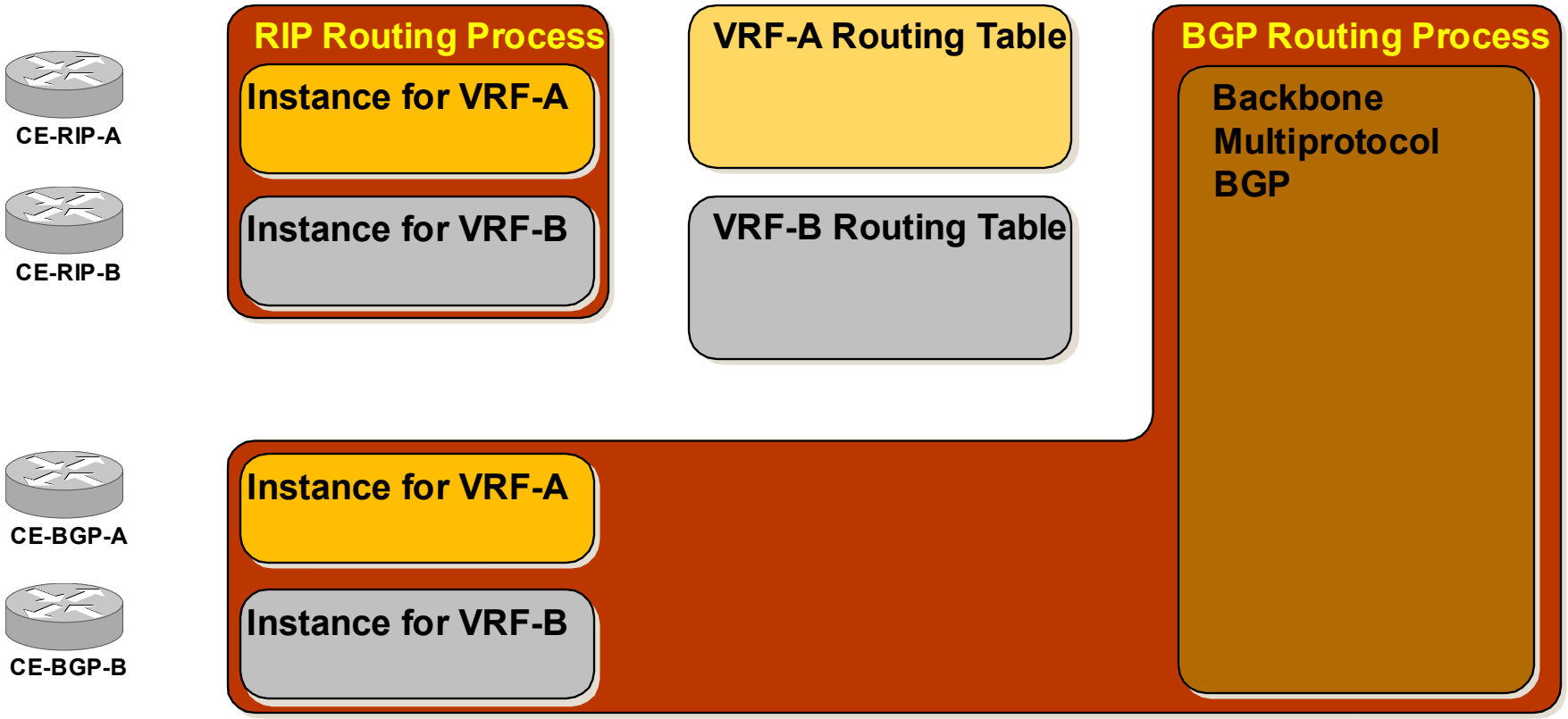
VPN-Aware Routing Protocols

- Routing context = routing protocol run in one VRF
  - Supported by VPN-aware routing protocols:
    - External BGP (EBGP), OSPF, RIP version 2 (RIPv2), EIGRP, IS-IS, Static routes
  - Implemented as several instances of a single routing process (EBGP, RIPv2) or as several routing processes (OSPF)
  - Independent per-instance router variables for each instance

VRF Routing Table

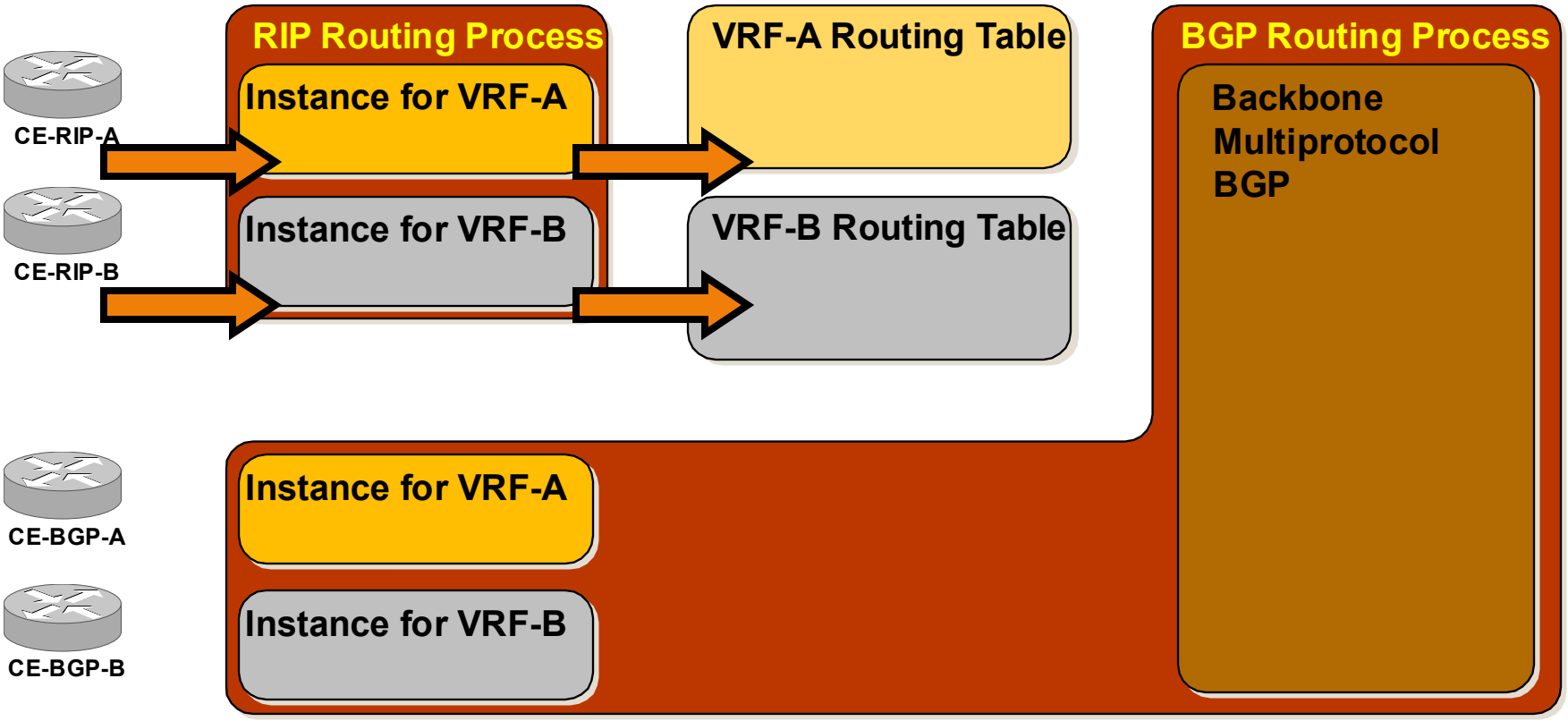
- Contains routes that should be available to a particular set of sites
- Analogous to standard Cisco IOS software routing table; supports same set of mechanisms
- VPN interfaces (physical interface, subinterfaces, logical interfaces) assigned to VRFs
  - Many interfaces per VRF
  - Each interface assignable to only one VRF

Routing Contexts, VRF, and MP-BGP Interaction: 1/9



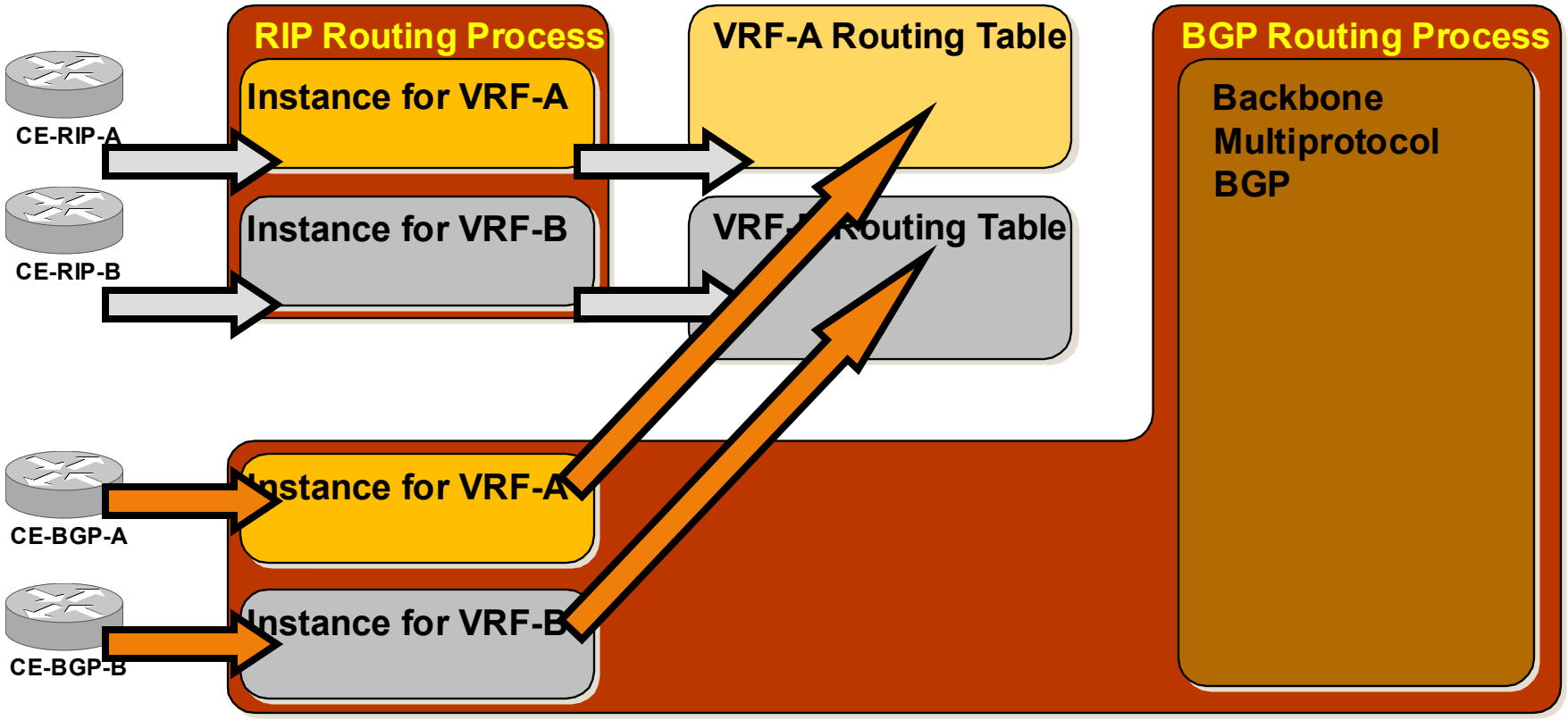
- Two VPNs attached to the same PE router
- Each VPN represented by a VRF (VRF-A and VRF-B)
- RIP and BGP running between PE and CE routers

Routing Contexts, VRF, and MP-BGP Interaction: 2/9



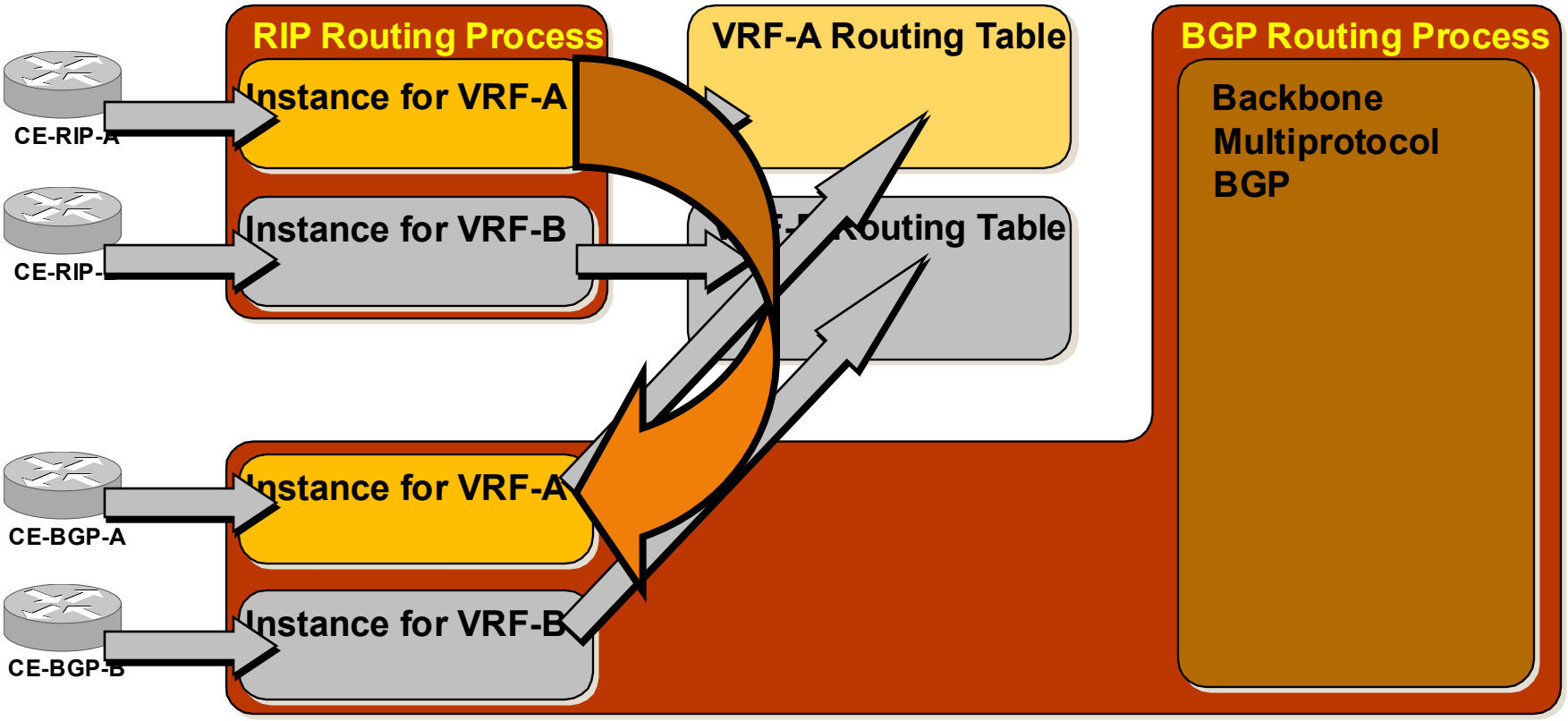
- RIP-speaking CE routers announce their prefixes to the PE router via RIP.
- Instance of RIP process associated with the VRF into which the PE-CE interface belongs collects the routes and inserts them into VRF routing table.

Routing Contexts, VRF, and MP-BGP Interaction: 3/9



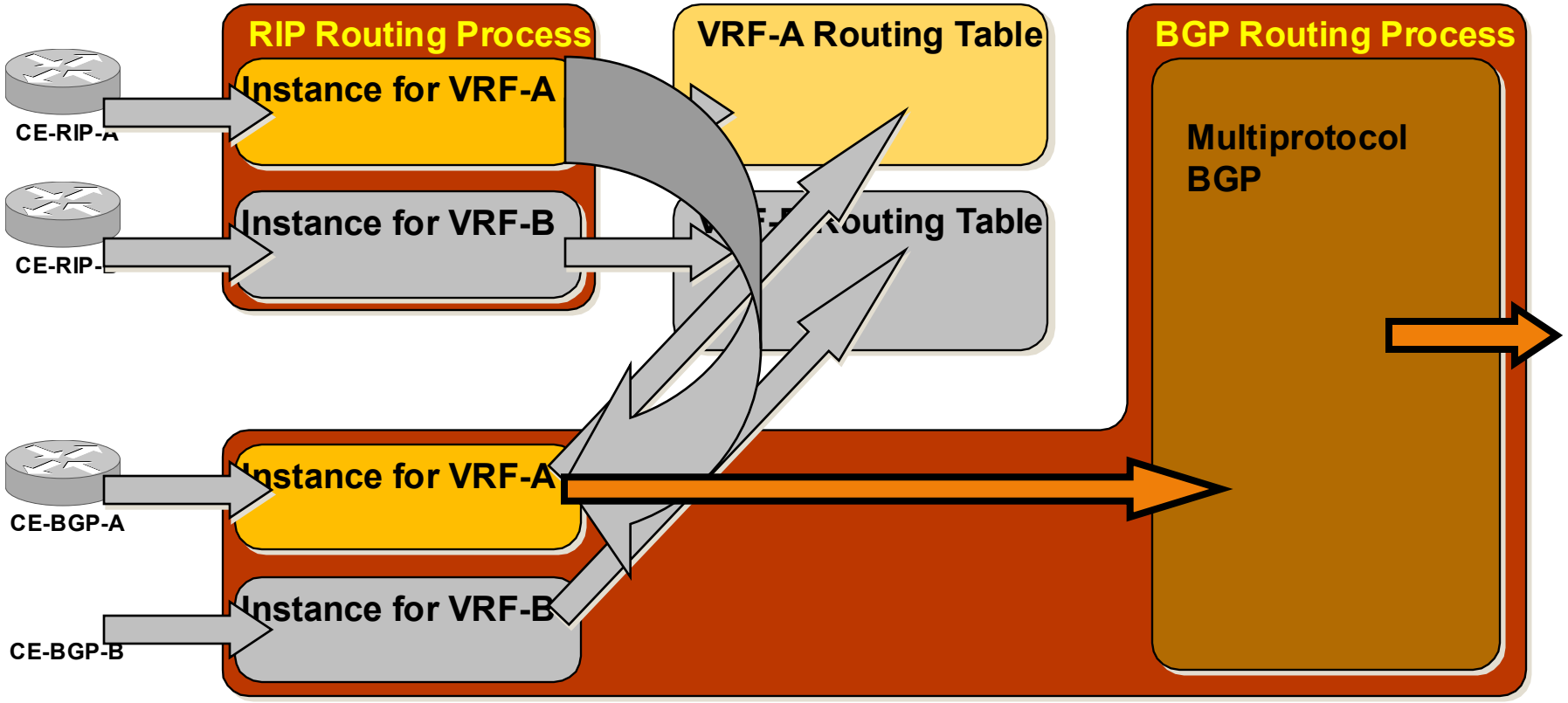
- BGP-speaking CE routers announce their prefixes to the PE router via BGP.
- Instance of BGP process associated with the VRF into which the PE-CE interface belongs collects the routes and inserts them into VRF routing table.

Routing Contexts, VRF, and MP-BGP Interaction: 4/9



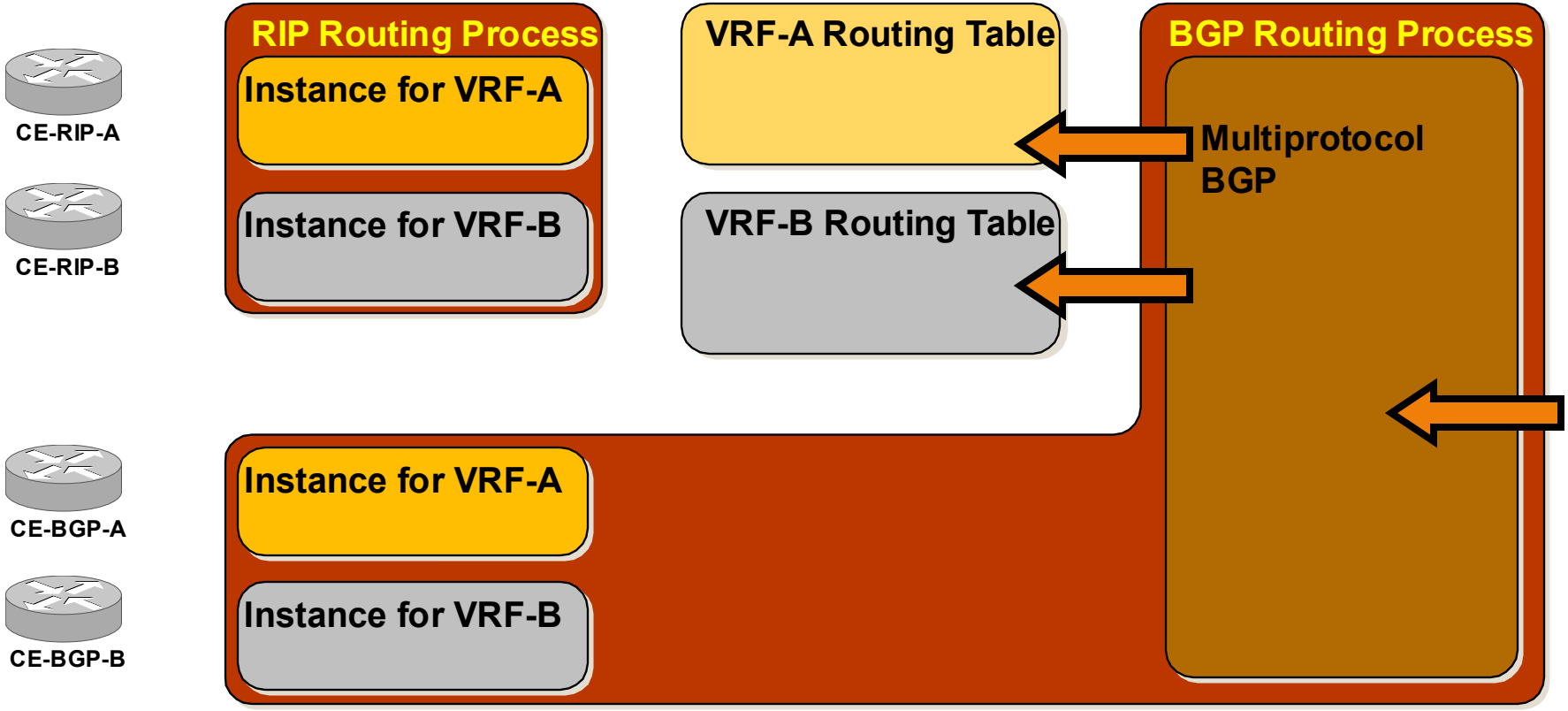
- **RIP routes entered in the VRF routing table are redistributed into BGP for further propagation into the MPLS VPN backbone.**
- **Redistribution between RIP and BGP has to be configured for proper MPLS VPN operation.**

Routing Contexts, VRF, and MP-BGP Interaction: 5/9



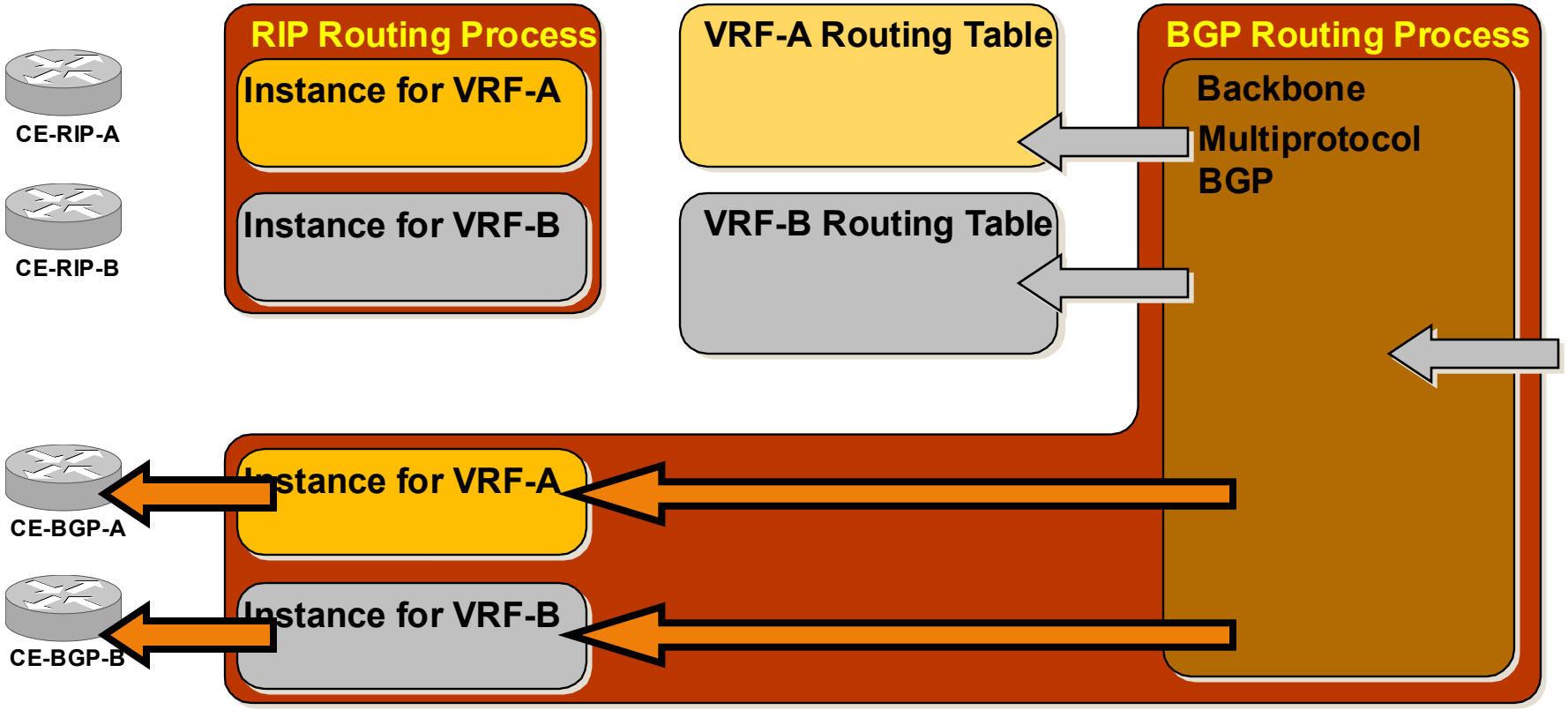
- Route distinguisher is prepended during route export to the BGP routes from VRF instance of BGP process to convert them into VPNv4 prefixes. Route targets are attached to these prefixes.
- VPNv4 prefixes are propagated to other PE routers.

Routing Contexts, VRF, and MP-BGP Interaction: 6/9



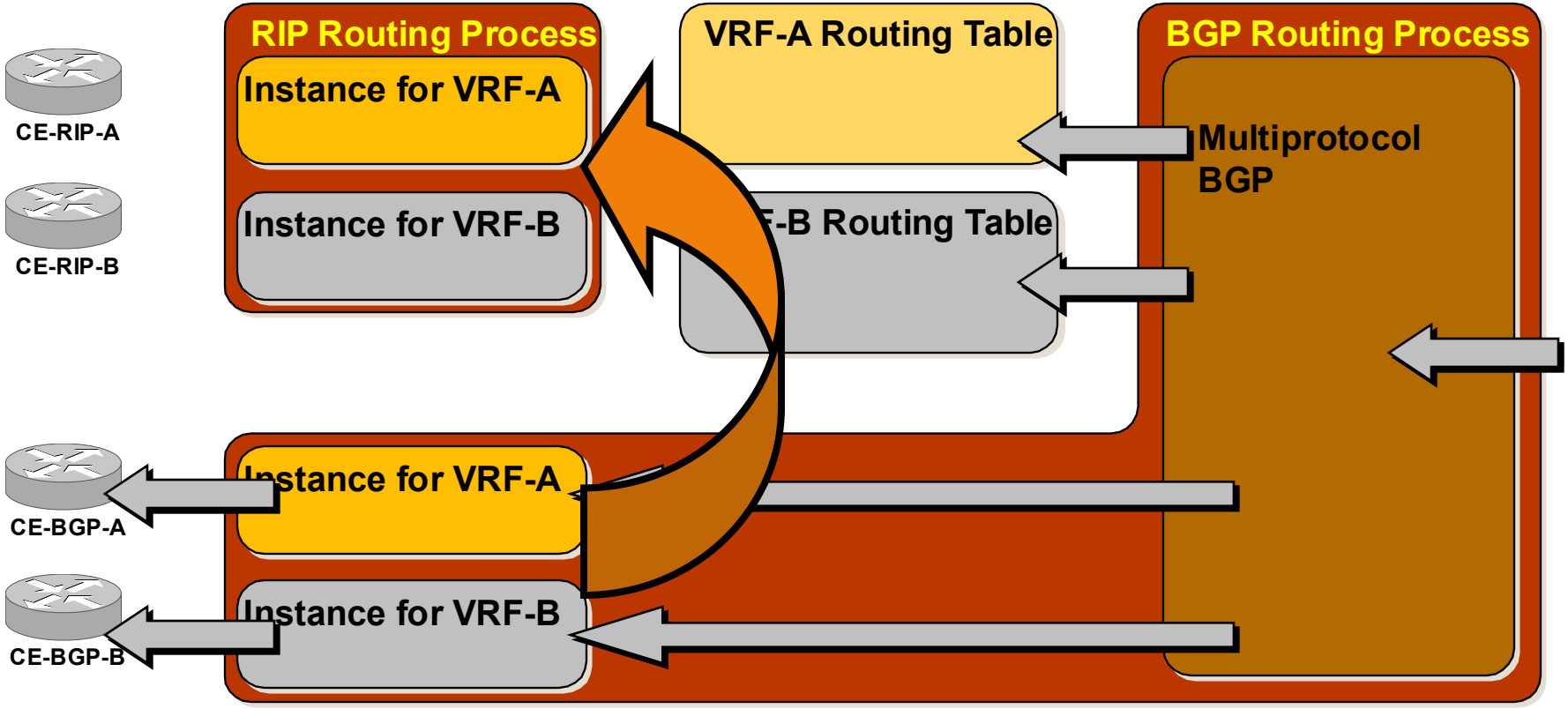
- VPNv4 prefixes are received from other PE routers.
- The VPNv4 prefixes are inserted into proper VRF routing tables based on their route targets and import route targets configured in VRFs.
- Route distinguisher is removed during this process.

Routing Contexts, VRF, and MP-BGP Interaction: 7/9



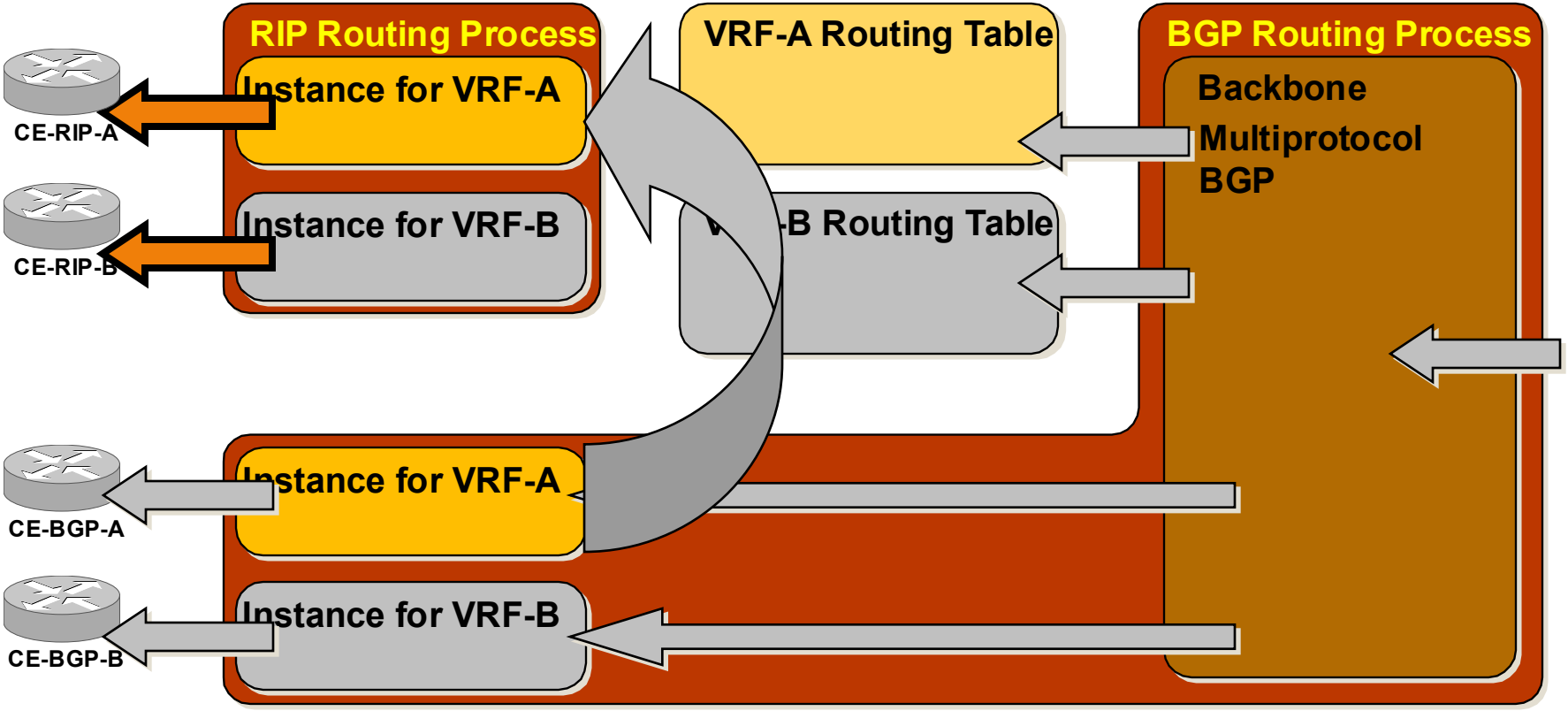
- **Routes received from backbone MP-BGP and imported into a VRF are forwarded as IPv4 routes to EBGP CE neighbors attached to that VRF.**

Routing Contexts, VRF, and MP-BGP Interaction: 8/9



- MP-IBGP routes imported into a VRF are redistributed into the instance of RIP configured for that VRF.
- Redistribution between BGP and RIP has to be configured for end-to-end RIP routing between CE routers.

Routing Contexts, VRF, and MP-BGP Interaction: 9/9



- **Routes redistributed from BGP into a VRF instance of RIP are sent to RIP-speaking CE routers.**



Traceroute in an MPLS VPN

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TTL Field in Labels

- Only the TTL field in the label at the top of the stack counts
- The outgoing TTL value is only a function of the incoming TTL value
- Outgoing TTL is one less than incoming TTL
- If outgoing TTL = 0, packet is not forwarded (not even stripped and forwarded as an ip packet)

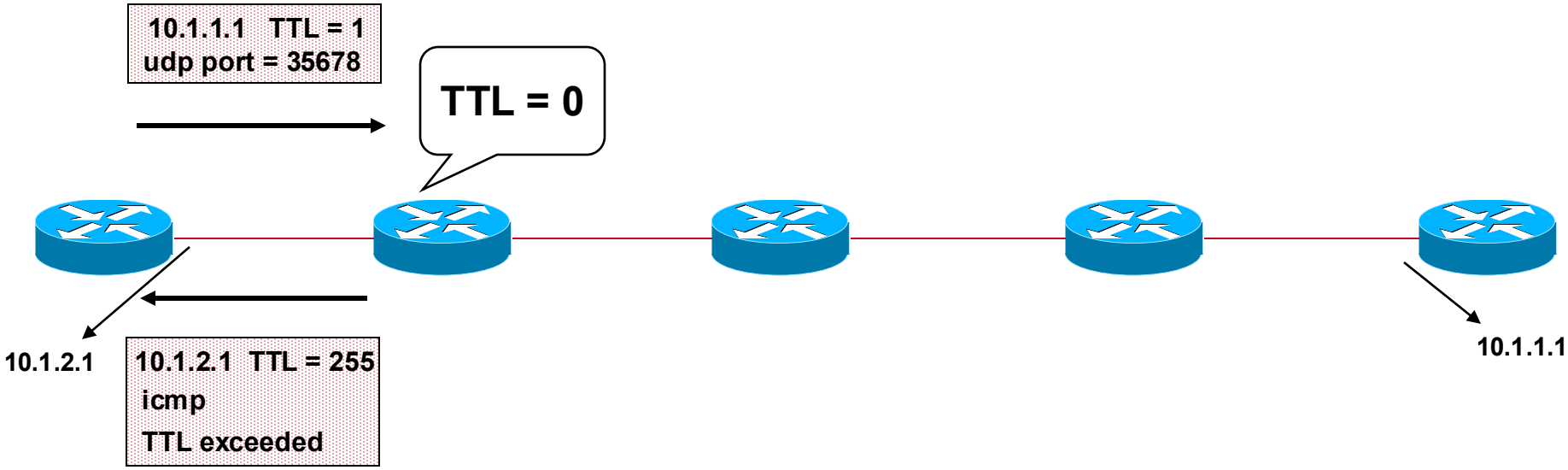
Copying of TTL field

- When an ip packet is first labeled, the TTL field is copied from the ip header to the MPLS header (after being decremented by 1)
- When the label stack is removed, the outgoing TTL value is copied to the TTL field in the ip header
  - Unless MPLS TTL > IP TTL

Traceroute in IP Network

all routers switch ip packets

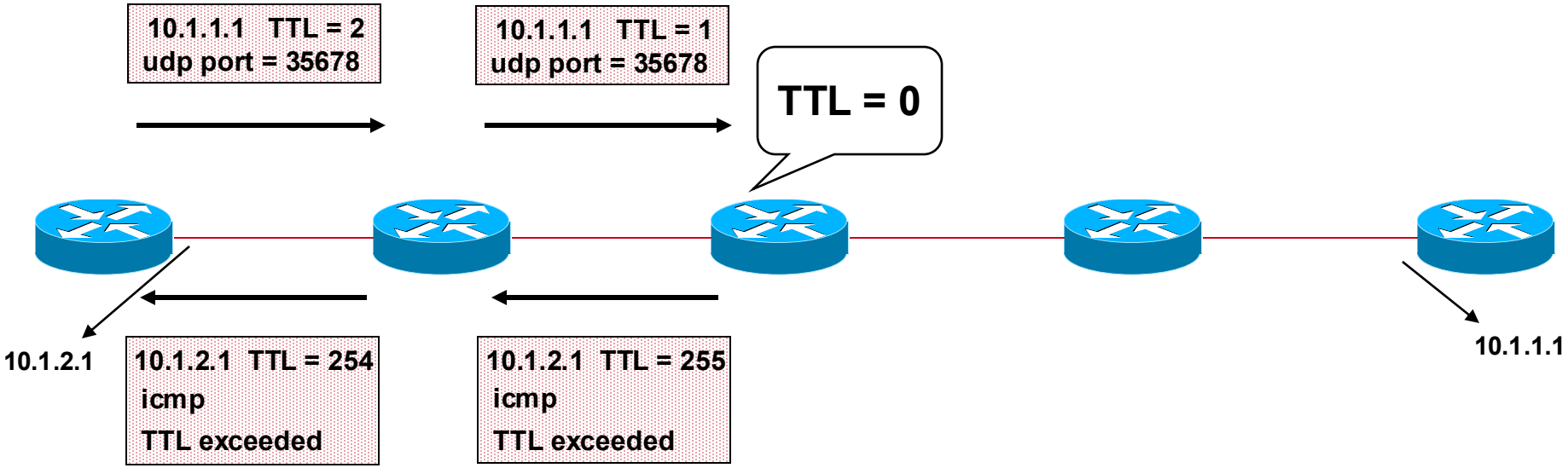
probe 1



Traceroute in IP Network

all routers switch ip packets

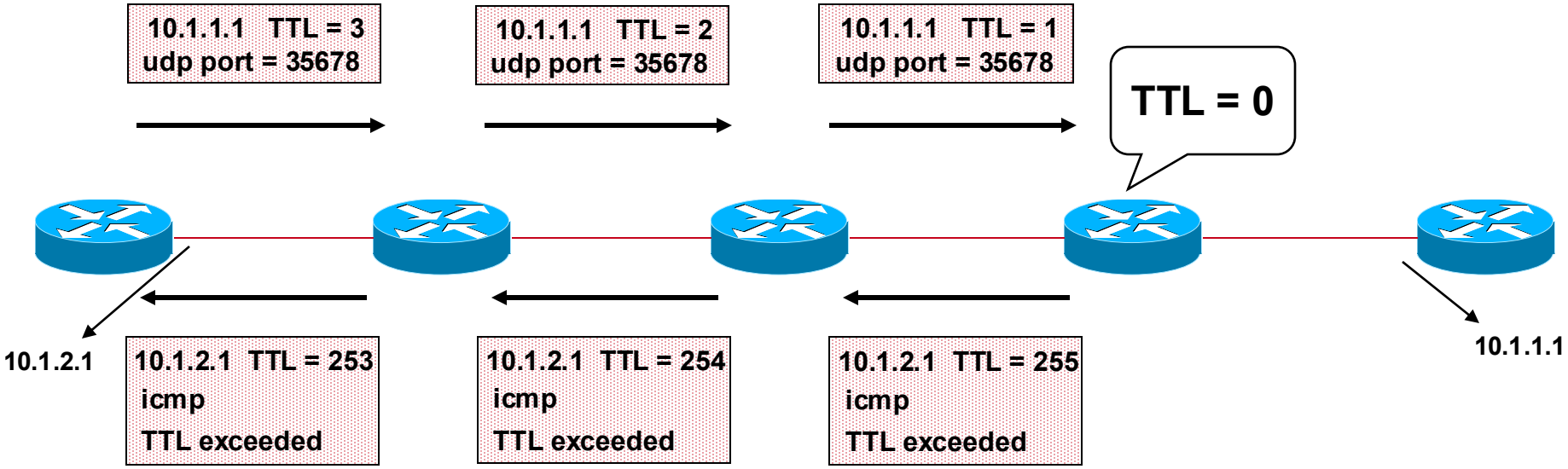
probe 2



Traceroute in IP Network

all routers switch ip packets

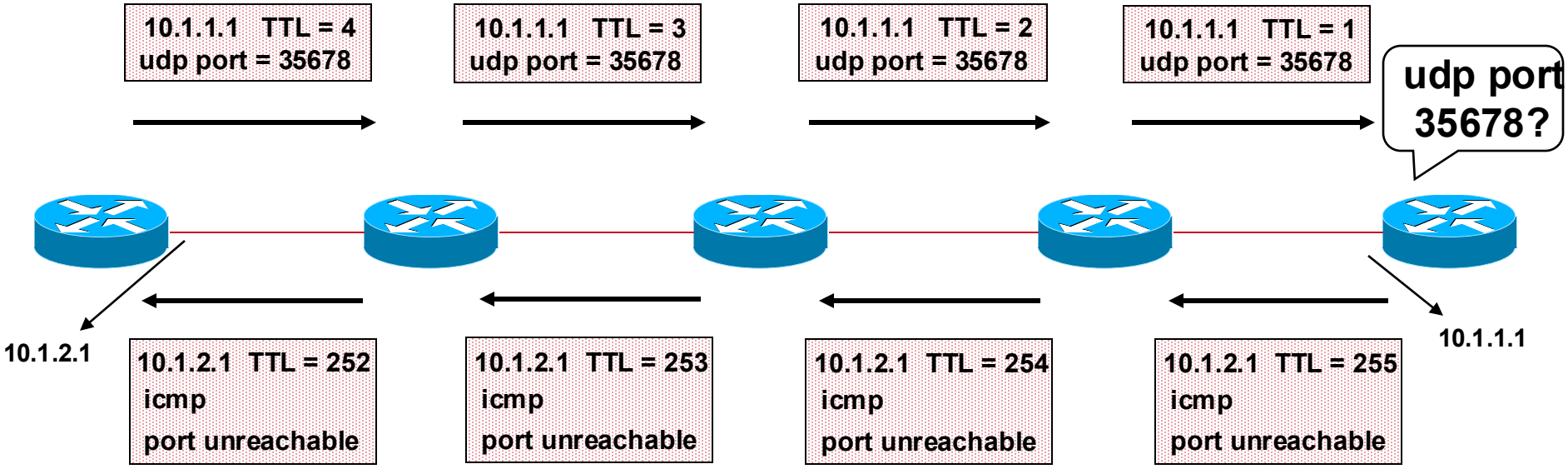
probe 3



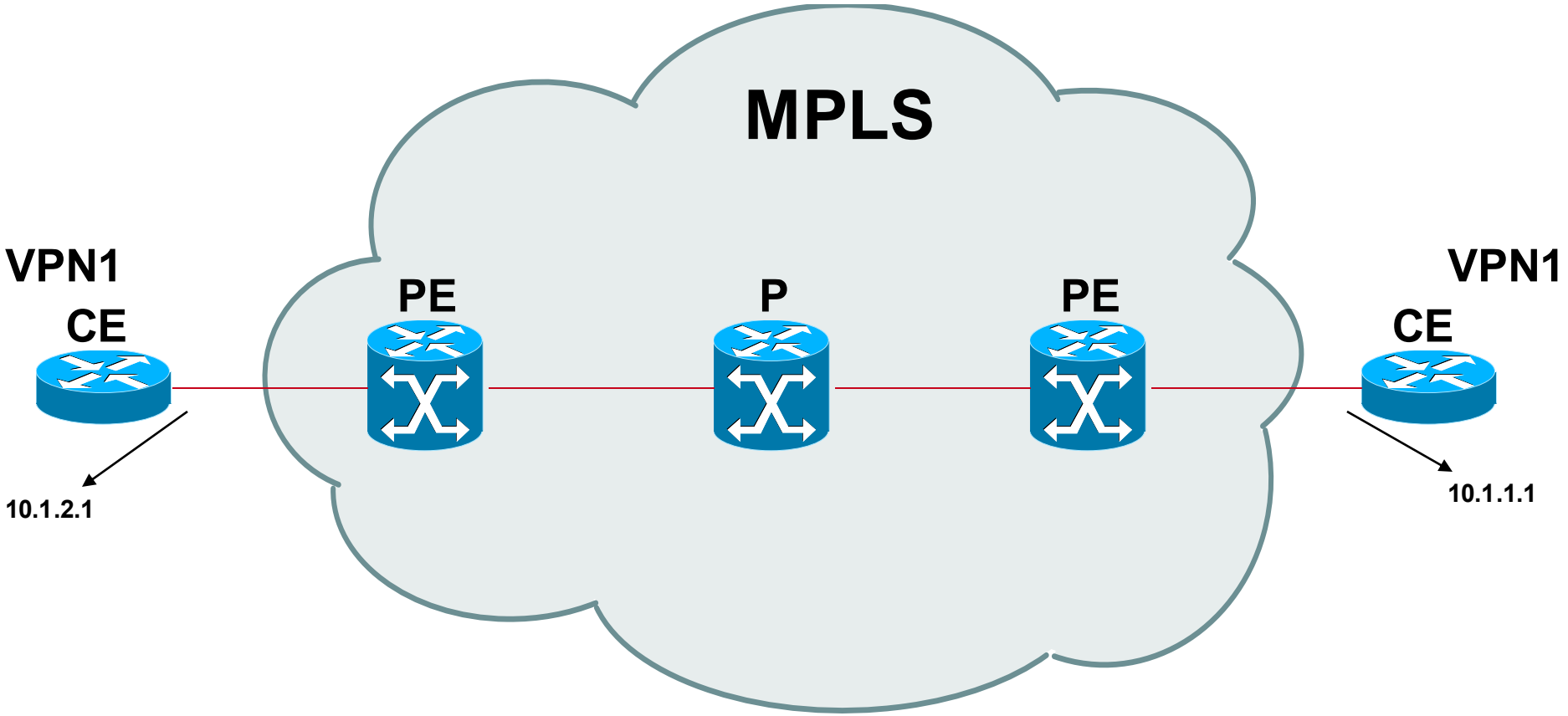
Traceroute in IP Network

all routers switch ip packets

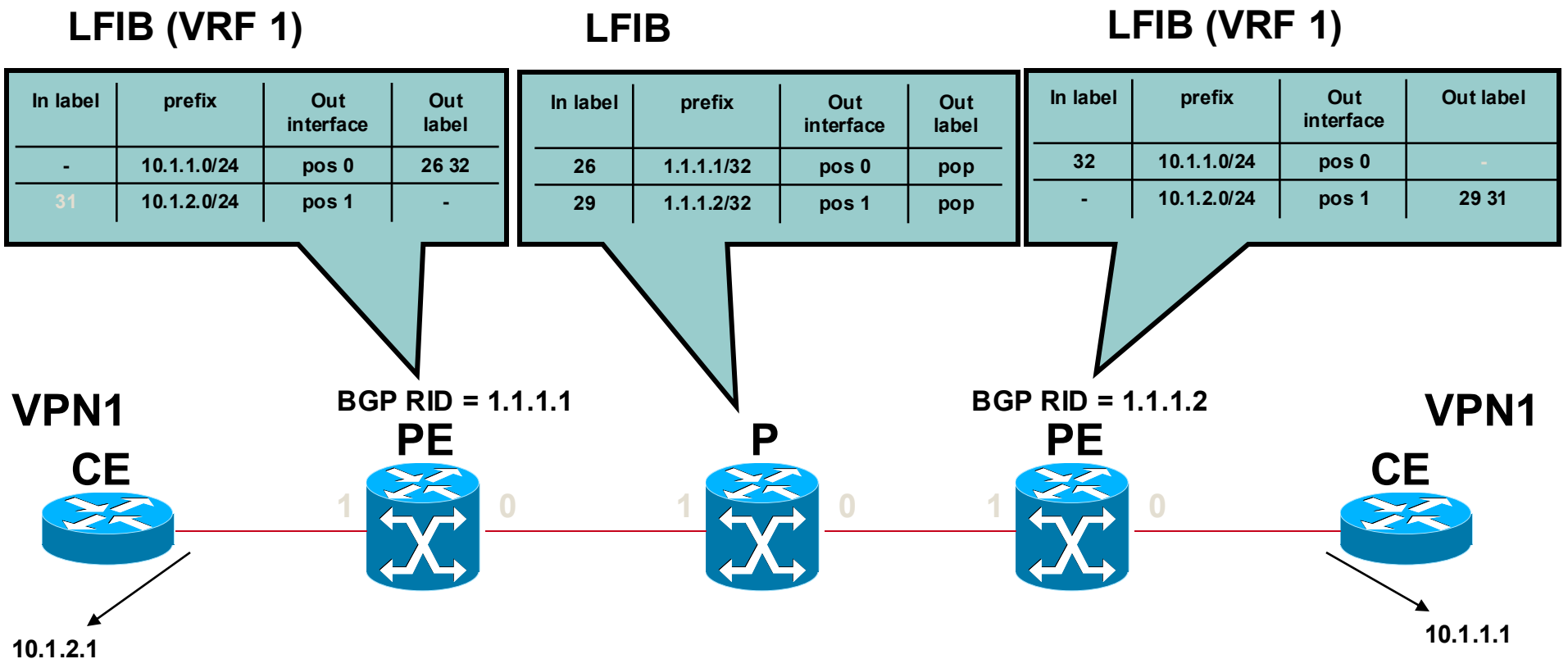
probe 4



Traceroute in MPLS VPN Network

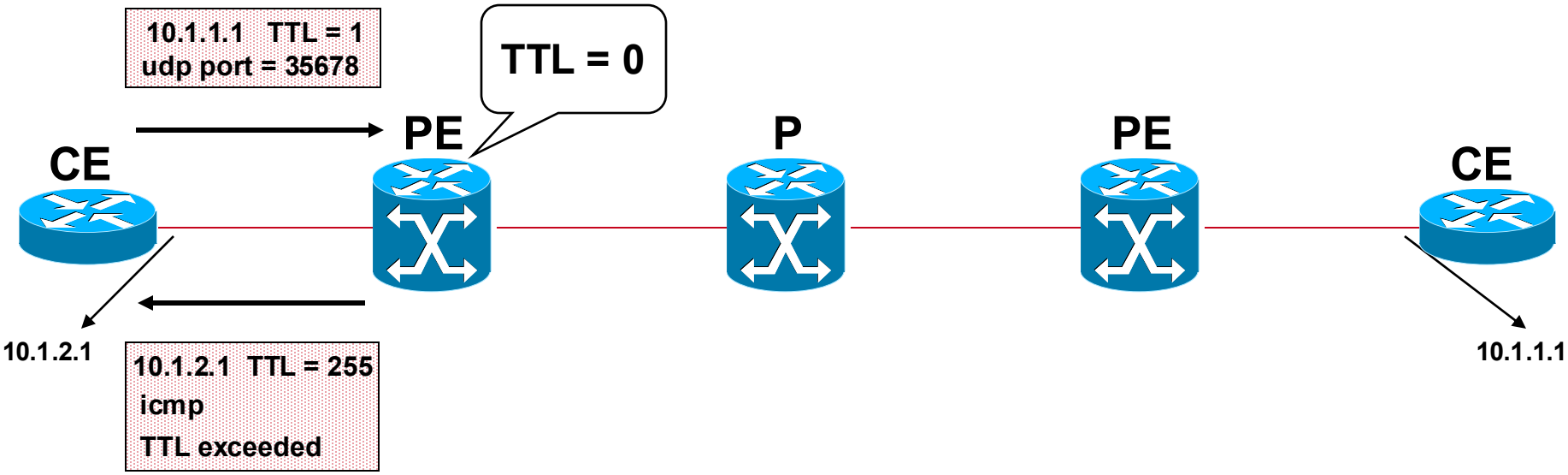


Traceroute in MPLS VPN Network



Traceroute in MPLS VPN Network

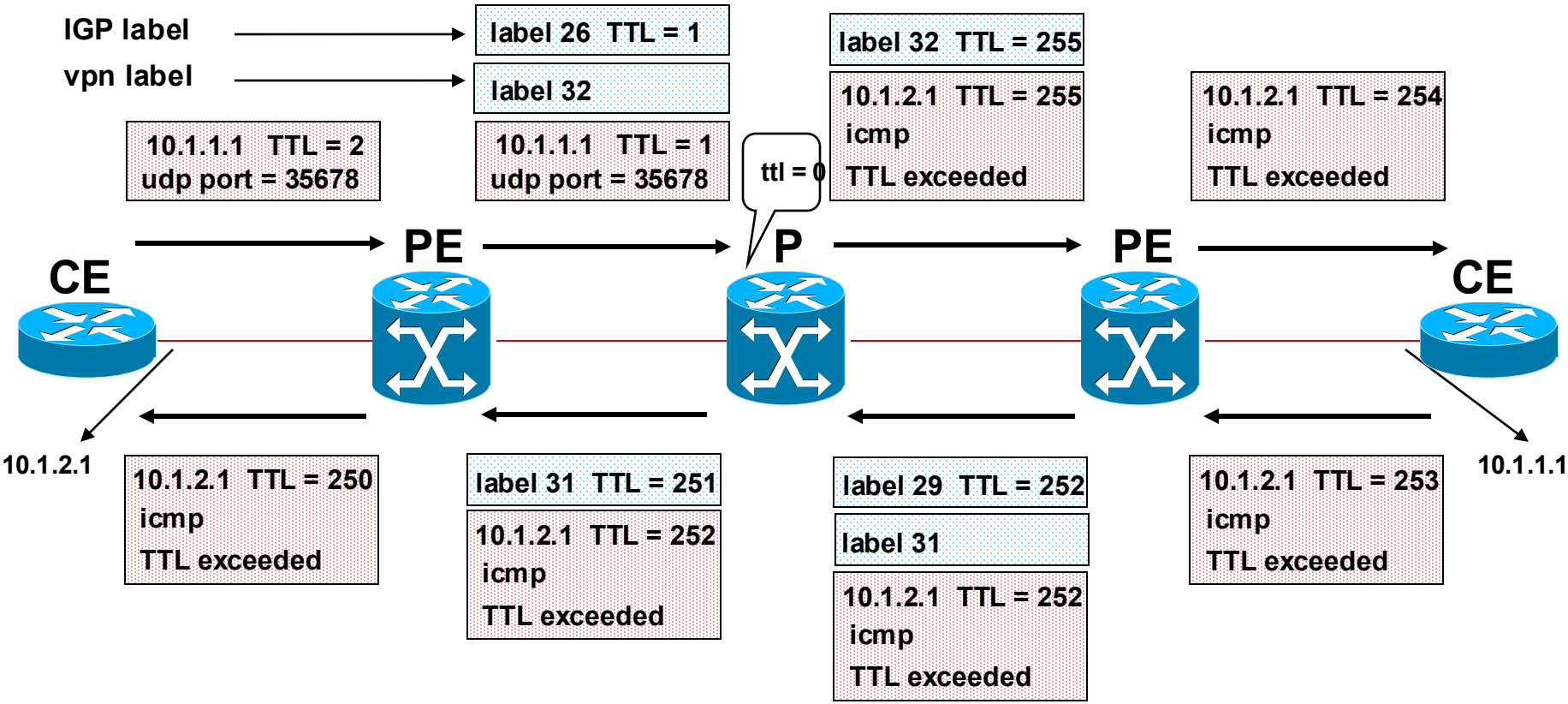
probe 1



Traceroute in MPLS VPN Network

probe 2

tag-switching ip propagate-ttl on PE routers (default behaviour)



Traceroute : Example

```
singularity#traceroute 10.1.2.5

Type escape sequence to abort.
Tracing the route to 10.1.2.5

  1 10.1.1.26 [MPLS: Label 12310 Exp 0] 0 msec 0 msec 4 msec
  2 10.1.1.2 [MPLS: Label 12313 Exp 0] 0 msec 0 msec 0 msec
  3 10.1.1.6 [MPLS: Label 12312 Exp 0] 0 msec 4 msec 0 msec
  4 10.1.1.30 0 msec *  0 msec

singularity#traceroute vrf one 10.1.4.5

Type escape sequence to abort.
Tracing the route to 10.1.4.5

  1 10.1.1.26 [MPLS: Labels 12310/29 Exp 0] 0 msec 0 msec 4 msec
  2 10.1.1.2 [MPLS: Labels 12313/29 Exp 0] 0 msec 0 msec 0 msec
  3 10.1.1.6 [MPLS: Labels 12312/29 Exp 0] 0 msec 0 msec 0 msec
  4 10.1.4.6 [MPLS: Label 29 Exp 0] 0 msec 0 msec 4 msec
  5 10.1.4.5 0 msec *  0 msec
```

MPLS

MPLS VPN

TTL Commands

- forwarding -> tag ip propagate-ttl forwarded (default)
  - ip ttl is copied to label ttl
- not forwarding- > no tag ip propagate-ttl forwarded
  - MPLS ttl is set to 255
  - MPLS Core is “invisible” in traceroutes
- forwarding locally -> tag ip propagate-ttl local
  - IP packets coming from the CE are imposed with a label carrying ttl = 255
  - IP packets originated on PE within the vrf get label ttl = ip ttl
  - Result :
    - a traceroute from vrf on PE to a vrf destination, will still show all the hops : provider sees all hops in the MPLS cloud
    - a traceroute from CE to a VRF destination show the MPLS cloud as one ip hop : customer does not see the number of hops in the MPLS cloud

No Tag IP Propagate-ttl Forwarded

- Set on LSR doing ip->mpls, so on PEs

Traceroute CE to CE

tag ip propagate-ttl forwarded

```
moorcock#traceroute 10.1.5.2

  1 10.1.4.2 4 msec 0 msec 4 msec
  2 10.1.1.26 [MPLS: Labels 12310/30 Exp 0] 4 msec 4 msec 8 msec
  3 10.1.1.2 [MPLS: Labels 12313/30 Exp 0] 8 msec 4 msec 4 msec
  4 10.1.1.6 [MPLS: Labels 12312/30 Exp 0] 8 msec 8 msec 4 msec
  5 10.1.4.6 [MPLS: Label 30 Exp 0] 4 msec 4 msec 4 msec
  6 10.1.4.5 4 msec *  4 msec
```

local PE

P

P

P

remote PE

remote CE

no tag ip propagate-ttl forwarded

```
moorcock#traceroute 10.1.5.2

  1 10.1.4.2 0 msec 0 msec 4 msec
  2 10.1.4.5 4 msec *  4 msec
```

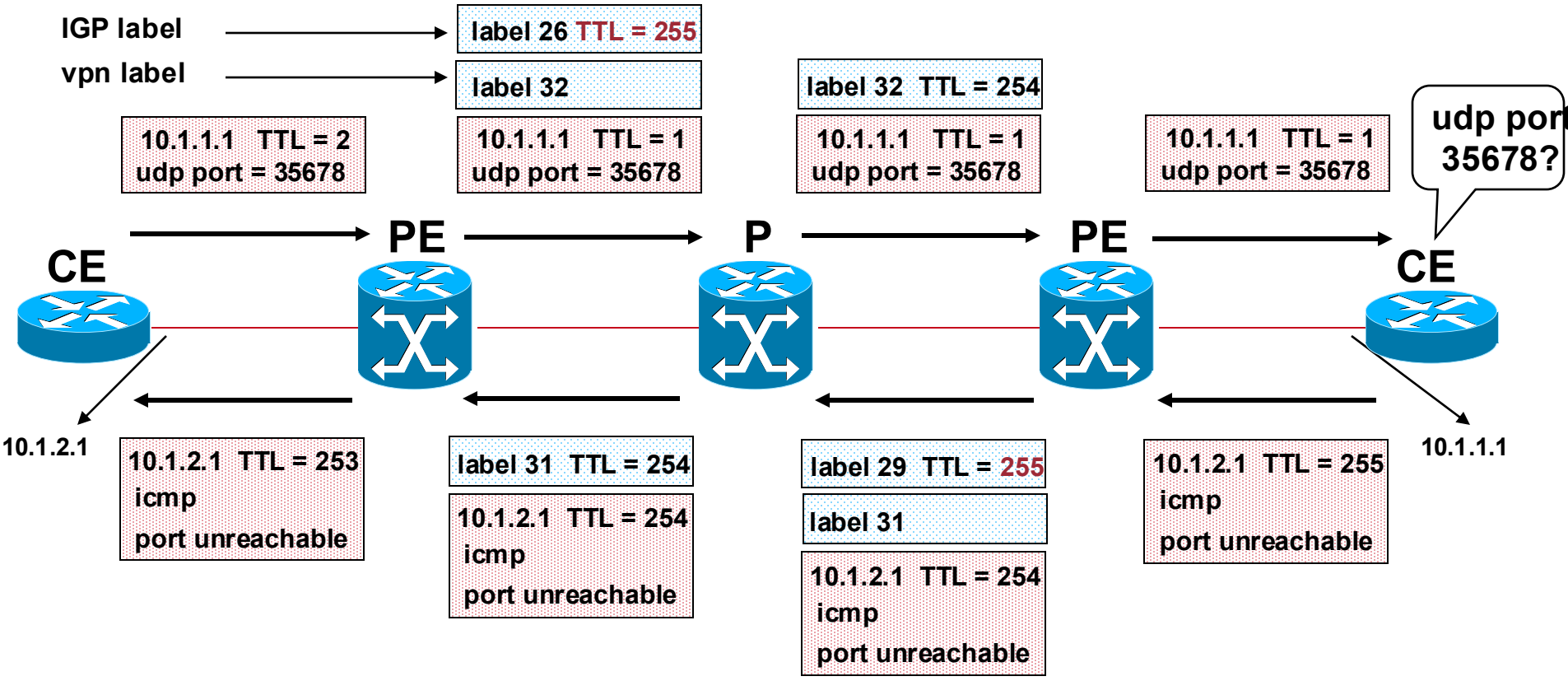
local PE

remote CE

Traceroute in MPLS VPN Network

probe 2

no tag-switching ip propagate-ttl on PE routers





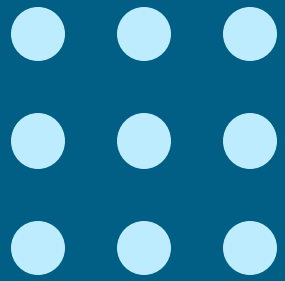
PE-CE Routing Protocols

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Configuring PE-CE Routing Protocols

- PE-CE routing protocols are configured for individual VRFs.
- Per-VRF routing protocols can be configured in two ways:
  - There is only one BGP or RIP process per router, per-VRF parameters are specified in routing contexts, which are selected with the address family command.
  - A separate OSPF process has to be started for each VRF.

3.1

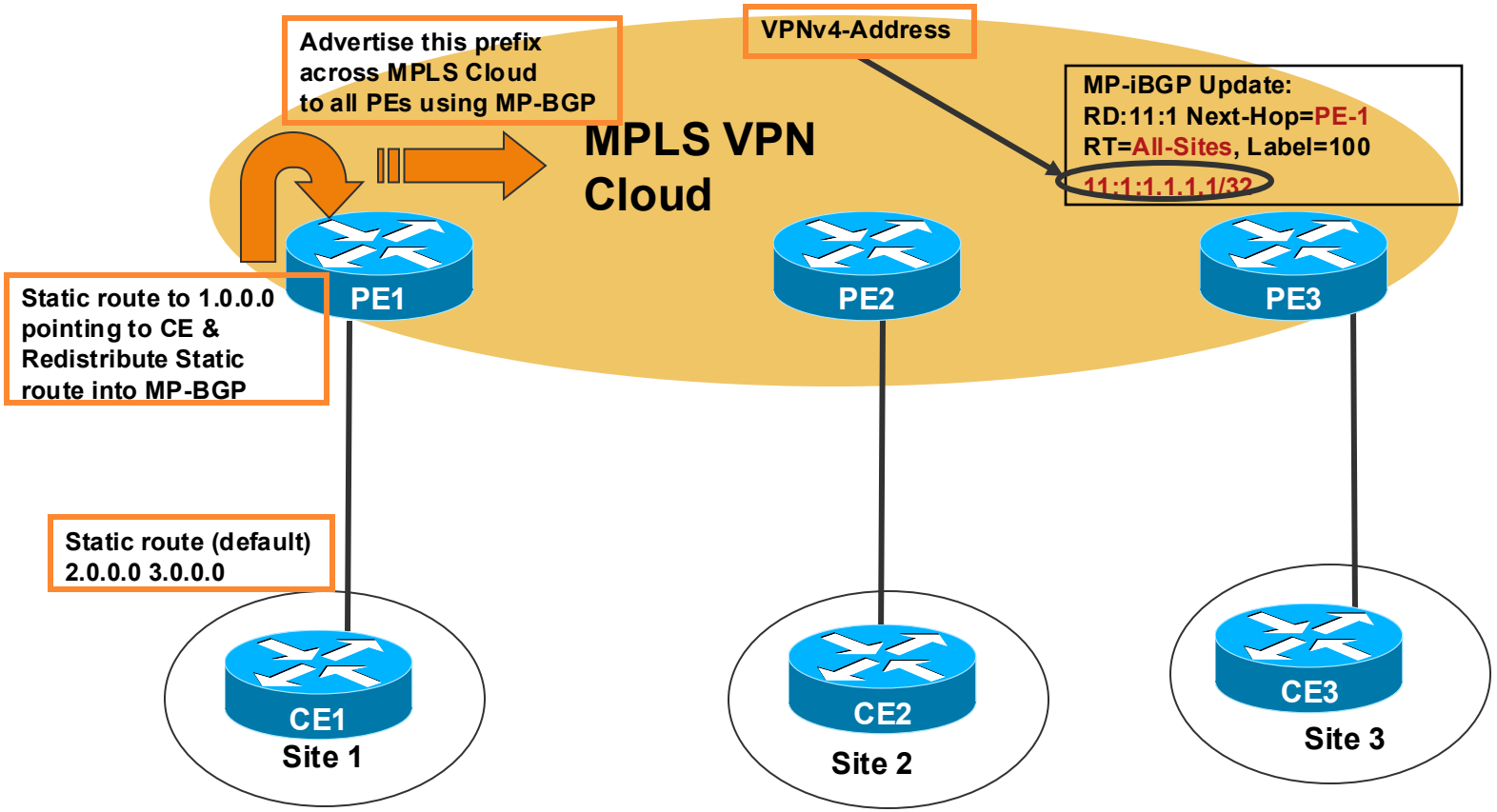


Static Routing

Static Routing

- Highly scalable and secured
- Ideal for single attached sites (one way out)
- For multi-homed sites, static routing may not provide re-routing capabilities in case of certain failure scenarios – highly discouraged
- For every net in a given site, static configuration is required on PE
- For traffic leaving the site (single homed), a static default route could be used on CE
- Can co-exist with dynamic routing protocols

Static PE-CE:


Advertise this prefix
across MPLS Cloud
to all PEs using MP-BGP
VPNv4-Address
MP-iBGP Update:
RD:11:1 Next-Hop=PE-1
RT=All-Sites, Label=100
11:1:1.1.1.1/32
MPLS VPN
Cloud
PE1
PE2
PE3
Static route to 1.0.0.0
pointing to CE &
Redistribute Static
route into MP-BGP
Static route (default)
2.0.0.0 3.0.0.0
CE1
Site 1
CE2
Site 2
CE3
Site 3

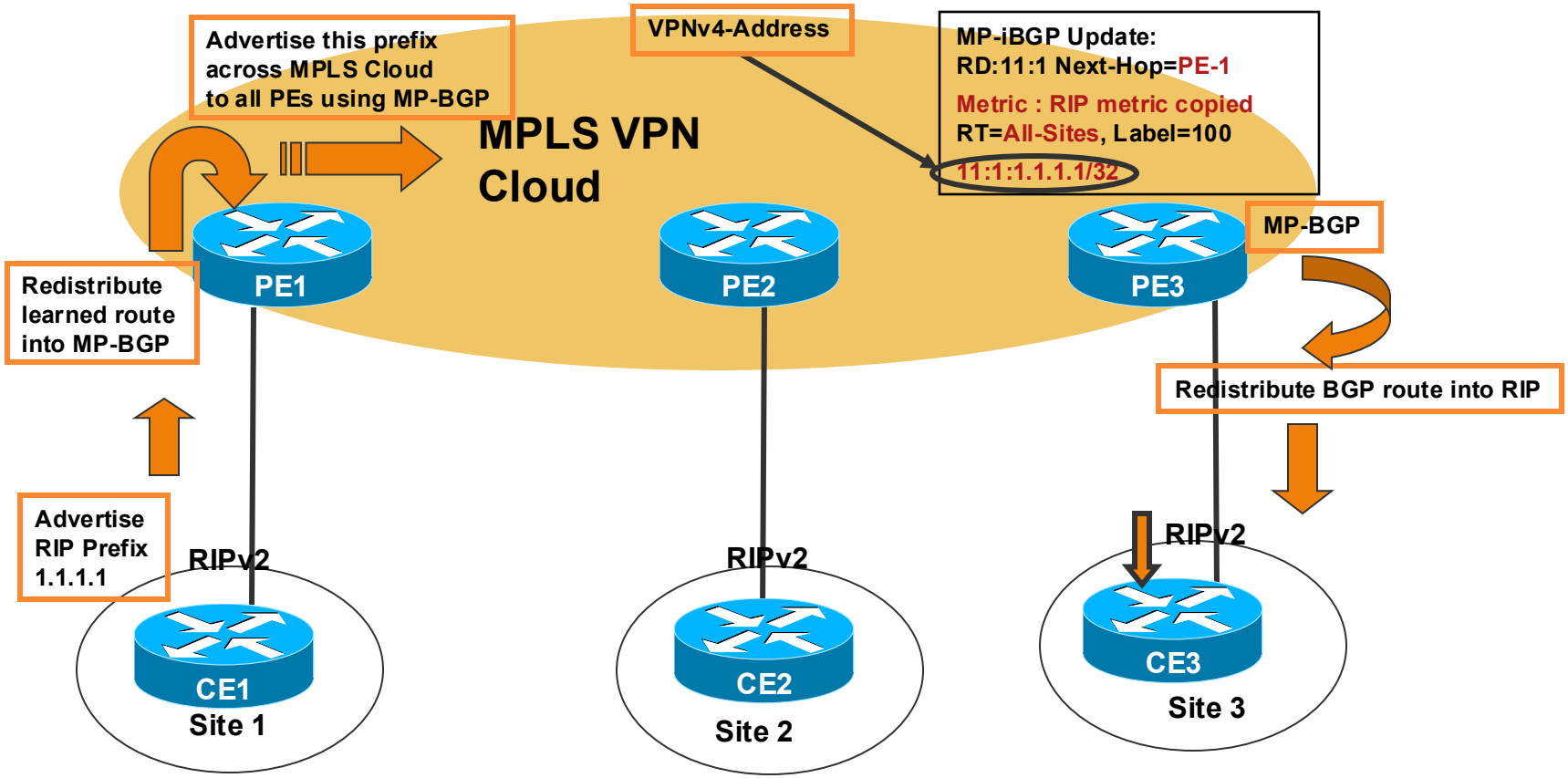
3.2

RIPv2 Routing

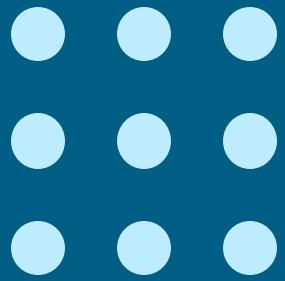
RIPv2

- Provides a simple distance vector protocol to exchange routing information
- Hop count limit of 15
- Could be used at smaller sites where hop-count is much less than 15
- Supports multi-homed sites with re-routing capabilities

RIPv2 PE-CE:


Advertise this prefix
across MPLS Cloud
to all PEs using MP-BGP
VPNv4-Address
MP-iBGP Update:
RD:11:1 Next-Hop=PE-1
Metric : RIP metric copied
RT=All-Sites, Label=100
11:1:1.1.1.1/32
MPLS VPN
Cloud
MP-BGP
PE1
PE2
PE3
Redistribute
learned route
into MP-BGP
Redistribute BGP route into RIP
Advertise
RIP Prefix
1.1.1.1
RIPv2
RIPv2
RIPv2
CE1
CE2
CE3
Site 1
Site 2
Site 3

3.3

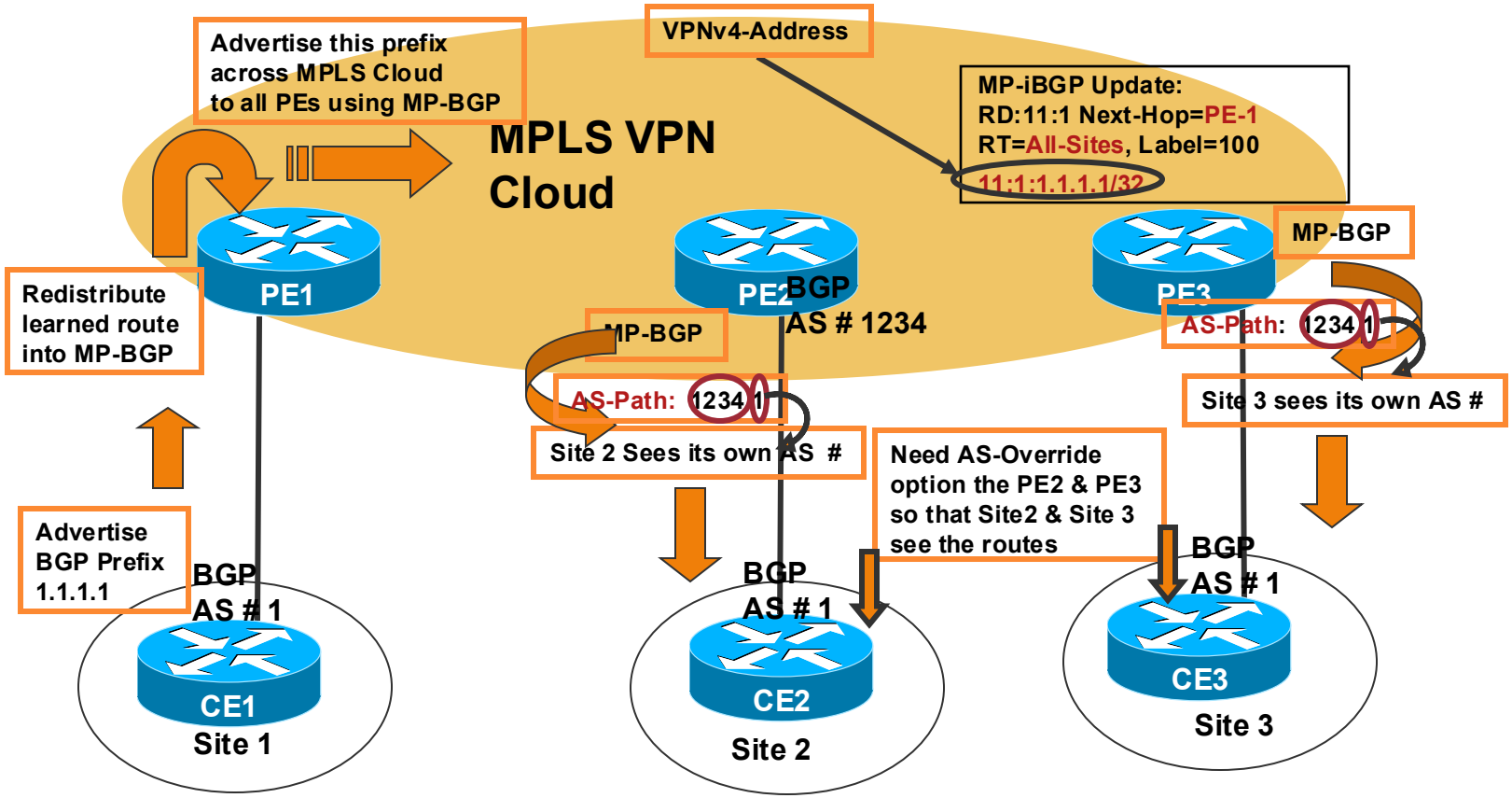


BGP Routing

BGP PE-CE Requirements

- Enterprise Customers choosing BGP as PE-CE protocol to peer with the Service Provider in MPLS-VPNs network should look into the following:
  - Same AS number at all the customer locations
  - Different AS number at the Customer Locations
  - With Back Doors
  - Private AS number vs. Public AS #
  - Multi-homing and Loadbalancing options
  - Customer sites connected to two or more Providers
- BGP is one of the widely used PE-CE protocol in the MPLS-VPN environment

BGP PE-CE: Same AS at Customer Sites



- Customer Site 1 sends the BGP routes to PE1
- PE1 advertises to all the iBGP-VPNv4 peers
- Remote PEs (PE2 & PE3) redistributes those routes into the local BGP peer at their sites
- Site 2 & 3 CE routers will not accept the routes as they see their own AS in the AS-Path
- MPLS-VPN routers need to be configured with the “AS-OVERRIDE” option so that remote CE Routers will not see their own AS in AS PATH

BGP PE-CE:
Same AS at Customer Sites

PE is sending the update to remote CEs for 1.1.1.1/32

BGP(4): 140.0.0.201 send UPDATE (chgflags: 0x820) 1.1.1.1/32, next 140.0.0.202, metric 0, path 1, extended community RT:1:1

CE receives the update and ignores the Update due to its own AS in the AS-Path

BGP(0): 140.0.0.202 rcv UPDATE about 1.1.1.1/32 -- DENIED due to: AS-PATH contains our own AS;

With "**AS-Override**" option configured on the PE Router(s)

PE is now sending the update to remote CEs for 1.1.1.1/32 with "prepend" option

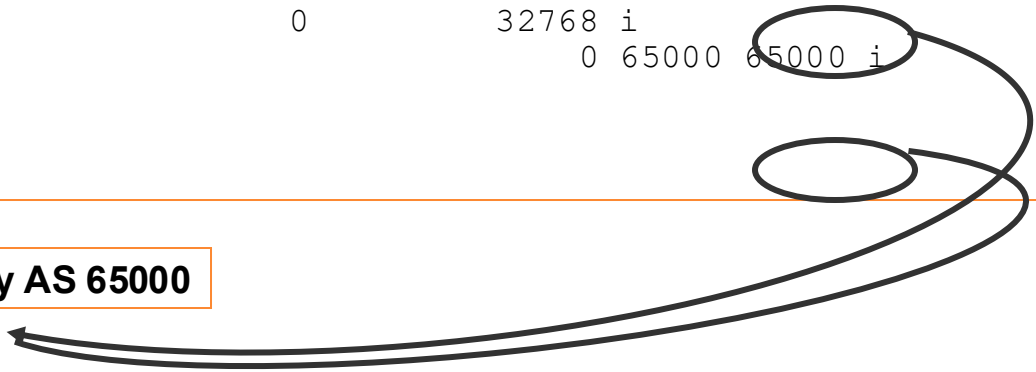
BGP(4): 140.0.0.201 send UPDATE (**prepend**, chgflags: 0x820) 1.1.1.1/32, next 140.0.0.202, metric 0, path 1, extended community RT:1:1

CE now receives and accepts the route

BGP(0): 140.0.0.202 rcvd 1.1.1.1/32
BGP(0): Revise route installing 1 of 1 routes for 1.1.1.1/32 -> 140.0.0.202(main) to main IP table

BGP PE-CE: Same AS at Customer Sites

```
CE2#show ip bgp
   Network          Next Hop            Metric LocPrf Weight Path
*> 1.1.1.1/32       140.0.0.202                              0 65000 65000 i
*> 2.2.2.2/32       0.0.0.0                       0         32768 i
*> 3.3.3.3/32       140.0.0.202                              0 65000 65000 i
```



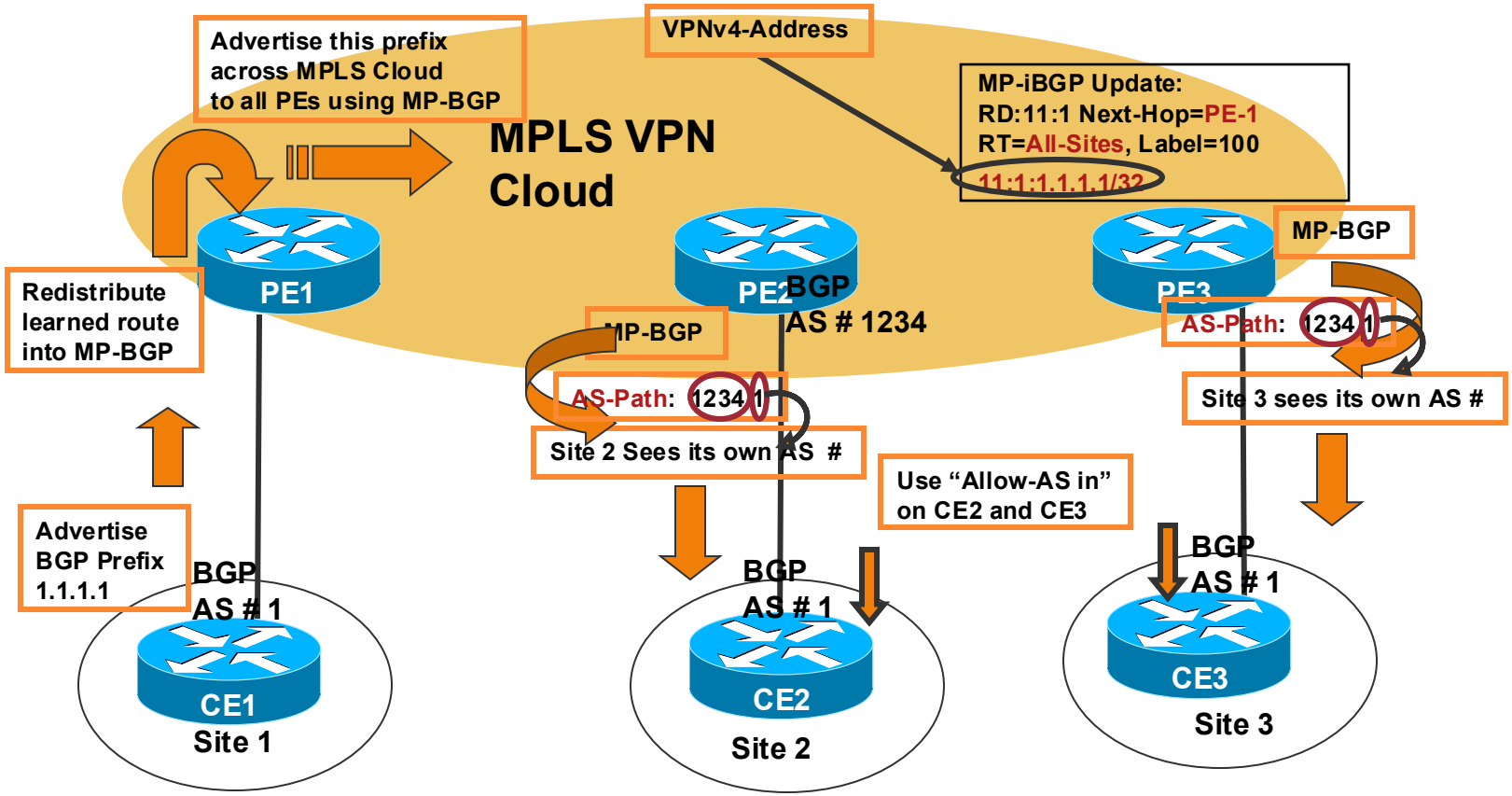
AS 1 is replaced by AS 65000

PE Router Configuration

```
!
router bgp 65000
 no bgp default ipv4-unicast
 bgp log-neighbor-changes
 neighbor 11.11.11.11 remote-as 65000
 neighbor 11.11.11.11 update-source Loopback0
 neighbor 13.13.13.13 remote-as 65000
 neighbor 13.13.13.13 update-source Loopback0
 !
 address-family vpnv4
 neighbor 11.11.11.11 activate
 neighbor 11.11.11.11 send-community extended
 neighbor 13.13.13.13 activate
 neighbor 13.13.13.13 send-community extended
 exit-address-family
```

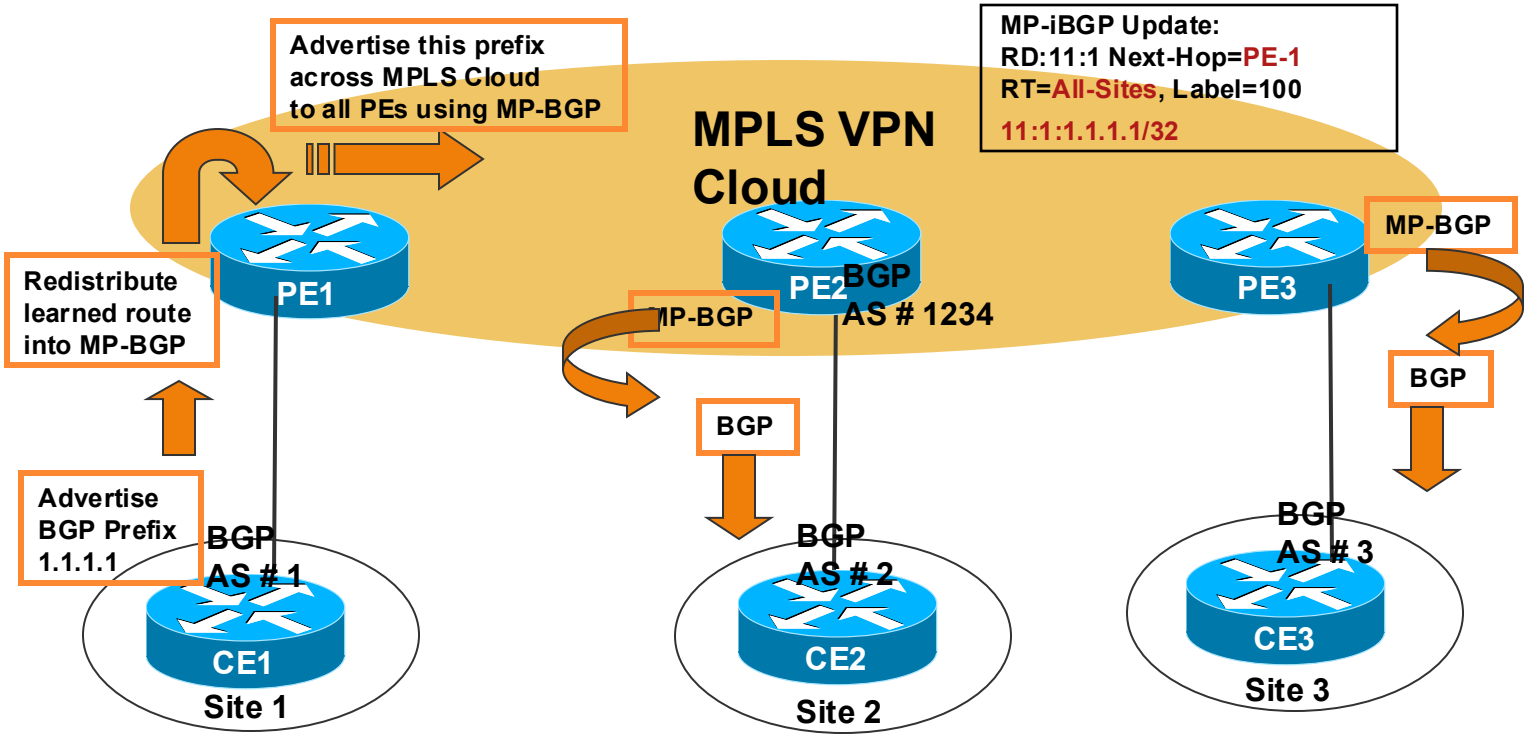
```
!
 address-family ipv4 vrf BGP-PE-CE
 redistribute connected
 neighbor 140.0.0.201 remote-as 1
 neighbor 140.0.0.201 activate
 neighbor 140.0.0.201 as-override
 no synchronization
 exit-address-family
!
```

BGP PE-CE: Same AS at Customer Sites



- Alternate approach is to use "Allow-AS in" on Ces facing the PEs
- CEs bypass the AS based loop detection mechanism
- Acceptable number of AS occurrences within the AS-Path could be defined, to detect loops

BGP PE-CE: Different AS at Customer Sites



- Customer Site 1 sends the BGP routes to PE1
- PE1 advertises to all the iBGP-VPNv4 peers
- Remote PEs (PE2 & PE3) redistributes those routes into the local BGP peer at their sites
- Site 2 & 3 CE routers will receive and install the routes from the PEs in the MPLS Core

BGP PE-CE: Different AS at Customer Sites

```
CE2#show ip bgp
   Network          Next Hop      Metric LocPrf Weight Path
*> 1.1.1.1/32       140.0.0.202                          0 65000  1 i
```

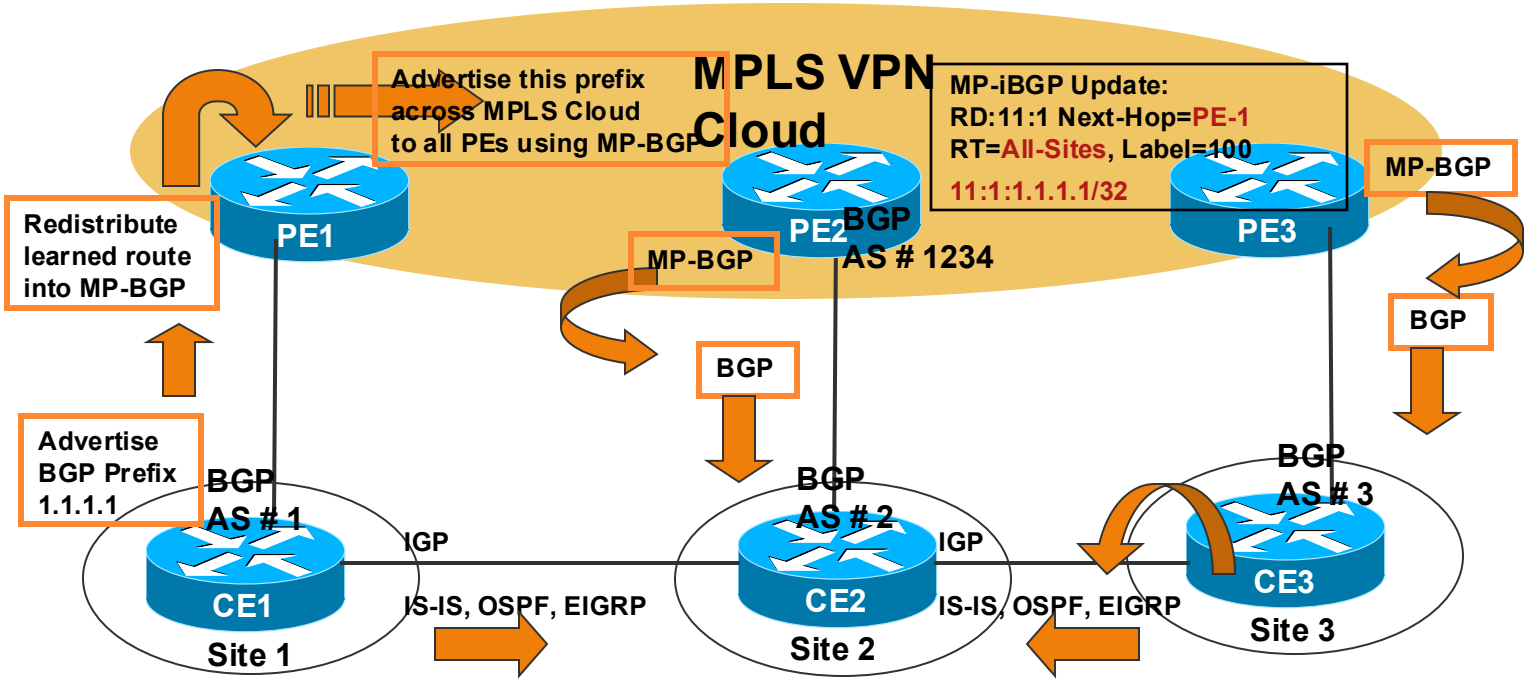
```
CE2#show ip route
B       1.1.1.1 [20/0] via 140.0.0.202, 00:03:14
```

The complete AS Path traversed by the IPv4 Prefix

```
CE3#show ip bgp
   Network          Next Hop      Metric LocPrf Weight Path
*> 1.1.1.1/32       140.0.0.130                          0 65000  1 i
```

```
CE3#show ip route
B       1.1.1.1 [20/0] via 140.0.0.130, 00:00:44
```

BGP PE-CE: With Back Door Connections



- Mechanisms explained earlier are still valid
- When the MP-BGP route on either PE2 or PE3 sent to CE2 or CE3 for CE1—the route will be learned as an external eBGP route with Admin. distance of 20
- If via the back door the same route comes via either OSPF, IS-IS or EIGRP, Admin distance will come into the consideration
- Since eBGP has Admin. Distance of 20—MPLS-Core will be used to reach the destination

BGP PE-CE: With Back Door Connections

```
CE2#show ip route 1.1.1.1
Routing entry for 1.1.1.1/32
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 140.0.0.194 on Ethernet0/0, 00:00:09 ago
```

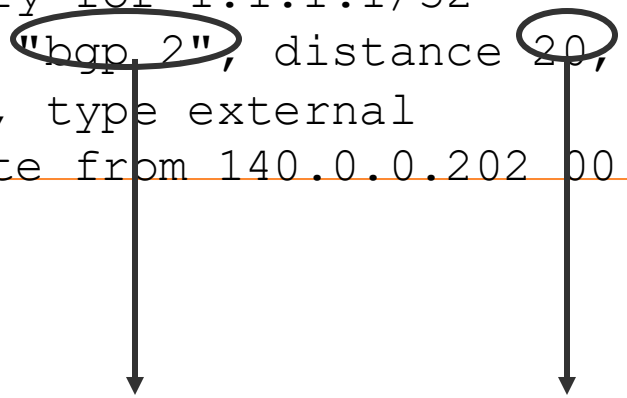
We see 1.1.1.1/32 being learned via the backdoor path [OSPF]

With the BGP session being UP, we see 1.1.1.1/32 being learned via BGP and installed into the RIB as a eBGP route

```
%BGP-5-ADJCHANGE: neighbor 140.0.0.202 Up
RT: closer admin distance for 1.1.1.1, flushing 1 route
RT: add 1.1.1.1/32 via 140.0.0.202, bgp metric [20/0]
```

BGP PE-CE:
With Back Door Connections

```
CE2#show ip route 1.1.1.1
Routing entry for 1.1.1.1/32
  Known via "bgp 2", distance 20, metric 0
  Tag 65000, type external
  Last update from 140.0.0.202 00:02:05 ago
```



Route Learned via MPLS-Core as expected

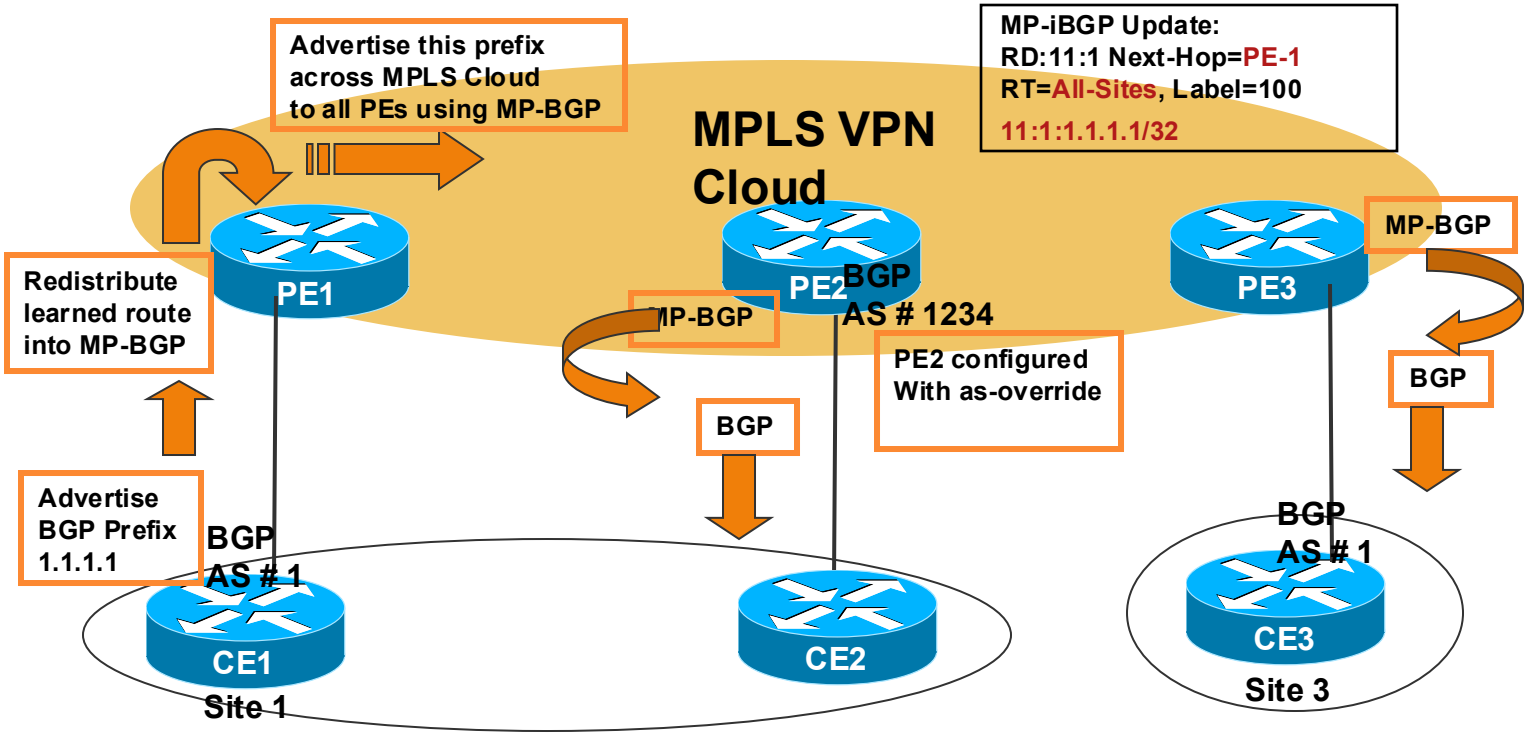
BGP PE-CE: Private AS vs. Public AS

- Enterprises that get connected to the Service Provider MPLS-VPN Network can use a BGP Private-AS number
- The Private BGP-AS number will be hidden from the Global Internet
- The Customer BGP AS will only be used to connect the different customer entities

BGP PE-CE: MCS

- Certain customers have a mission critical site example a hub that needs to be connected to the MPLS VPN Service Provider to get maximum uptime
- Such sites need added mechanisms to provide redundancy and at the same time avoid looping.

BGP PE-CE: Loop in a MCS

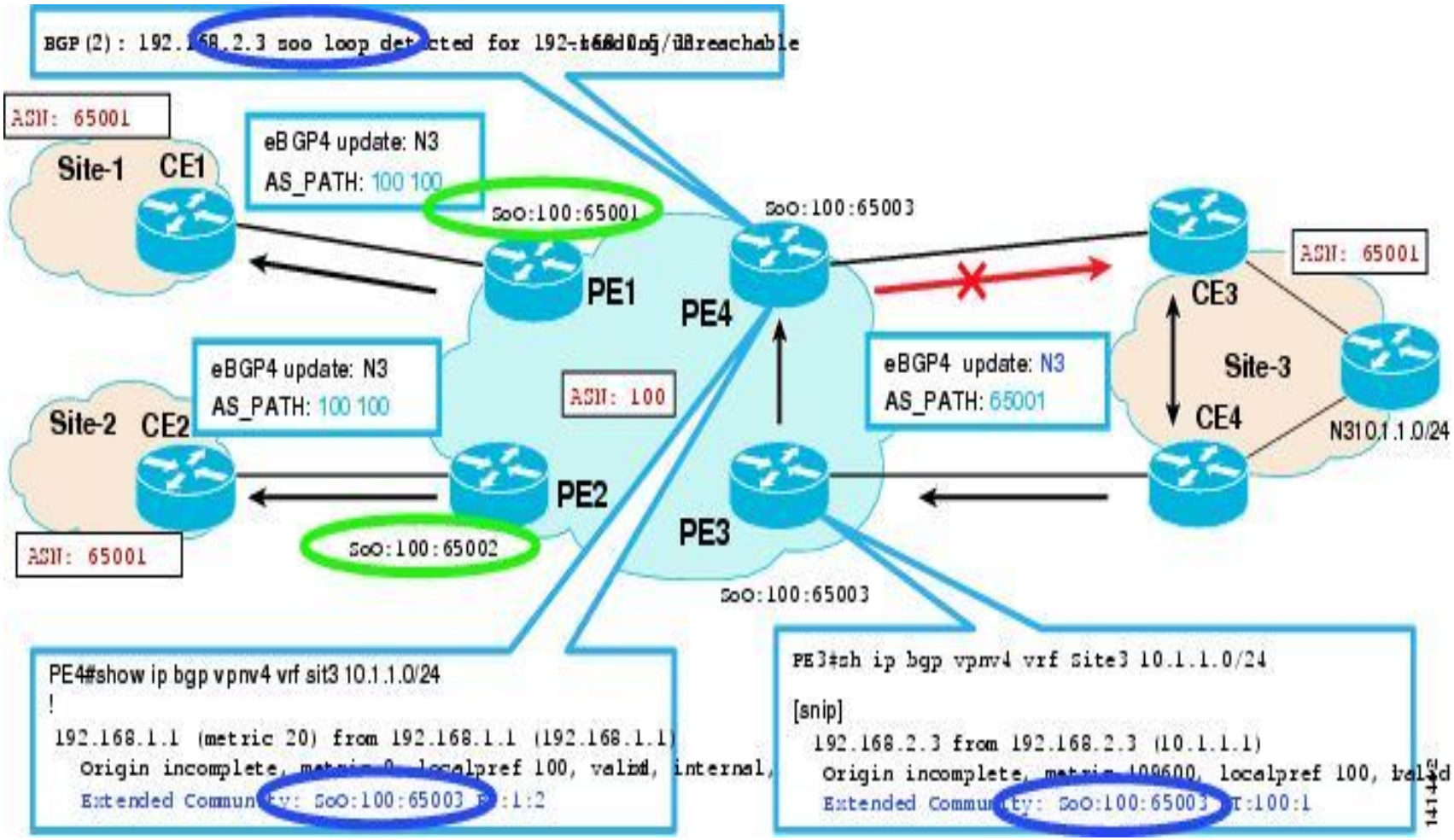


- Customer Site 1 sends the BGP routes to PE1
- PE1 advertises to all the iBGP-VPNv4 peers
- Remote PE (PE2) redistributes those routes into the local BGP peer at their sites because it has as-override configured so it will change the AS Path.
- CE2 will accept this route because it doesn't see its AS in the AS-Path and hence we have created a loop in the routing. Why? Because CE2 would have the same route via an IGP and basis AD which route will win ????

BGP PE-CE: Site-of-Origin

- In order to solve this loop problem, SOO was created.
- Site-of-Origin is an extended community attribute that is attached to a route received from a CE router before being advertised into VPNv4 address family.
- The receiving PE router matches the received SOO value with the one configured locally.
- If the values match, receiving PE doesn’t send that route update to the connected CE.
- SOO is configured per neighbor.

BGP PE-CE: Site-of-Origin



3.4

OSPF Routing

OSPF PE-CE Requirements

- In MPLS-VPN environment, PE Routers are part of the OSPF AREA 0 (SuperBackbone)
- If OSPF Domain has Area 0 routers, then at least one of those MUST be a CE router and MUST have an Area 0 link to at least one PE Router
- PE MUST support one OSPF instance for each OSPF domain to which it attaches
- If PE is running OSPF as its IGP, the instance of OSPF running must be separate and independent from OSPF as PE-CE

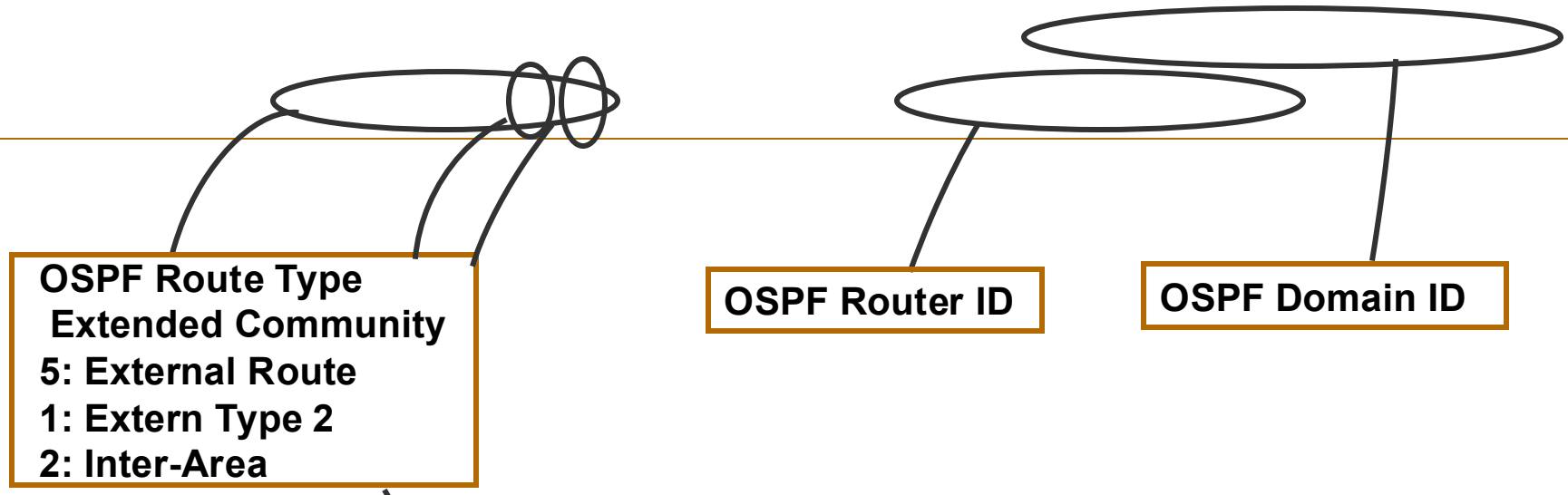
OSPF PE-CE—MP-BGP Extensions

Extensions to MP-BGP to Carry OSPF Routes:

- OSPF Router-ID
  - Each OSPF Instance has a Unique Rtr-ID
- Domain-Identifier
  - 8-Byte value that is a valid BGP EXT Community attribute
  - Identifies the how the prefix originated
- Route-Type
  - 8-Byte value includes Area number, OSPF Route-Type (Intra-Inter-External-NSSA)
- DN Bit
  - Used for Loop prevention for Type 3 LSAs
  - MUST be set when it is sent by PE to CE
- Route-Tags
  - MUST be included in the Type 5 LSAs that PE originates as a result of receiving external routes

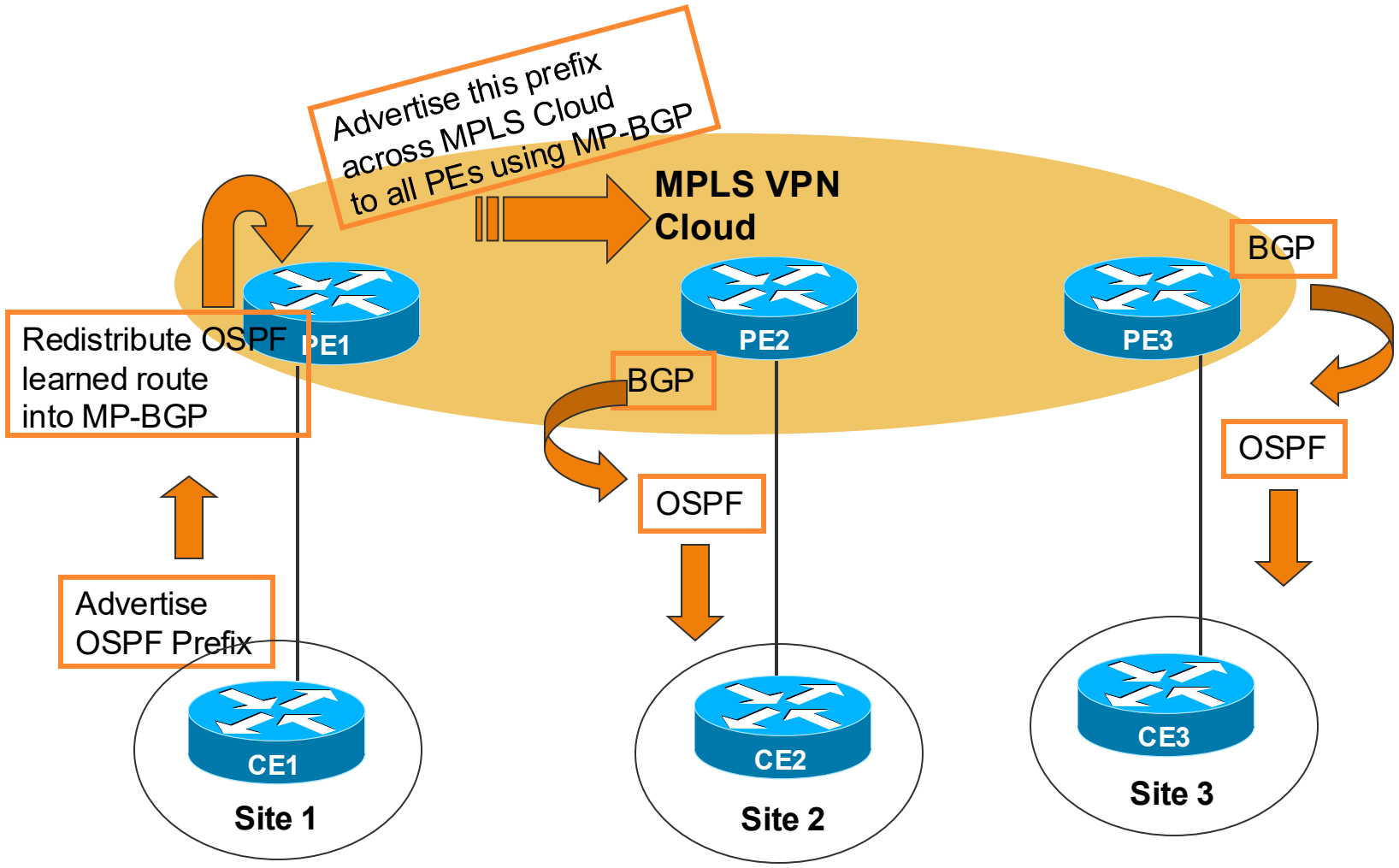
OSPF PE-CE—MP-BGP Extensions

```
PE12# show ip bgp vpnv4 all 22.22.22.22
BGP routing table entry for 12:1:22.22.22.22/32, version 88
   140.0.0.201 (via OSPF-Sham) from 0.0.0.0 (12.12.12.12)
      Origin incomplete, metric 10, localpref 100, weight 32768,
  valid, sourced, best
      Extended Community: RT:1:1 OSPF DOMAINID:0x0005:0x000000010200
        OSPF RT:0.0.0.0:5:1 OSPF ROUTER ID:12.12.12.12:512
```




```
PE11# show ip bgp vpnv4 all 2.2.2.2
OSPF RT:0.0.0.0:2:0 OSPF ROUTER
ID:12.12.12.12:512
```

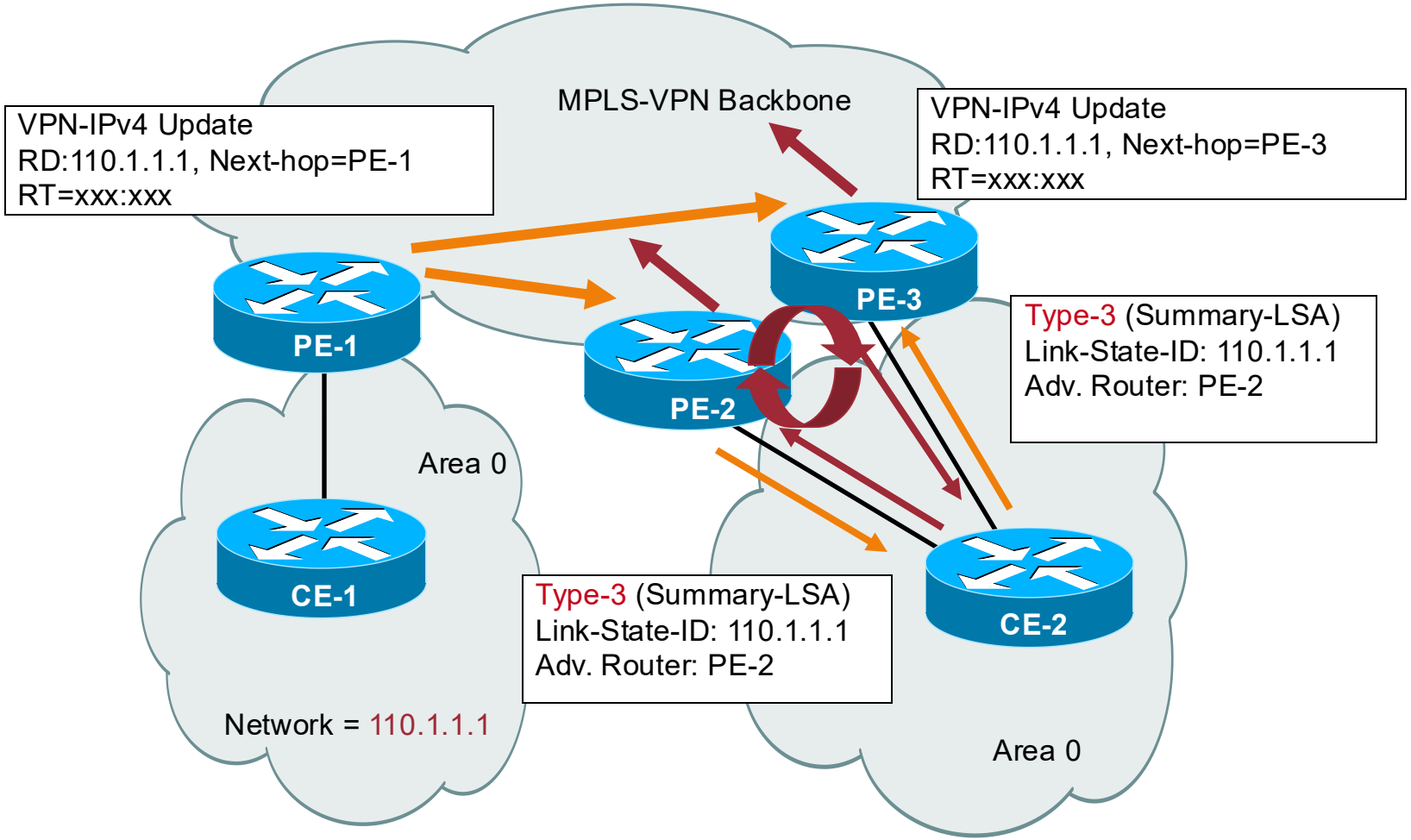
OSPF PE-CE Operation



OSPF PE-CE Operation : Down BIT and Tag

- The MPLS VPN model has 3 levels of hierarchy and this poses some challenges during redistribution and flooding which did not exit in the 2 level model of OSPF.
- In the normal OSPF the area prefixes are advertised in the backbone and vice-versa using summary type 3 LSAs
- The fact that only backbone summary LSA are considered or that LSA generation is suppressed in some cases we never have flooding loops.
- These techniques are not sufficient in this model.

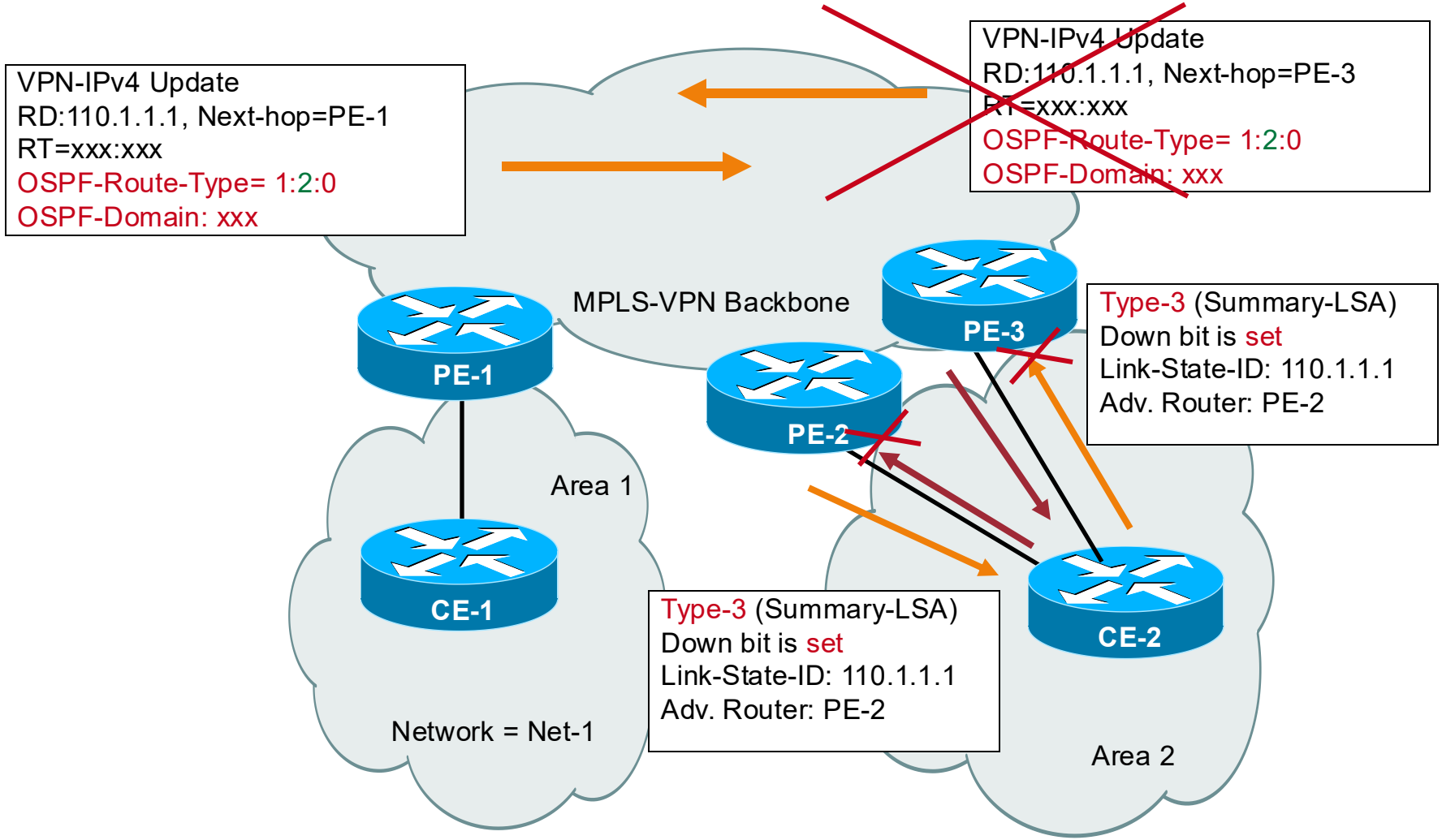
OSPF - Routing Loop



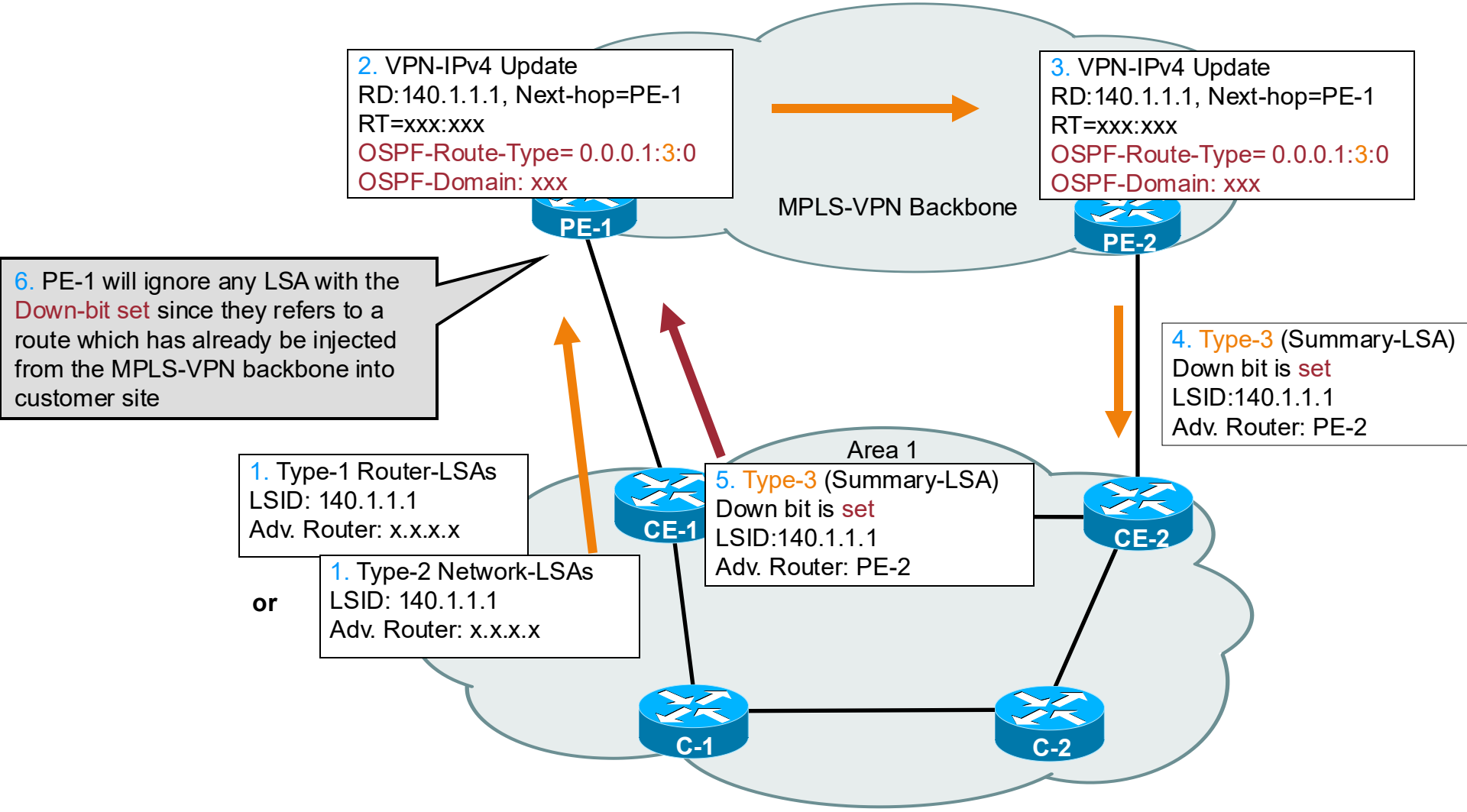
OSPF Down Bit and Tag

- Down bit and Tag are used to prevent looping and on by default.
- The Down bit is bit 7 in the option field of an LSA
- The Down bit is set by the PE on the LSAs describing routes coming from the BGP redistribution and need to get propagated.
- When a PE receives a route from the CE, it will generate a summary LSA into the MPLS-VPN BB if and only if the Down-bit is not set on that LSA
- Down bit processing is only done on PE
- Down-bit check can be disabled by configuration

OSPF - Routing Loop avoidance



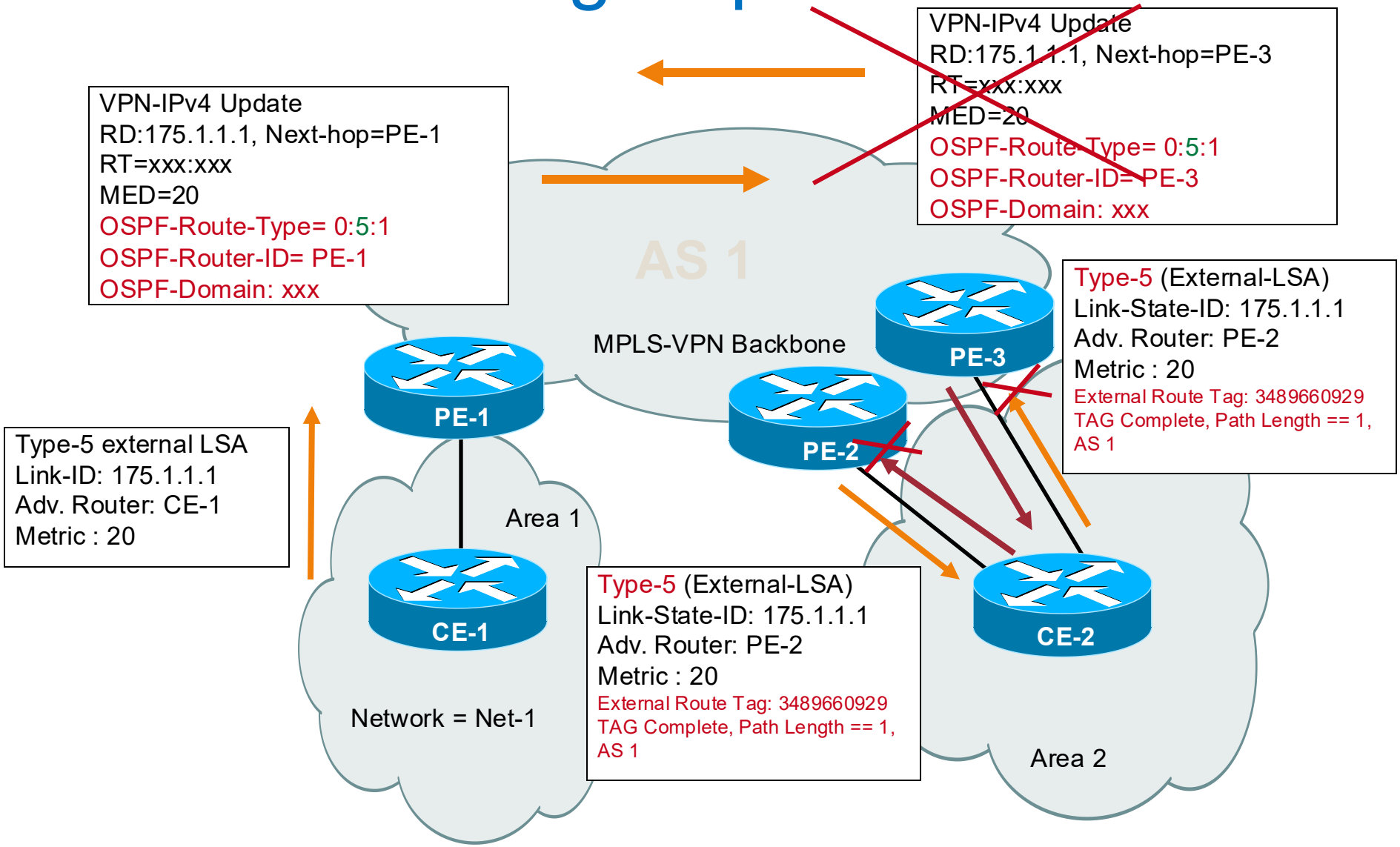
OSPF - Down bit example



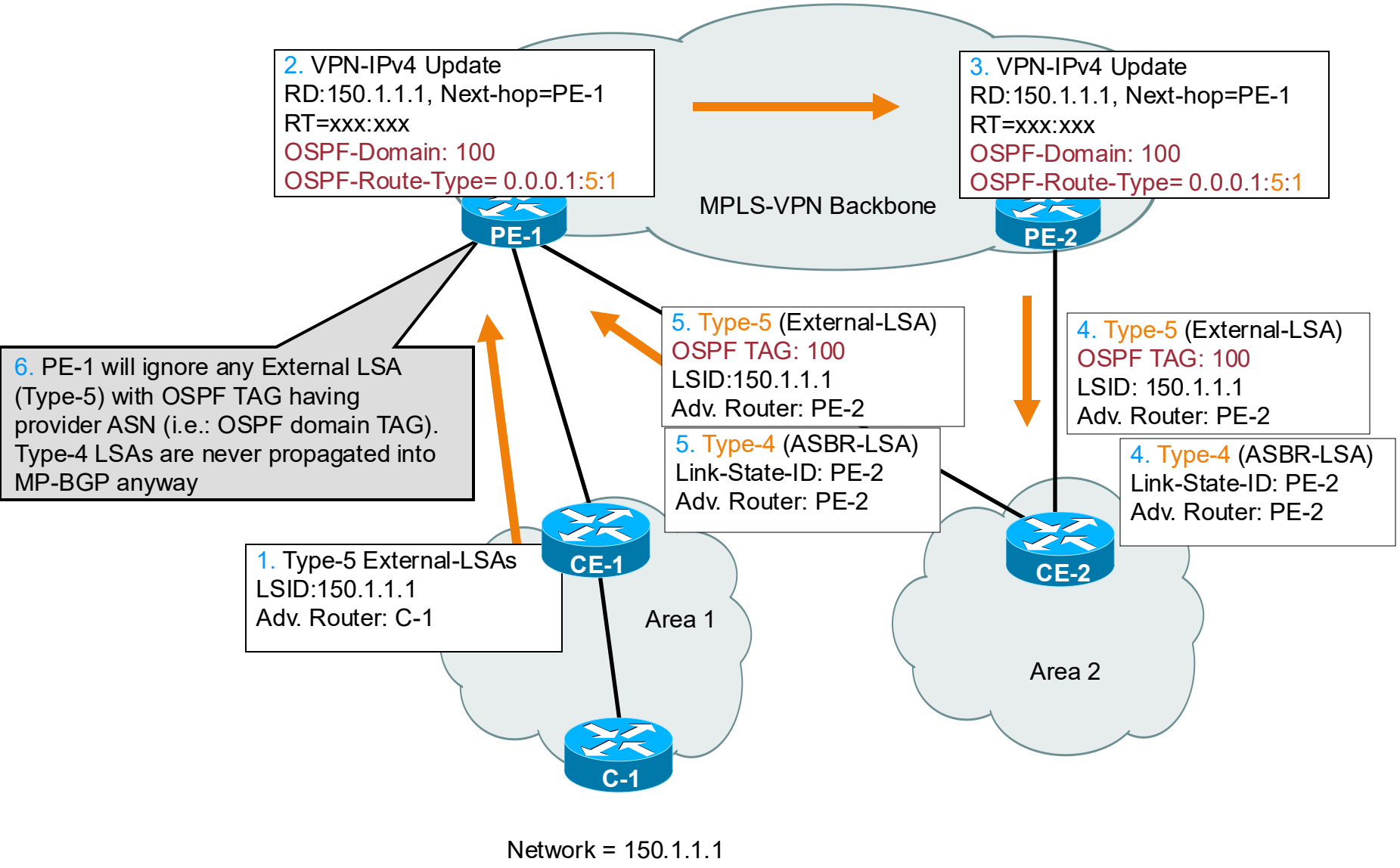
OSPF - VPN Route Tag

- The TAG is the equivalent of the down bit for type 5 and used to prevent looping
- The Tag value may be configured or computed automatically for a VPN and is the same across the same OSPF domain.
- Each OSPF domain should have a unique TAG value
- The Tag is a 32 bit field in the type 5 LSA.
- The Tag is set on the Type 5 LSA by the PE to indicate the flooding from a higher to a lower level
- If a tag match is found on a Type 5 LSA, this LSA will not be considered for route computation on a PE.

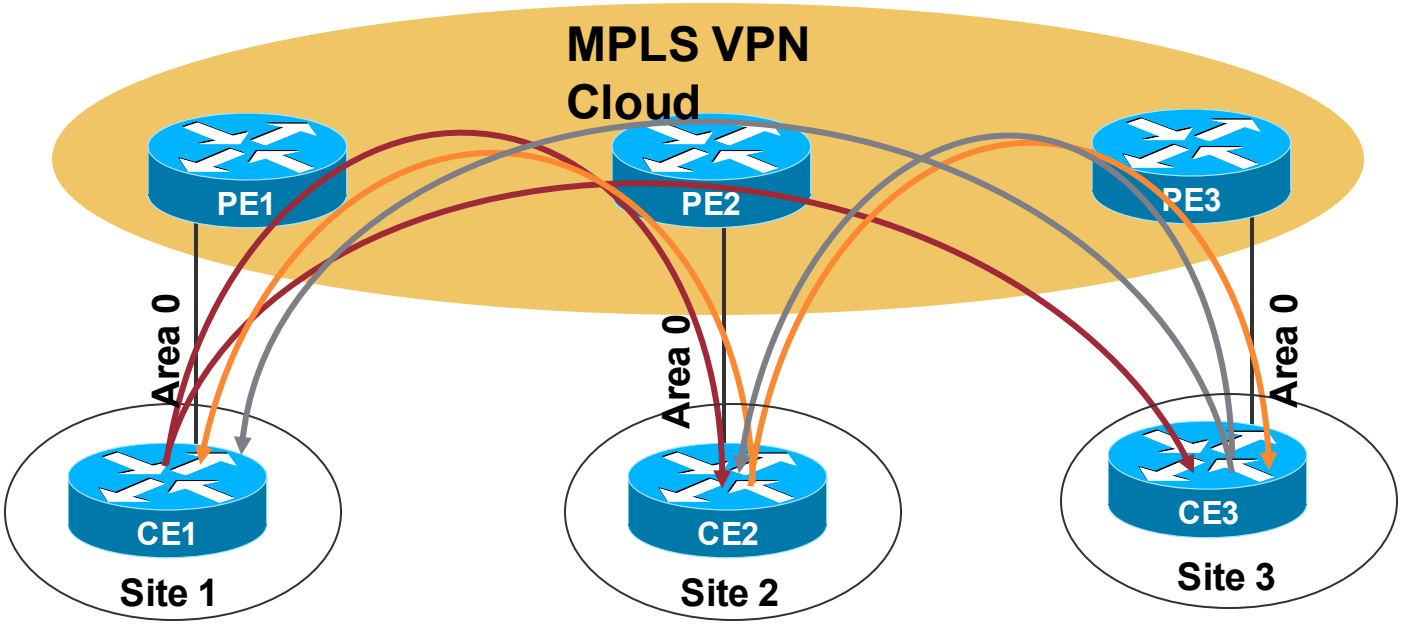
OSPF - Routing loop avoidance



OSPF - Tag example



Customer Sites Are All in Area 0



- MPLS VPN Cloud is OSPF area 0 (Super Backbone)
- PE Routers act as a ABR (Area Border Routers)
- PE Routers redistribute OSPF routes into MPLS-VPN cloud as a Summary 3 LSA

Customer Sites Are All in Area 0

```
interface Loopback0
ip address 12.12.12.12
    255.255.255.255
!
ip vrf OSPF-PECE
 rd 12:1
 route-target export 1:1
 route-target import 1:1
!
mpls label protocol ldp
!
router ospf 1 vrf OSPF-PECE
 router-id 12.12.12.12
 redistribute bgp 65000 subnets
 network 140.0.0.200 0.0.0.3 area 0
```

```
router bgp 65000
 no bgp default ipv4-unicast
 neighbor 11.11.11.11 remote-as 65000
 neighbor 11.11.11.11 update-source Loopback0
!
 address-family vpnv4
 neighbor 11.11.11.11 activate
 neighbor 11.11.11.11 send-community extended
exit-address-family
 !
 address-family ipv4 vrf OSPF-PECE
 redistribute connected
 redistribute ospf 1 vrf OSPF-PECE
 no synchronization
 exit-address-family
```

Customer Sites Are All in Area 0

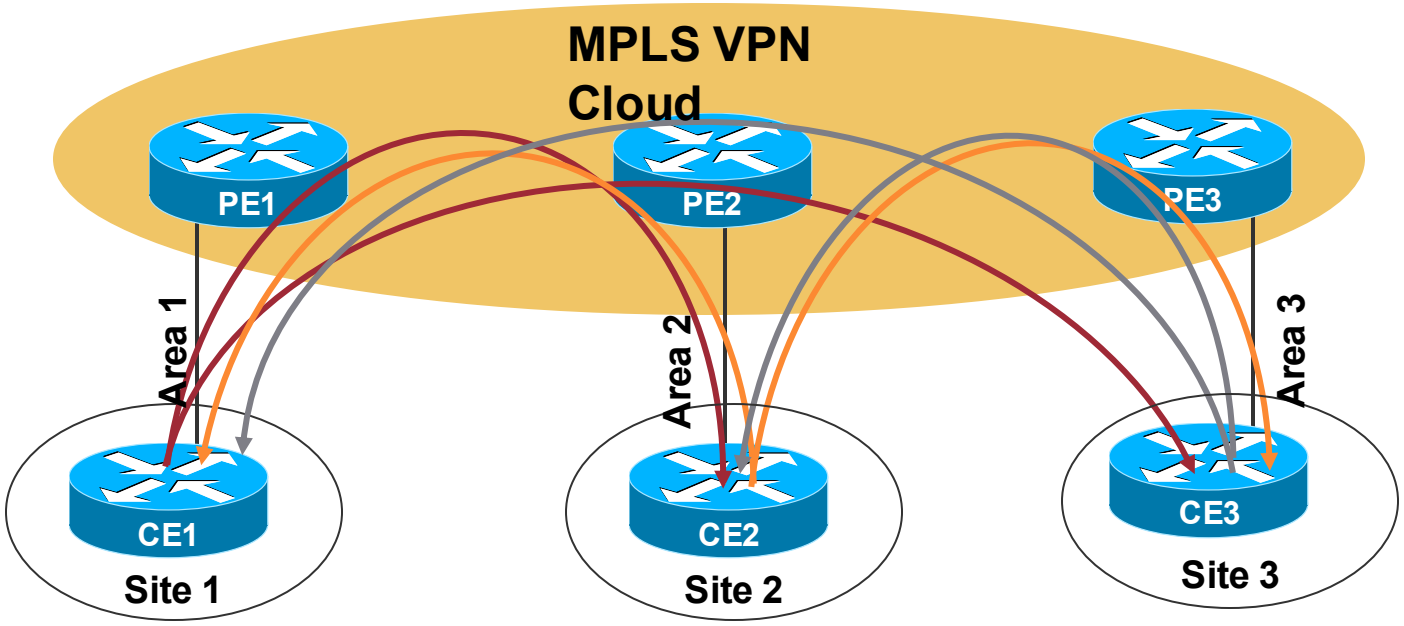
```
CE1#show ip route ospf
     O IA    2.2.2.2 [110/97] via 140.0.0.2, 00:18:26, Serial2/0
     O IA    3.3.3.3 [110/97] via 140.0.0.2, 00:18:11, Serial2/0
```

```
CE1#show ip route 2.2.2.2
Known via "ospf 1", distance 110, metric 97, type inter area
  Last update from 140.0.0.2 on Serial2/0, 00:20:50 ago
```

```
CE1#show ip route 3.3.3.3
Known via "ospf 1", distance 110, metric 97, type inter area
  Last update from 140.0.0.2 on Serial2/0, 00:22:55 ago
```

Summary LSA from the MPLS VPN Backbone

Customer Sites Are in Different Areas



- Customer Sites are in different areas
- MPLS-VPN backbone is a Super Backbone

Customer Sites Are in Different Areas

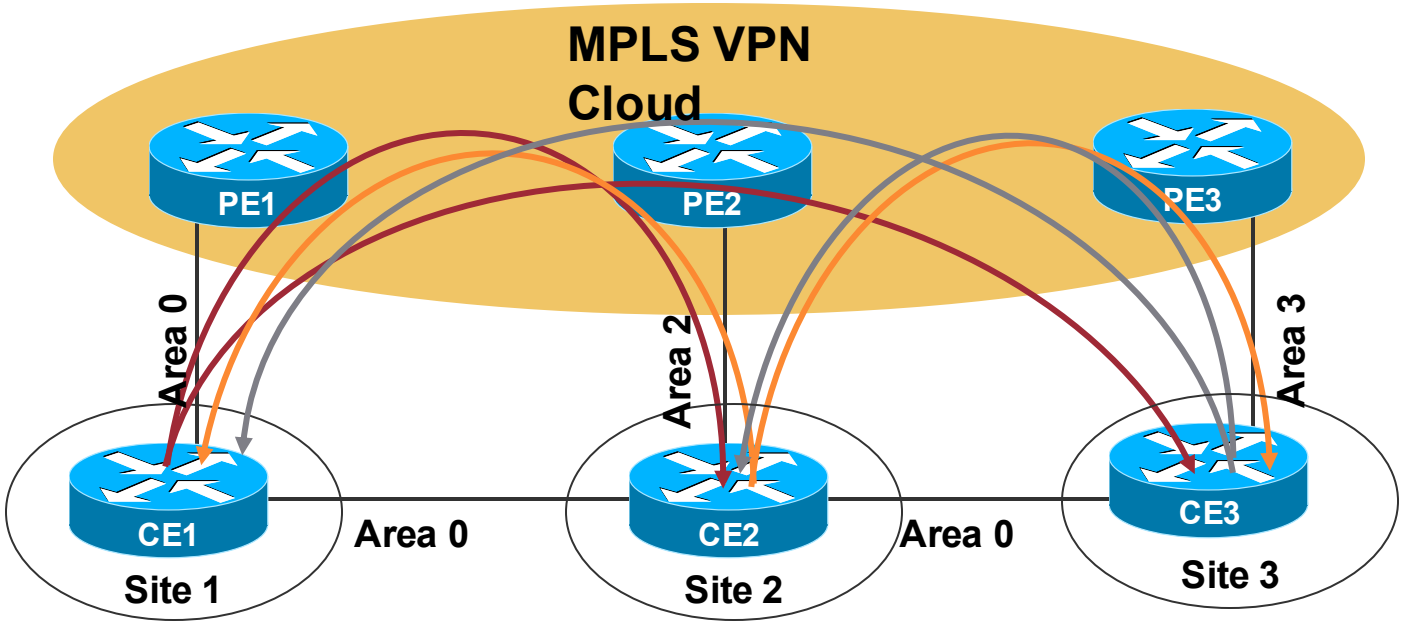
```
CE1#show ip route ospf
O IA    2.2.2.2 [110/97] via 140.0.0.2, 00:01:02, Serial2/0
O IA    3.3.3.3 [110/97] via 140.0.0.2, 00:01:02, Serial2/0
```

```
CE1#show ip route 2.2.2.2
Known via "ospf 1", distance 110, metric 97, type inter area
  Last update from 140.0.0.2 on Serial2/0, 00:04:02 ago
```

```
CE1#show ip route 3.3.3.3
Known via "ospf 1", distance 110, metric 97, type inter area
  Last update from 140.0.0.2 on Serial2/0, 00:04:08 ago
```

Summary LSA from the MPLS VPN Backbone

Customer Sites With Back Door Links



- Customer Sites have the Back Door links
- If the backdoor links are in the same area as the PE-CE link(s), then the backdoor links will be preferred over the MPLS-Core
- This is due to PE routers will advertise the Summary-route to CEs causing the Intra-Area routes preferred over the Inter-Area routes

Customer Sites With Back Door Links

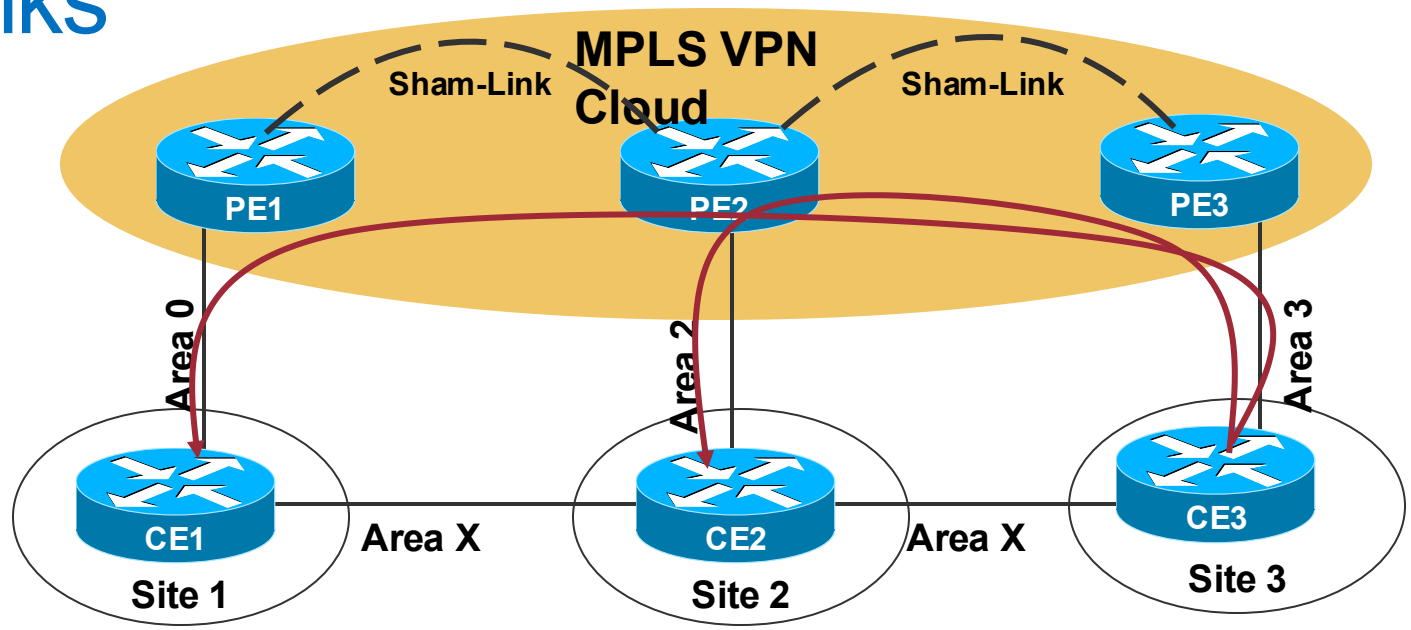
```
CE1#show ip route ospf
O    2.2.2.2 [110/101] via 140.0.0.193, 00:00:52, Ethernet0/0
O    3.3.3.3 [110/201] via 140.0.0.193, 00:00:52, Ethernet0/0
```

```
CE1#show ip route 2.2.2.2
Known via "ospf 1", distance 110, metric 101, type intra area
  Last update from 140.0.0.193 on Ethernet0/0, 00:08:55 ago
```

```
CE1#show ip route 3.3.3.3
Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 140.0.0.193 on Ethernet0/0, 00:09:00 ago
```

Remote Customer site routes are learned via the Back Door links rather than through the MPLS-VPNs network

Sham-Link Support for the Back Door Links



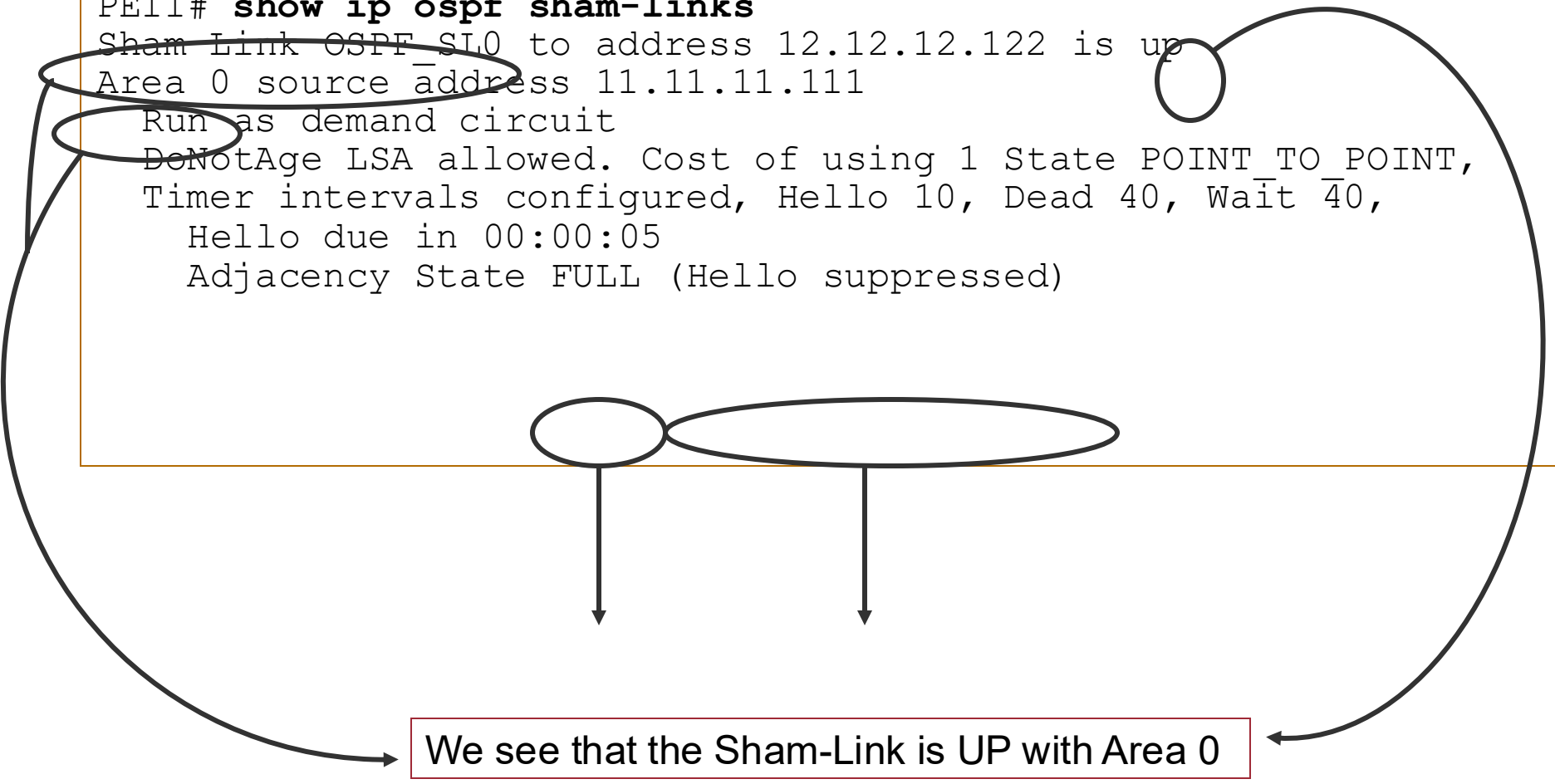
- If Customer sites over Backdoor links belong to same OSPF area, the path over a backdoor link will always be selected because OSPF prefers INTRA Area paths
- PE routers advertise OSPF Routes learned over the VPN backbone as INTER Area paths
- With Sham-Link, we can create additional OSPF Intra-Area (logical) link between ingress and egress VRFs on the relevant PE routers.
- Sham-Link seen as an INTRA-Area link between PE routers, an OSPF Adjacency is created and DB Exchange occurs across the link.
- With Metric manipulation MPLS VPN backbone can be made preferable

Requirements for Creating Sham-Links

- Before you create a sham-link between PE routers in an MPLS VPN, you must:
- Configure a separate /32 address on the remote PE so that OSPF packets can be sent over the VPN backbone to the remote end of the sham-link
- The /32 address must meet the following criteria:
  - Belong to a VRF
  - Not be advertised by OSPF
  - Be advertised by BGP

Checking the Status of Sham-Links

```
PE11# show ip ospf sham-links
Sham Link OSPF_SL0 to address 12.12.12.122 is up
Area 0 source address 11.11.11.111
  Run as demand circuit
  DoNotAge LSA allowed. Cost of using 1 State POINT_TO_POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
    Hello due in 00:00:05
    Adjacency State FULL (Hello suppressed)
```



We see that the Sham-Link is UP with Area 0

Checking the Status of Sham-Links

```
CE1#show ip route ospf
O    2.2.2.2 [110/98] via 140.0.0.2, 00:12:41, Serial2/0
O    3.3.3.3 [110/99] via 140.0.0.2, 00:12:41, Serial2/0
```

```
CE1#show ip route 2.2.2.2
Known via "ospf 1", distance 110, metric 98, type intra area
Last update from 140.0.0.2 on Serial2/0, 00:11:26 ago
```

```
CE1#show ip route 3.3.3.3
Known via "ospf 1", distance 110, metric 99, type intra area
Last update from 140.0.0.2 on Serial2/0, 00:14:31 ago
```

With Sham-links in place, the CE routes are being learned via the MPLS-VPN Backbone network.

Sham-Links Configuration Example

```
!
interface Loopback1
 ip vrf forwarding OSPF-Sham
 ip address 11.11.11.111 255.255.255.255
!
router ospf 1 vrf OSPF-Sham
 router-id 11.11.11.11
 log-adjacency-changes
 area 0 sham-link 11.11.11.111 12.12.12.122
 redistribute bgp 65000 subnets
 network 140.0.0.0 0.0.0.3 area 0
!
```

```
!
interface Loopback1
 ip vrf forwarding OSPF-Sham
 ip address 12.12.12.122 255.255.255.255
!
router ospf 1 vrf OSPF-Sham
 router-id 12.12.12.12
 log-adjacency-changes
area 0 sham-link 12.12.12.122 11.11.11.111
 redistribute bgp 65000 subnets
 network 140.0.0.200 0.0.0.3 area 0
!
```

```
PE11#show ip ospf sham-links
Sham Link OSPF_SL0 to address 12.12.12.122 is up
Area 0 source address 11.11.11.111
  Run as demand circuit
  DoNotAge LSA allowed. Cost of using 1 State POINT_TO_POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
    Hello due in 00:00:00
    Adjacency State FULL (Hello suppressed)
```

```
PE12#show ip ospf sham-links
Sham Link OSPF_SL0 to address 11.11.11.111 is up
Area 0 source address 12.12.12.122
  Run as demand circuit
  DoNotAge LSA allowed. Cost of using 1 State POINT_TO_POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
    Hello due in 00:00:09
    Adjacency State FULL (Hello suppressed)
```

3.5

EIGRP Routing

EIGRP PE-CE Goals

- Impose little requirements or no restrictions on customer networks
  - CE and C routers are NOT required to run newer code (CE/C upgrades recommended for full SoO functionality)
  - Customer sites may be same or different Autonomous Systems
  - Customer sites may consist of several connections to the MPLS VPN backbone
  - Customer sites may consist of one or more connections not part of the MPLS VPN backbone **("backdoor" links)**

EIGRP PE-CE: Operation

- CE runs EIGRP as before where as PE runs EIGRP-VRF process per VRF/AS
- EIGRP routes are distributed to customer sites via MP-iBGP on the MPLS-VPN backbone
- There are no EIGRP adjacencies or EIGRP updates in MPLS/VPN backbone
- EIGRP information is carried across MPLS/VPN backbone by MP-BGP in new extended communities (set and used by PEs)

EIGRP PE-CE: New Extended Communities

- Cost Community attribute can be applied at various points in the MP-BGP best-path calculation
- An Opaque Extended Community Type: 0x43

Type	Usage	Value
8800	EIGRP General Route Information	Flags + Tag
8801	EIGRP Route Metric Information + AS	AS + Delay
8802	EIGRP Route Metric Information	Reliability + Hop + BW
8803	EIGRP Route Metric Information	Reserve + Load + MTU
8804	EIGRP Ext. Route Information	Remote AS + Remote ID
8805	EIGRP Ext. Route Information	Remote Protocol+ Remote Metric

EIGRP PE-CE: Looking for Cost Communities

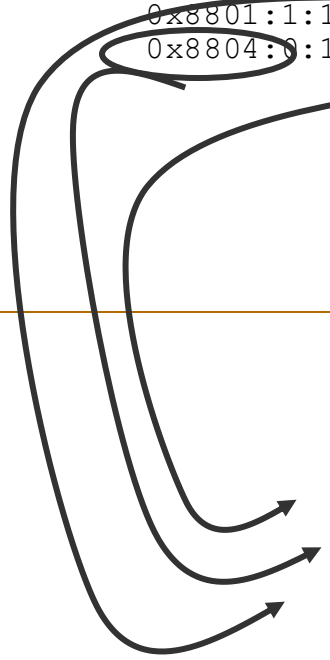
```
PE11#show ip bgp vpnv4 all 1.1.1.1
BGP routing table entry for 11:1:1.0.0.0/8, version 7
Paths: (1 available, best #1, table EIGRP-Same-AS)
   140.0.0.1 (via EIGRP-Same-AS) from 0.0.0.0 (11.11.11.11)
     Origin incomplete, metric 1889792, localpref 100, weight 32768, valid, sourced, best
     Extended Community: RT:1:1
       Cost:pre-bestpath:128:1889792 (default-2145593855) 0x8800:32768:0
       0x8801:1:640000 0x8802:65281:1249792 0x8803:65281:1500
```

We see that EIGRP Attributes of
Delay + BW + Hop Count + Reliability + MTU
are carried via MP-BGP Extended Community

Value 128 represents that route is originated internal to EIGRP domain

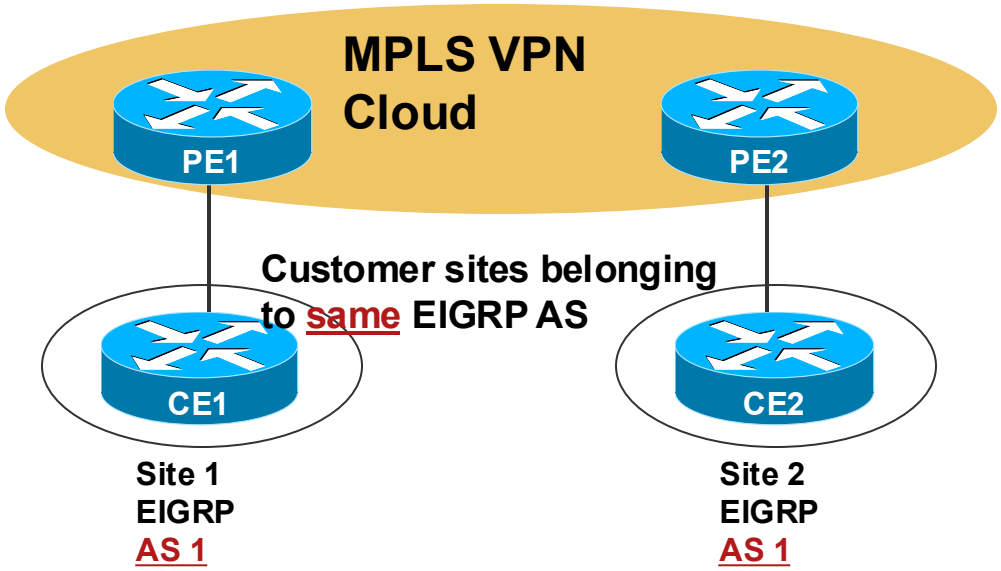
EIGRP PE-CE: Looking for Cost Communities

```
PE11#show ip bgp vpnv4 all 111.0.0.0
BGP routing table entry for 11:1:111.0.0.0/8, version 25
Paths: (1 available, best #1, table EIGRP-Same-AS)
   12.12.12.12 (metric 10) from 12.12.12.12 (12.12.12.12)
      Origin incomplete, metric 2274048, localpref 100, valid, internal, best
      Extended Community: RT:1:1
        Cost:pre-bestpath:129:2274048 (default-2145209599) 0x8800:0:0
        0x8801:1:1024256 0x8802:65281:1249792 0x8803:65281:1500
        0x8804:0:168430900 0x8805:4:1
```



If the route is external to EIGRP AS, we see a value of 129
We also see two additional pieces of information in the Cost Community value:
0x8804 includes External-AS + External Originator ID
0x8805 includes External Protocol + External Metric

Customer Sites in the Same EIGRP AS



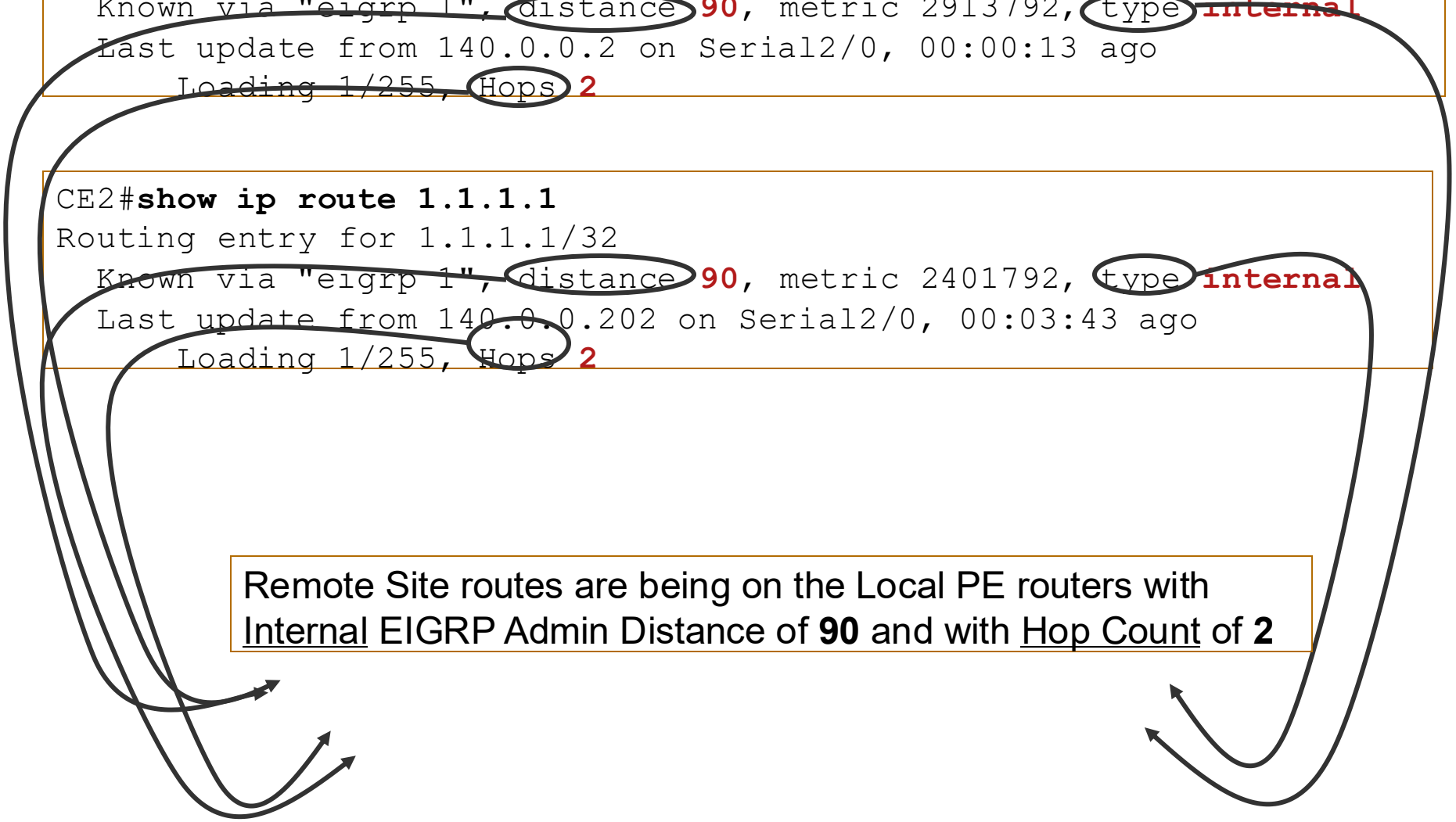
- As CE-Sites are in the same-AS, the routes will be learned with normal EIGRP attributes
- MP-BGP running on the PEs will carry the EIGRP attributes natively
  - EIGRP AS #
  - EIGRP Metrics
  - As part of the BGP update
- Customer sites will see the remote sites as part of their normal EIGRP domain

Customer Sites in the Same EIGRP AS

```
CE1#show ip route 2.2.2.2
Routing entry for 2.2.2.2/32
  Known via "eigrp 1", distance 90, metric 2913792, type internal
  Last update from 140.0.0.2 on Serial2/0, 00:00:13 ago
     Loading 1/255, Hops 2
```

```
CE2#show ip route 1.1.1.1
Routing entry for 1.1.1.1/32
  Known via "eigrp 1", distance 90, metric 2401792, type internal
  Last update from 140.0.0.202 on Serial2/0, 00:03:43 ago
     Loading 1/255, Hops 2
```

Remote Site routes are being on the Local PE routers with Internal EIGRP Admin Distance of **90** and with Hop Count of **2**



Customer Sites in the Same EIGRP AS

```
PE11#show ip eigrp vrf EIGRP-Same-AS topology 1.1.1.1   255.255.255.255
IP-EIGRP topology entry for 1.1.1.1/32
  State is Passive, Query origin flag is 1, 1 Successor(s), FD is 1889792
  Routing Descriptor Blocks:
  140.0.0.1 (Serial2/0), from 140.0.0.1, Send flag is 0x0
      Composite metric is (1889792/128256), Route is Internal
      Vector metric:
        Minimum bandwidth is 2048 Kbit
        Total delay is 25000 microseconds
        Reliability is 255/255
        Load is 1/255
        Minimum MTU is 1500
        Hop count is 1
```

```
PE11#show ip eigrp vrf EIGRP-Same-AS topology 2.2.2.2   255.255.255.255
IP-EIGRP topology entry for 2.2.2.2/32
  State is Passive, Query origin flag is 1, 1 Successor(s), FD is 2401792
  Routing Descriptor Blocks:
  0.0.0.0, from 0.0.0.0, Send flag is 0x0
      Composite metric is (2401792/0), Route is Internal (VPNv4 Sourced)
      Vector metric:
        Minimum bandwidth is 2048 Kbit
        Total delay is 45000 microseconds
        Reliability is 255/255
        Load is 1/255
        Minimum MTU is 1500
        Hop count is 1
```

PE Router is carrying all the EIGRP attributes in the learned via MP-BGP

1.1.1.1/32 is locally learned via EIGRP from CE1

2.2.2.2/32 is learned via MP-BGP from remote-PE and redistributed into the EIGRP-VRF on local Router

Customer Sites in the Same EIGRP AS

Configuration Example

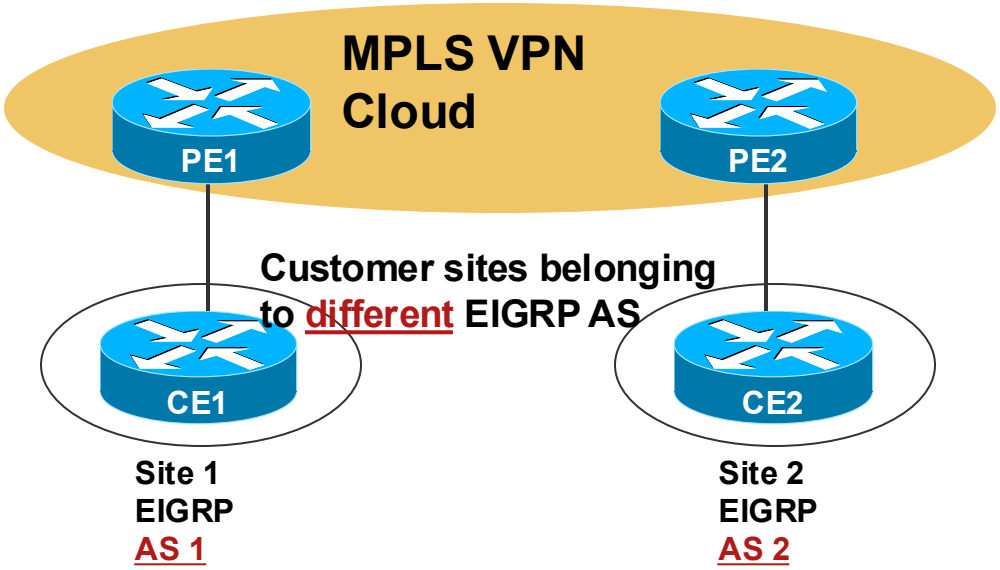
PE 1

```
ip vrf EIGRP-Same-AS
 rd 11:1
 route-target export 1:1
 route-target import 1:1
!
router eigrp 100
 !
 address-family ipv4 vrf EIGRP-Same-AS
 redistribute bgp 65000 metric 10000 1 255 1
1500
 network 140.0.0.0
 no auto-summary
 autonomous-system 1
 exit-address-family
!
router bgp 65000
 no bgp default ipv4-unicast
 bgp log-neighbor-changes
 neighbor 12.12.12.12 remote-as 65000
 neighbor 12.12.12.12 update-source Loopback0
 !
 address-family vpnv4
 neighbor 12.12.12.12 activate
 neighbor 12.12.12.12 send-community extended
 exit-address-family
 !
 address-family ipv4 vrf EIGRP-Same-AS
 redistribute eigrp 1
 no synchronization
 exit-address-family
!
```

PE 2

```
ip vrf EIGRP-Same-AS
 rd 12:1
 route-target export 1:1
 route-target import 1:1
!
router eigrp 100
 !
 address-family ipv4 vrf EIGRP-Same-AS
 redistribute bgp 65000 metric 10000 1 255 1
1500
 network 140.0.0.0
 no auto-summary
 autonomous-system 1
 exit-address-family
!
router bgp 65000
 no bgp default ipv4-unicast
 bgp log-neighbor-changes
 neighbor 11.11.11.11 remote-as 65000
 neighbor 11.11.11.11 update-source Loopback0
 !
 address-family vpnv4
 neighbor 11.11.11.11 activate
 neighbor 11.11.11.11 send-community extended
 exit-address-family
 !
 address-family ipv4 vrf EIGRP-Same-AS
 redistribute eigrp 1
 no synchronization
 exit-address-family
!
```

Customer Sites in Different EIGRP AS

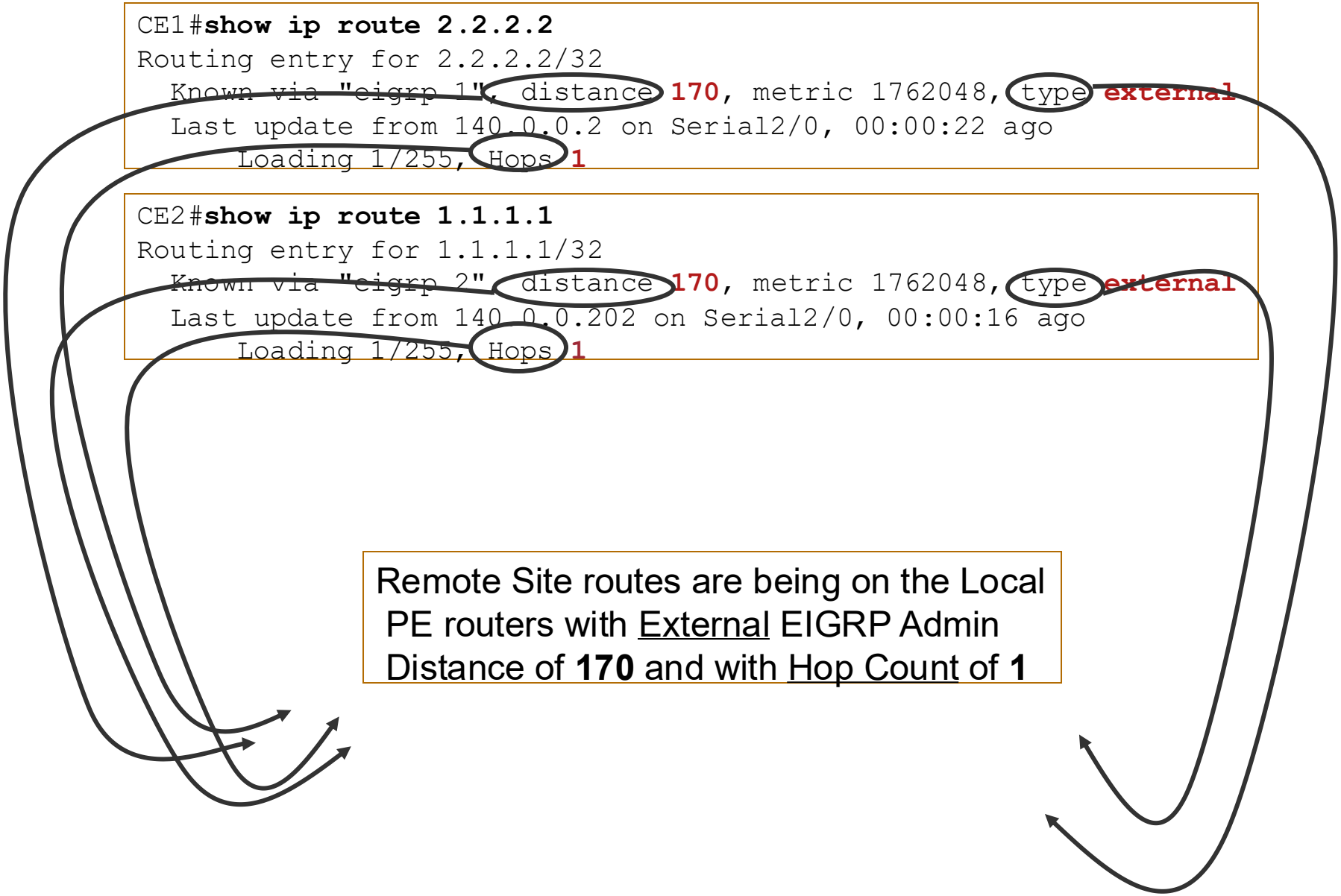


- Customer sites are in different EIGRP AS
- CE Sites will learn the remote-CE-site routes as external routes
- This is normal behavior due to the different EIGRP AS
- MP-BGP on the PE routers will carry the EIGRP routes with their normal attributes

Customer Sites in Different EIGRP AS

```
CE1#show ip route 2.2.2.2
Routing entry for 2.2.2.2/32
  Known via "eigrp 1", distance 170, metric 1762048, type external
  Last update from 140.0.0.2 on Serial2/0, 00:00:22 ago
     Loading 1/255, Hops 1
```

```
CE2#show ip route 1.1.1.1
Routing entry for 1.1.1.1/32
  Known via "eigrp 2", distance 170, metric 1762048, type external
  Last update from 140.0.0.202 on Serial2/0, 00:00:16 ago
     Loading 1/255, Hops 1
```



Remote Site routes are being on the Local PE routers with External EIGRP Admin Distance of **170** and with Hop Count of **1**

Customer Sites in Different EIGRP AS

```
PE11#show ip eigrp vrf EIGRP-Diff-AS topology 1.1.1.1 255.255.255.255
IP-EIGRP topology entry for 1.1.1.1/32
  State is Passive, Query origin flag is 1, 1 Successor(s), FD is
1889792
  Routing Descriptor Blocks:
  140.0.0.1 (Serial2/0), from 140.0.0.1, Send flag is 0x0
      Composite metric is (1889792/128256), Route is Internal
      Vector metric:
        Minimum bandwidth is 2048 Kbit
        Total delay is 25000 microseconds
        Reliability is 255/255
        Load is 1/255
        Minimum MTU is 1500
        Hop count is 1
```

```
PE11# show ip eigrp vrf EIGRP-Diff-AS topology 2.2.2.2
255.255.255.255
IP-EIGRP topology entry for 2.2.2.2/32
  State is Passive, Query origin flag is 1, 1 Successor(s), FD is
256256
  Routing Descriptor Blocks:
  0.0.0.0, from Redistributed, Send flag is 0x0
      Composite metric is (256256/0), Route is External
      Vector metric:
        Minimum bandwidth is 10000 Kbit
        Total delay is 10 microseconds
        Reliability is 255/255
        Load is 1/255
        Minimum MTU is 1500
        Hop count is 0
      External data:
        Originating router is 140.0.0.2 (this system)
        AS number of route is 65000
        External protocol is BGP, external metric is 2401792
        Administrator tag is 0 (0x00000000)
```

1.1.1.1/32 is locally learned via EIGRP from CE1

2.2.2.2/32 is learned via MP-BGP from remote-PE and redistributed into the EIGRP-VRF on local Router. This is an external route from the EIGRP domain and the info. carried in the EIGRP-VRF topology.

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Customer Sites in Different EIGRP AS

Configuration Example

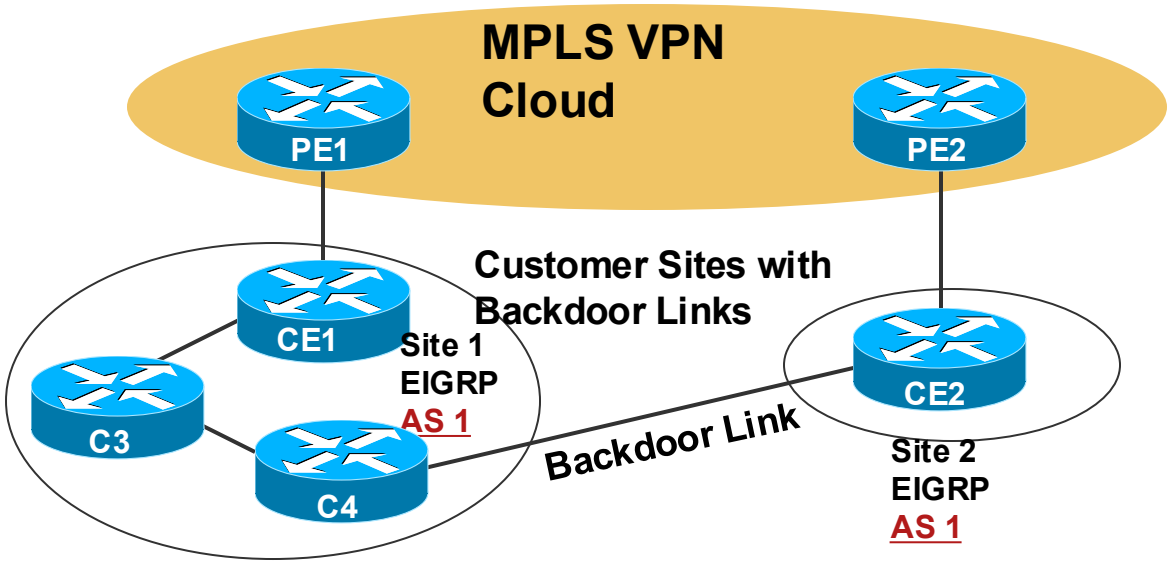
PE 1

```
ip vrf EIGRP-Diff-AS
 rd 11:1
 route-target export 1:1
 route-target import 1:1
!
router eigrp 100
 !
 address-family ipv4 vrf EIGRP-Diff-AS
 redistribute bgp 65000 metric 10000 1 255 1 1500
 network 140.0.0.0
 autonomous-system 1
 exit-address-family
!
router bgp 65000
 no bgp default ipv4-unicast
 bgp log-neighbor-changes
 neighbor 12.12.12.12 remote-as 65000
 neighbor 12.12.12.12 update-source Loopback0
 !
 address-family vpnv4
 neighbor 12.12.12.12 activate
 neighbor 12.12.12.12 send-community extended
 exit-address-family
 !
 address-family ipv4 vrf EIGRP-Diff-AS
 redistribute eigrp 1
 no synchronization
 exit-address-family
!
```

PE 2

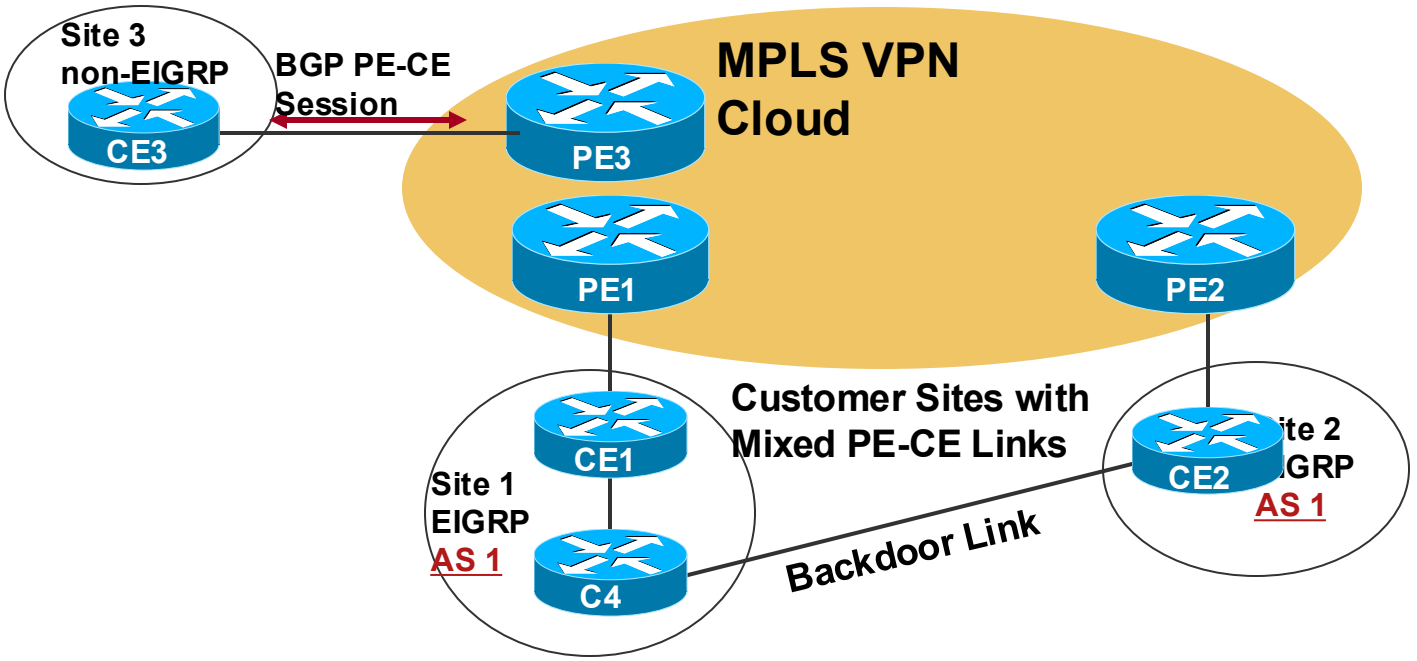
```
ip vrf EIGRP-Diff-AS
 rd 12:1
 route-target export 1:1
 route-target import 1:1
!
router eigrp 100
 !
 address-family ipv4 vrf EIGRP-Diff-AS
 redistribute bgp 65000 metric 10000 1 255 1 1500
 network 140.0.0.0
 autonomous-system 2
 exit-address-family
!
router bgp 65000
 no bgp default ipv4-unicast
 bgp log-neighbor-changes
 neighbor 11.11.11.11 remote-as 65000
 neighbor 11.11.11.11 update-source Loopback0
 !
 address-family vpnv4
 neighbor 11.11.11.11 activate
 neighbor 11.11.11.11 send-community extended
 exit-address-family
 !
 address-family ipv4 vrf EIGRP-Diff-AS
 redistribute eigrp 2
 no synchronization
 exit-address-family
 !
```

Customer Sites With Backdoor Links



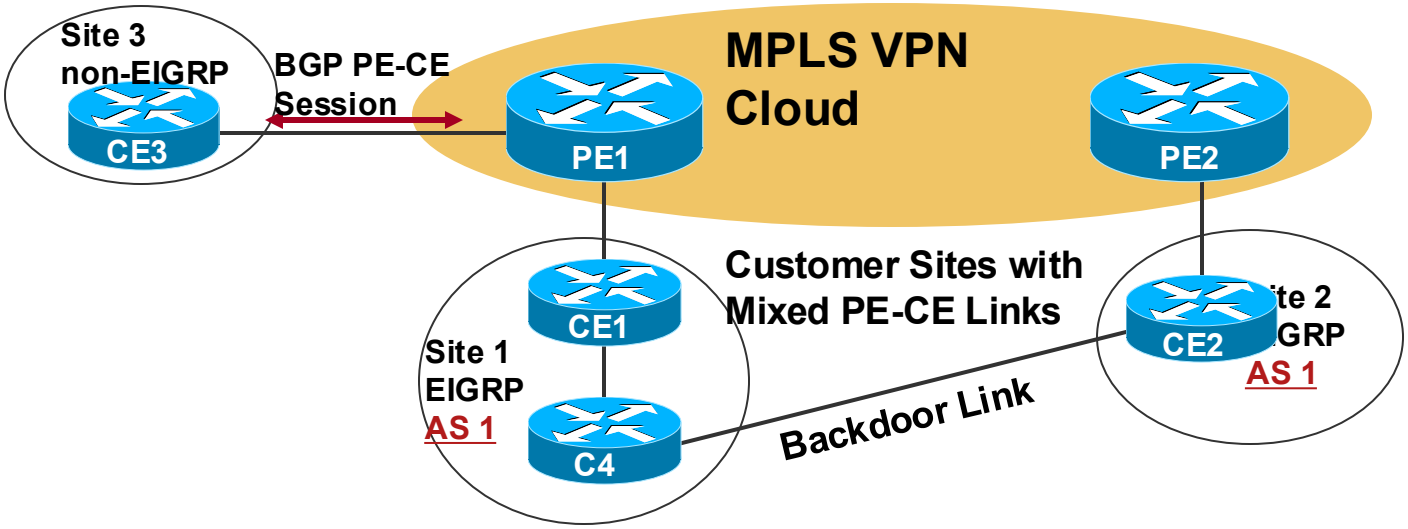
- Customer wants to use the MPLS-VPN core for the Sites connectivity
- Use the Back-door links in case of a failure (they usually are low-speed links)
- Use EIGRP attributes on the backdoor link for the Sites Connectivity (for example: delay)
- Everything should work as expected in case of a loss of connectivity through the MPLS-VPN Core

Customer Sites Have One or More Non-EIGRP Site(s)



- In this, Site 3 is connected to a remote PE rather than to the local-PE where the EIGRP Site is connected
- **Goal**: Remains the same—Non-EIGRP sites to have the reachability to EIGRP sites vice versa

Customer Sites Have One or More Non-EIGRP Site(s)



- If a Customer uses EIGRP and non-EIGRP as PE-CE protocol at different sites, we may see a persistent loop as the prefix originated from the Non-EIGRP site doesn't have the Cost-Community attribute
- The loop happens because:
  1. Initial prefix received doesn't have the SoO/Cost community
  2. Leaked into the EIGRP and then to the Remote-PE
  3. Remote-PE advertises it back to the source-PE with the SoO/Cost Community
  4. Since the SoO/Cost community route is preferred by BGP, the original route is replaced with the new one

Customer Sites Have One or More Non-EIGRP Site(s)

- CSCsa81039 addresses this problem
- Since EIGRP has the maximum hop count of 100, we can decrease this so that the loop will not be sustained for long period

```
router eigrp <AS #>
  address-family ipv4 vrf <name>
     metric maximum-hops <#>
```

Customer Sites Have One or More Non-EIGRP Site(s)

- For the proper routing to happen immediately, it is encouraged to use the following route-map.
- The route-map with the cost-community needs to be applied on the PE which is connected to the CE directly: (In our Example on PE3 towards CE3)

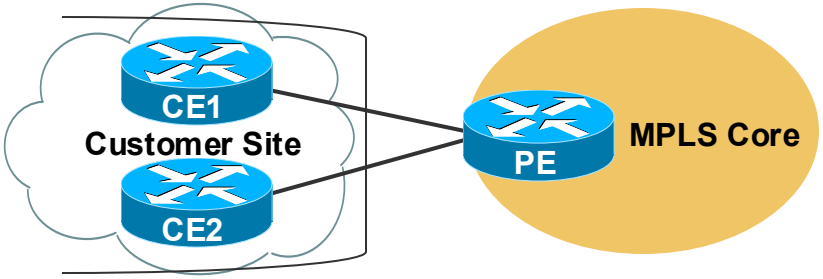
```
router bgp <X>
 address-family ipv4 vrf loop-with-non-EIGRP-PE-CE
 neighbor <ip-addr> route-map COST1 in
!
route-map COST1 permit 10
 set extcommunity cost pre-bestpath 1 100
```

3.6

Deployment Scenarios with BGP

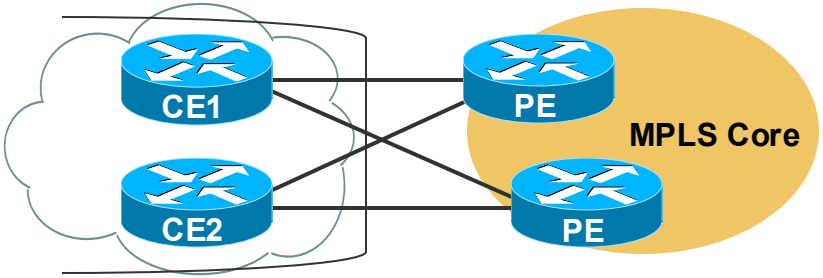
Multi-Homing and Load Balancing Options

- Customers may require as part of their SLAs to have their CE devices to be either dual-homed and/or do some form loadbalancing across the Service Provider Network for lower latency etc.
- Couple of Examples:



Customer Site is multi-homed via two different CE routers

Used for the redundancy and Fail-over scenarios



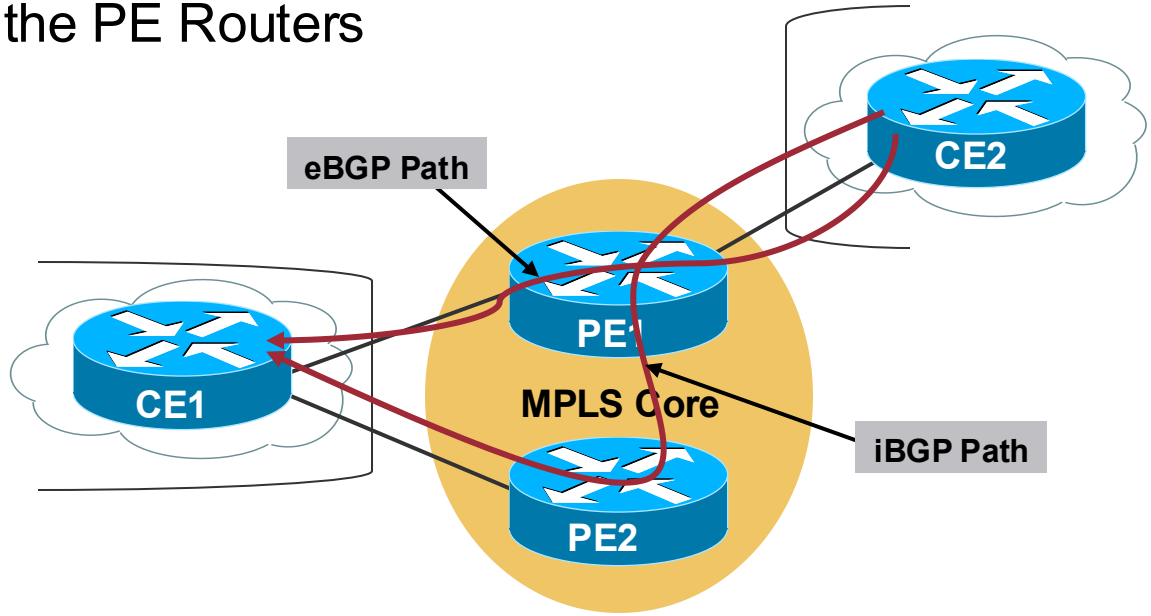
Customer Site is multi-homed via single CE to two different PEs or Two CEs meshed in with the two PEs

Used for the redundancy and Fail-over scenarios for Sites that are very Critical

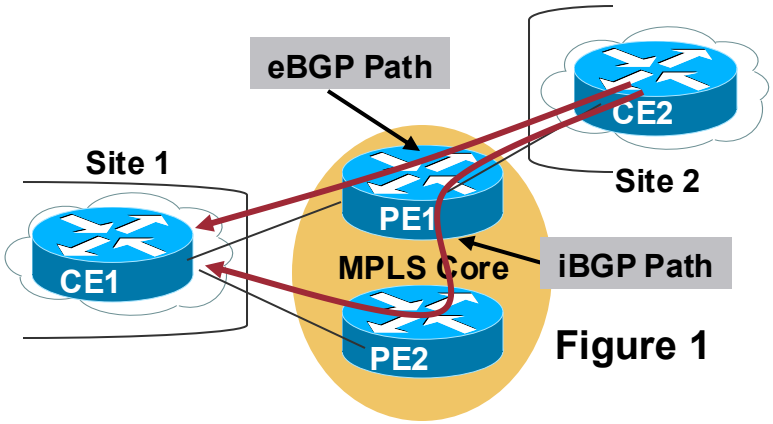
Multi-Homing and Load Balancing Options

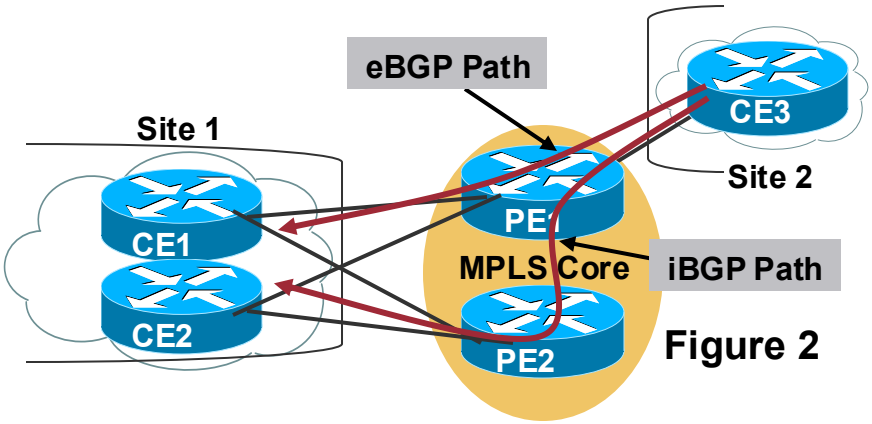
eiBGP Load Balancing:

- Customer wants their traffic to be loadbalanced across the MPLS Core
- Customer Site CE Router(s) is/are multi-homed and they want the traffic from remote sites to be load-balanced across the MPLS core both on the iBGP and eBGP paths
- This is called eiBGP Multipath and all the configuration is done on the PE Routers



Multi-Homing and Load Balancing Options





- **Figure 1**: CE1 is Dual-homed to PE1 and PE2. Traffic coming from CE2 will be loadbalanced by PE1 both on the iBGP path to PE2 and on the directly connected eBGP link to CE1
- **Figure 2**: Similar to Site 1 in Figure 1 but Site 1 has redundancy at the customer site routers
- In both the above cases, Site 1 and Site 2 are connected to the same PE router in the MPLS core

Multi-Homing and Load Balancing Options

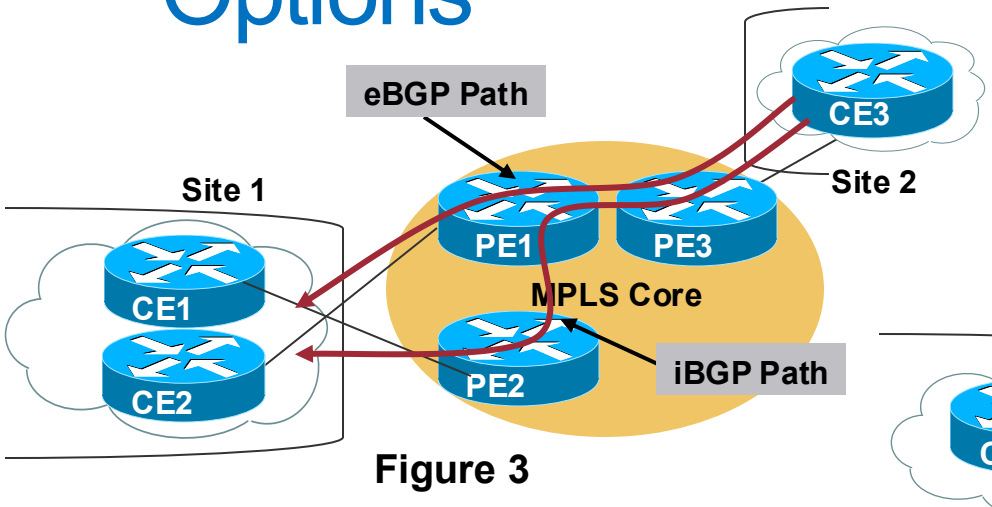


Figure 3

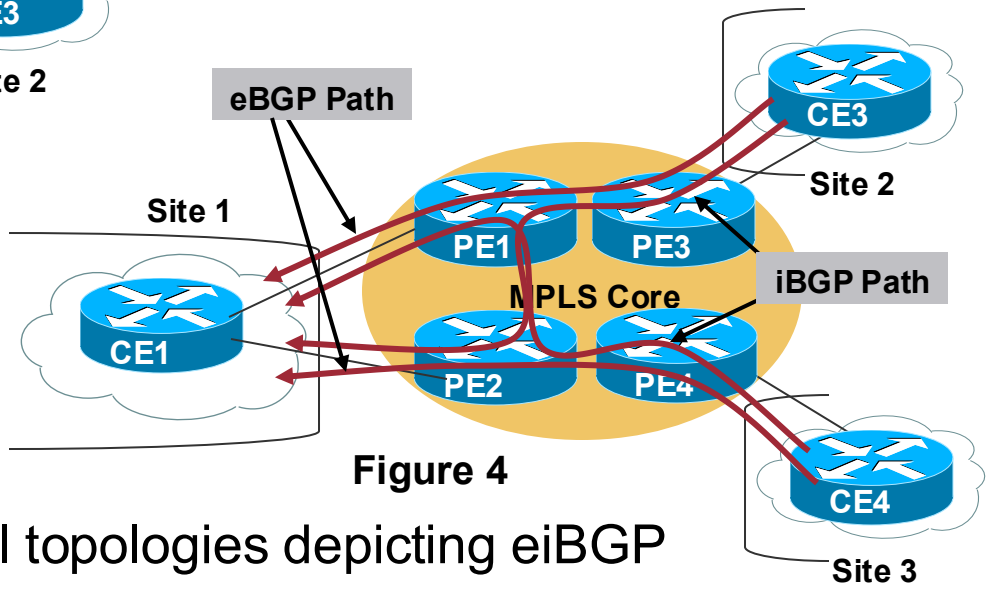
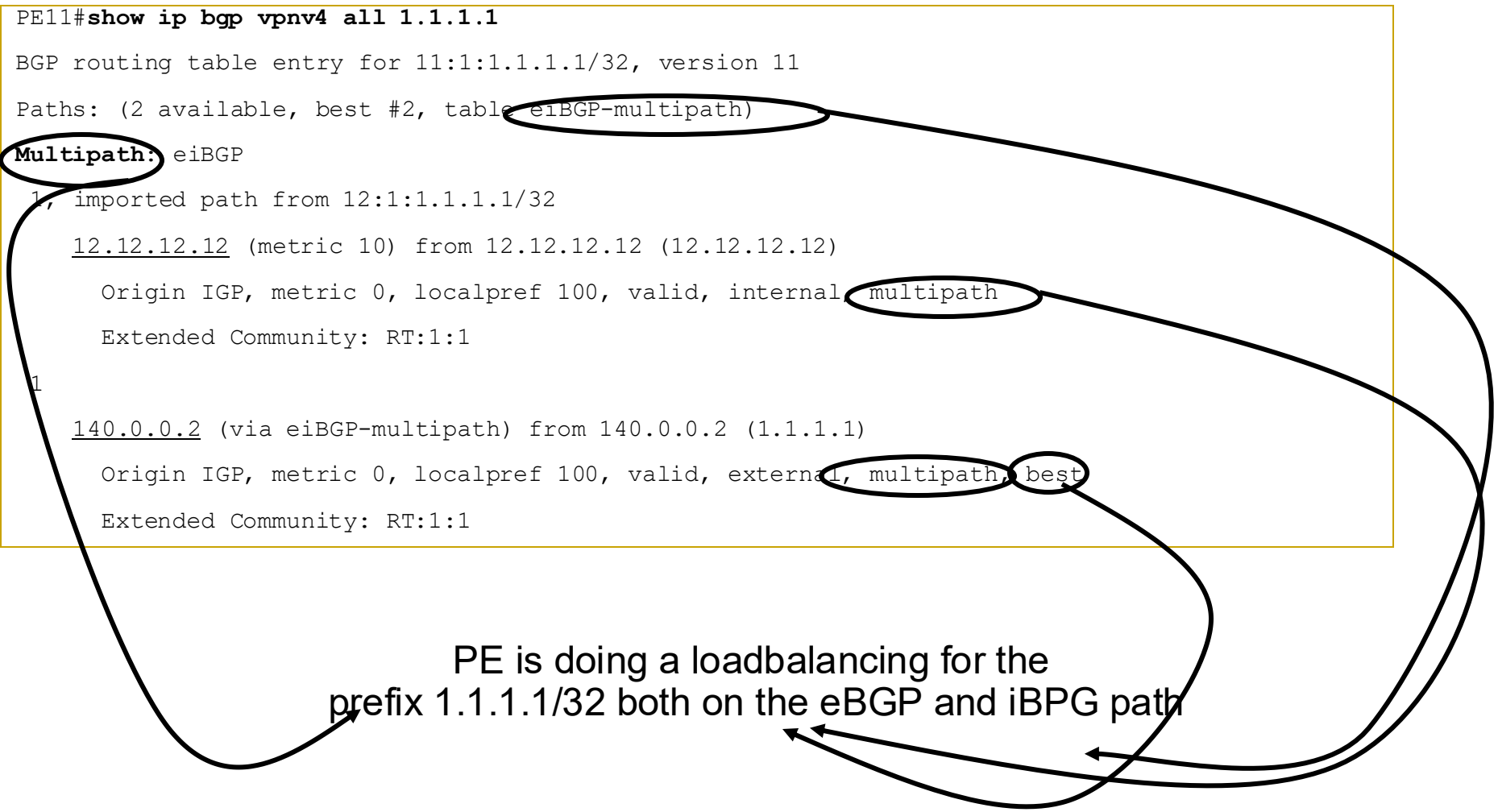


Figure 4

- Here are couple of additional topologies depicting eiBGP load balancing examples
- **Figure 3**: Site 1 is locally connected to PE1 and PE2 where as Site 2 is connected to PE3
- **Figure 4**: Multiple Customer remote Sites (Site 2, Site 3) trying to reach Site 1. Loadbalancing should happened with in the MPLS-VPN Core

eiBGP: At the PE Router

```
PE11#show ip bgp vpnv4 all 1.1.1.1
BGP routing table entry for 11:1:1.1.1.1/32, version 11
Paths: (2 available, best #2, table eiBGP-multipath)
Multipath: eiBGP
  1, imported path from 12:1:1.1.1.1/32
    12.12.12.12 (metric 10) from 12.12.12.12 (12.12.12.12)
      Origin IGP, metric 0, localpref 100, valid, internal, multipath
      Extended Community: RT:1:1
  1
    140.0.0.2 (via eiBGP-multipath) from 140.0.0.2 (1.1.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, multipath, best
      Extended Community: RT:1:1
```



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eiBGP: At the PE Router

```
PE11# show ip route vrf eiBGP-multipath 1.1.1.1

Routing entry for 1.1.1.1/32

  Known via "bgp 65000", distance 20, metric 0

  * 140.0.0.2, from 140.0.0.2, 00:20:14 ago

      AS Hops 1, BGP network version 0

    12.12.12.12 (Default-IP-Routing-Table), from 12.12.12.12, 00:20:01 ago

     AS Hops 1, BGP network version 0
```

```
PE11#show ip cef vrf eiBGP-multipath 1.1.1.1
1.1.1.1/32, version 13, epoch 0, per-destination sharing
  via 140.0.0.2, 0 dependencies, recursive
    next hop 140.0.0.2, Serial2/0 via 140.0.0.0/30 (eiBGP-multipath)
    valid adjacency
    tag rewrite with Se2/0, point2point, tags imposed {}
  via 12.12.12.12, 0 dependencies, recursive
    next hop 130.0.0.2, Ethernet0/0 via 12.12.12.12/32 (Default)
    valid adjacency
    tag rewrite with Et0/0, 130.0.0.2, tags imposed {35}
```

We see both the IP-VRF RIB and VRF-CEF are doing the loadbalancing for the prefix 1.1.1.1/32 reflecting the BGP-VPNv4-RIB.

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eiBGP: PE Router Configuration

```
router bgp 65000
 neighbor 12.12.12.12 remote-as 65000
 neighbor 12.12.12.12 update-source Loopback0
 neighbor 13.13.13.13 remote-as 65000
 neighbor 13.13.13.13 update-source Loopback0
 !
 address-family vpnv4
 neighbor 12.12.12.12 activate
 neighbor 12.12.12.12 send-community extended
 neighbor 13.13.13.13 activate
 neighbor 13.13.13.13 send-community extended
 exit-address-family
 !
 address-family ipv4 vrf eiBGP-multipath
 redistribute connected
 neighbor 140.0.0.2 remote-as 1
 neighbor 140.0.0.2 activate
 neighbor 140.0.0.2 as-override
 maximum-paths eibgp 2
 no synchronization
 exit-address-family
```

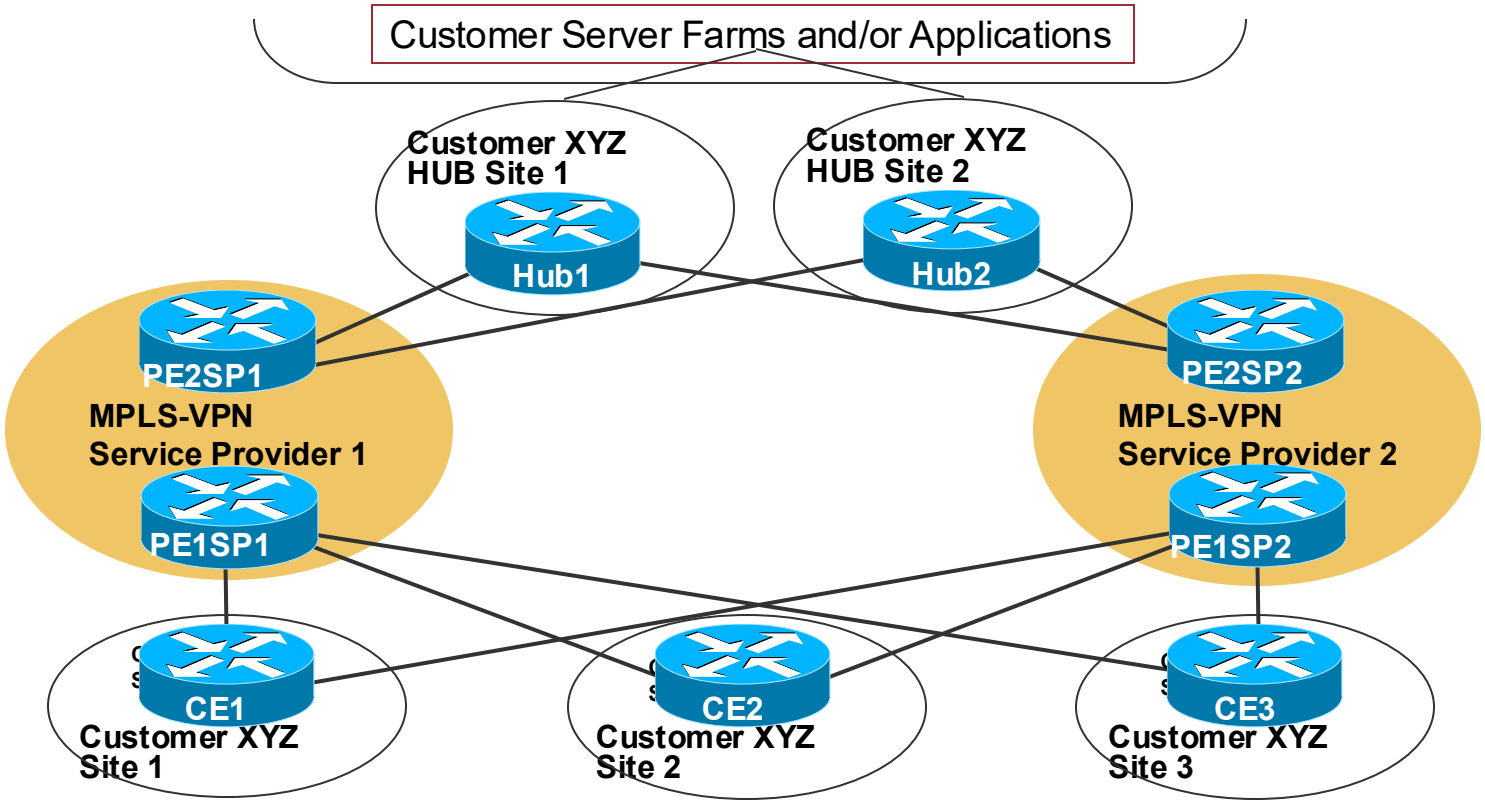
eiBGP Multipath is configured under the AF IPv4-VRF

There is no any additional configuration is needed on the Customer routers (CE)

A specific case where the customer wants their return traffic to be load balanced across the MPLS core using iBGP and eBGP paths

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Loadbalancing Across Dual Providers



- Customer sites are connected to two Service Providers
- Customer wants to do some form of load balancing across the dual SP Providers
- This is for all the traffic originating both from the Spoke Sites to Hub sites vice versa

Loadbalancing Across Dual Providers

```
CE2#show ip bgp 1.1.1.1
BGP routing table entry for 1.1.1.1/32, version 17
Paths: (2 available, best #1)
Multipath: eBGP
  65535 65535
    130.0.32.23 from 130.0.32.23 (23.23.23.23)
      Origin IGP, localpref 100, valid, external, multipath, best
  65535 65535
    130.0.2.13 from 130.0.2.13 (13.13.13.13)
      Origin IGP, localpref 100, valid, external, multipath
```

```
CE2#show ip route
B     1.1.1.1 [20/0] via 130.0.2.13, 00:28:21
              [20/0] via 130.0.32.23, 00:22:11
```

We see that the Remote site prefix is learned via eBGP using multipath at the Site 2. Both the AS Path Lengths have to match and have to be the same in order for the eBGP Multipath to work.

This require SPs to agree on the AS Path Policy for the given customer

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Loadbalancing Across Dual Providers

A Typical Configuration on the SP PE Router Will Look Like:

```
router bgp 12345
 address-family ipv4 vrf TEST
 redistribute connected
 neighbor 130.0.32.2 remote-as 65200
 neighbor 130.0.32.2 local-as 65535 no-prepend replace-as
 neighbor 130.0.32.2 activate
 neighbor 130.0.32.2 as-override
 no synchronization
 exit-address-family
!
```

With **Local-AS** option, the Customer Dual-SP will use the same AS number to peer with the customer CE routers

With **No-Prepend** option, local-as is not going to be prepended to update from eBGP peers

With **Replace-AS** option, replace real AS with local AS in the eBGP Updates

AS-Override option is used to replace the customer AS # with SP AS # when they use the Same AS at different customer Sites

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Loadbalancing Across Dual Providers

```
PE1# show ip bgp vpnv4 all 8.8.8.8
BGP routing table entry for 65000:65000:8.8.8.8/32, version 11
Paths: (1 available, best #1, table TEST)
  65200 65000
    130.0.11.1 (via boa) from 130.0.11.1 (1.1.1.1)
      Origin IGP, localpref 100, valid, external, best
      Extended Community: RT:65000:65000
```

```
BGP(0): 130.0.11.1 rcv UPDATE about 8.8.8.8/32 -- DENIED due to: AS-PATH contains our own AS
```

PE accepts the route matching with its own AS # due to the knob of **allowas-in** option

In the debug we see that the prefix is DENIED if the allowas-in option is not turned on.

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Loadbalancing Across Dual Providers

```
CE2#show ip bgp 8.8.8.8
BGP routing table entry for 8.8.8.8/32, version 11
Multipath: eBGP
  65535 65535 65000
    130.0.32.23 from 130.0.32.23 (23.23.23.23)
      Origin IGP, localpref 100, valid, external, multipath, best
  65535 65535 65000
    130.0.2.13 from 130.0.2.13 (13.13.13.13)
      Origin IGP, localpref 100, valid, external, multipath
```

```
CE2#show ip route 8.8.8.8
Routing entry for 8.8.8.8/32
  Known via "bgp 65200", distance 20, metric 0
  * 130.0.2.13, from 130.0.2.13, 00:00:17 ago
      AS Hops 3, BGP network version 0
      Route tag 65535
    130.0.32.23, from 130.0.32.23, 00:00:16 ago
      AS Hops 3, BGP network version 0
      Route tag 65535
```

The prefix 8.8.8.8/32 is being Load balanced using the eBGP Multipath

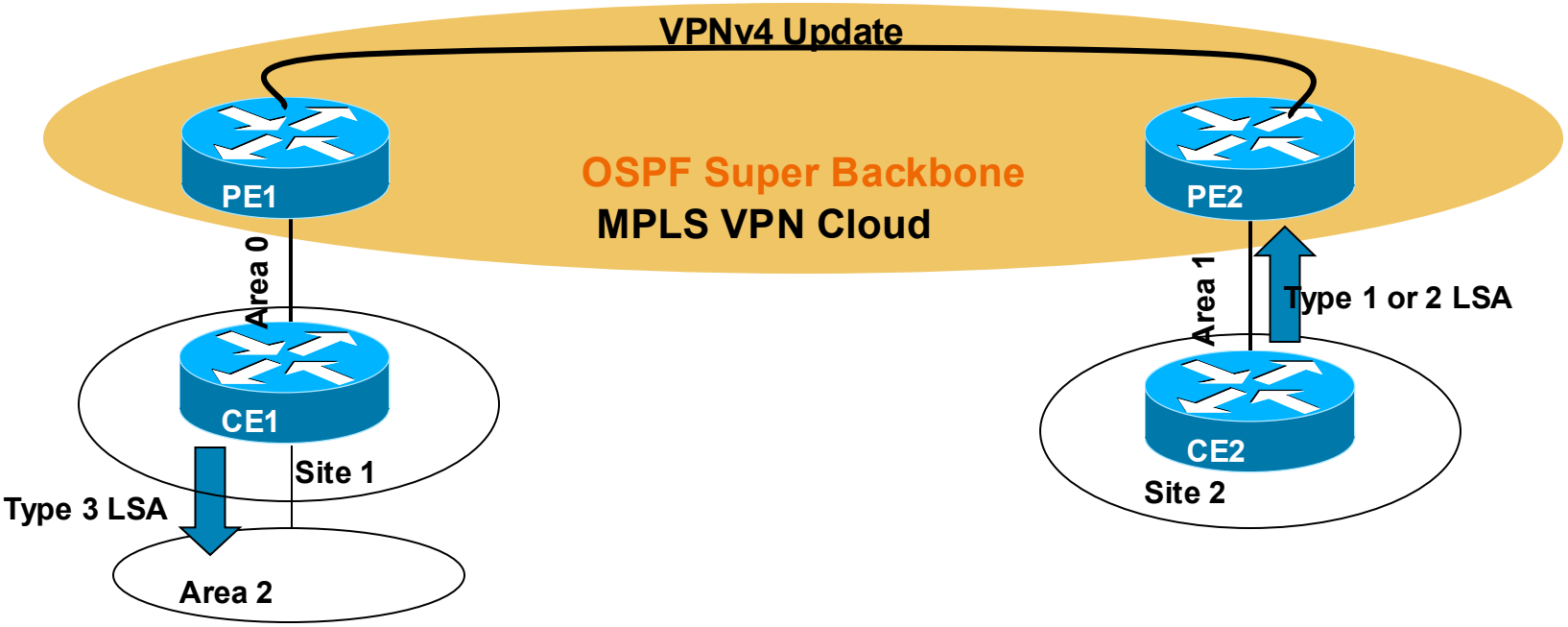
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3.6

Deployment Scenarios with OSPF

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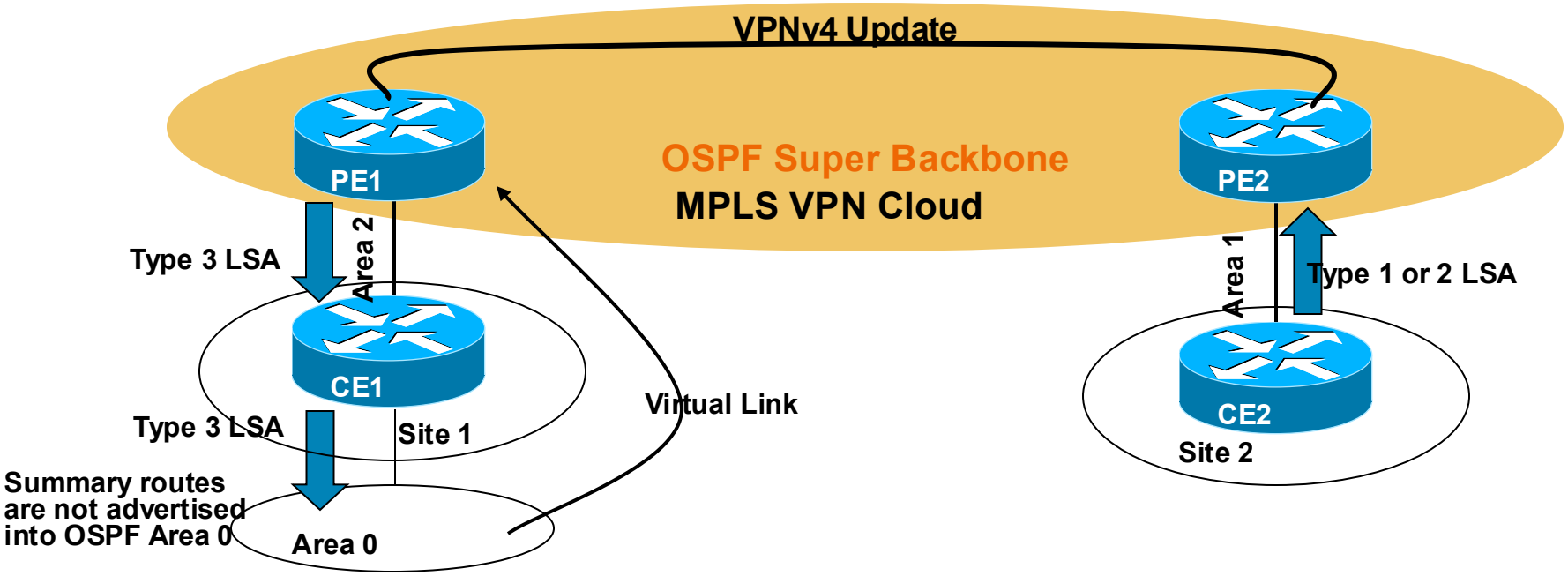
OSPF Area 0 Placement



- Placing of the Customer OSPF Area 0 is very important
- If a Customer Site has area 0, it MUST touch the MPLS-VPN PE Router
- In the above example, some site(s) consist of non-OSPF area 0 as well as area 0 Sites
- A type 1 or 2 LSA coming from non-OSPF area 0 will be flooded as a type 3 LSA

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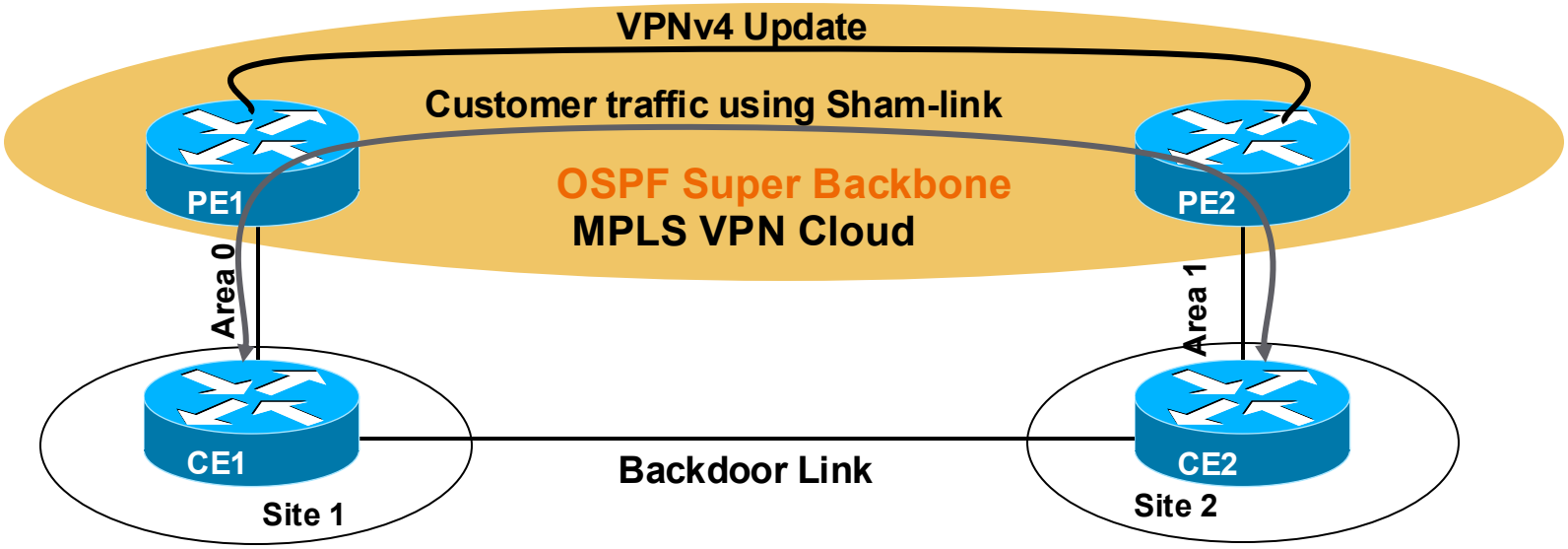
OSPF Area 0 Placement



- In the above topology, the customer OSPF area 0 is not directly connected to the PE
- A type 1 or 2 LSA, coming from MPLS-VPN superbackbone will be summarized and be sent as a Type 3 LSA into area 2.
- Then area 2 will send this type 3 LSA into AREA 0 below AND
- Summary routes are not advertised (back) into Area 0
- In this case, you need a Virtual-link to connect the Area 0
- Recommended to: Have the OSPF AREA 0 connected to the MPLS-VPN Core

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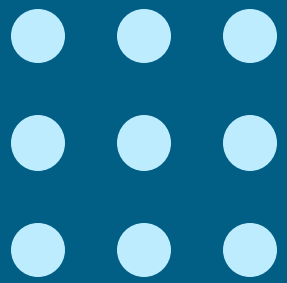
OSPF Area 0 Placement—Sham Link



- When Customer sites are connected via backdoor links – sham-links are recommended so that the MPLS-VPN core will be preferred rather than through the backdoor link
- The sham link is treated as a virtual-link: unnumbered, ptp, DC link
- The sham link is reported in the router LSA's type 1 originated by the two routers connecting to the sham link
- The MPLS VPN backbone or the backdoor link can be made preferred path by tweaking the metrics

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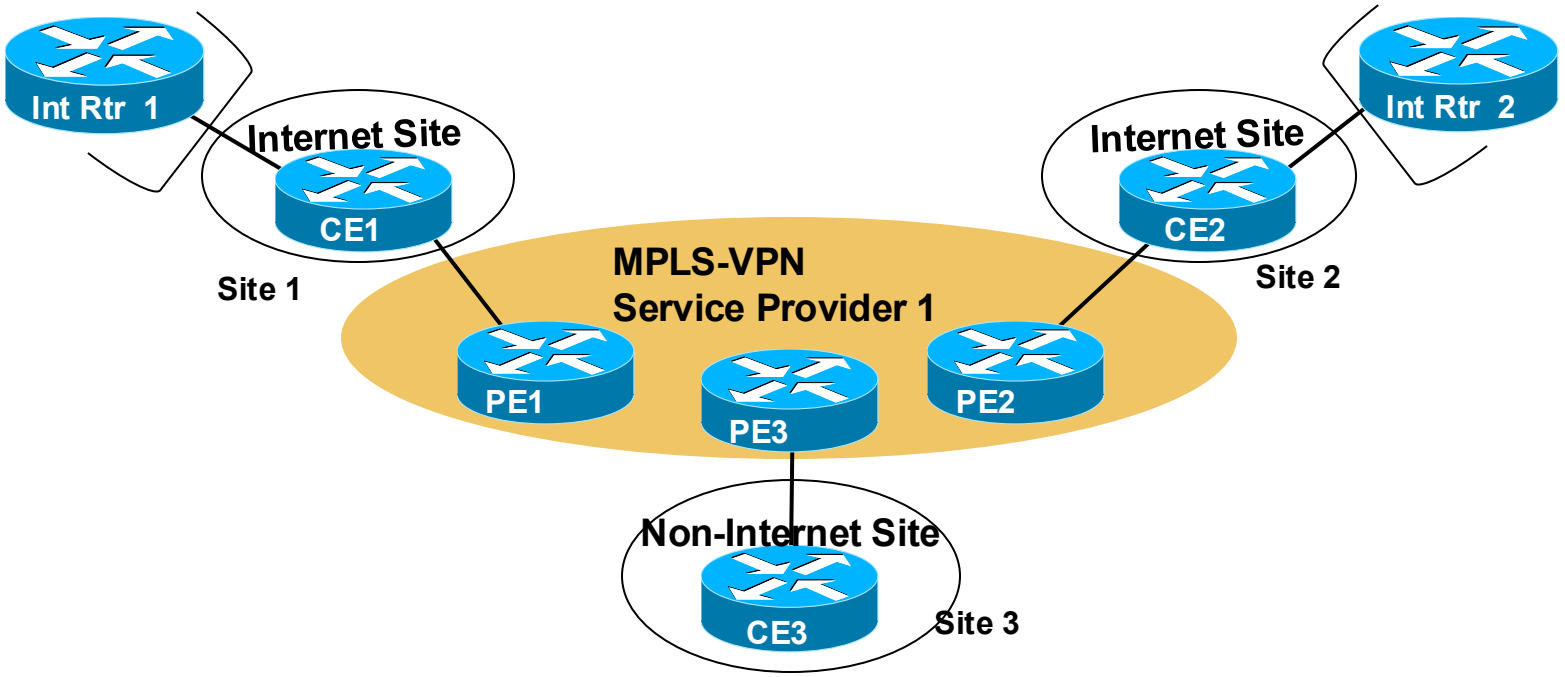
3.6



Deployment Scenarios with EIGRP

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Default-Gateway Using EIGRP PE-CE



- A typical deployment scenario with the Self-Managed MPLS network where big enterprises manage their own MPLS core
- Customer has peering to two ISPs at two different locations
- They want the default-route to get propagated into the non-Internet sites
- In the above example: Site 1 and Site 2 are Internet Connected Sites and Site 3 is not

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Default-Gateway Using EIGRP PE-CE

Note:

- PE needs the "**default-information originate**" under the IPv4-VRF-AF for the default-route to get propagated across the MPLS-Core which was learned via EIGRP
- By default, if a PE receives the same prefix from Customer site and also from MPLS-VPN backbone, then the metrics components which are carried in the BGP Extended communities are compared to the EIGRP metrics from the customer site
- In our Ex.: 0.0.0.0/0 is learned by PE2 from MPLS-VPN cloud via CE1-(Site 1)→ PE1 and also from the CE2 at Site 2
- Metric of the path determines which path is better

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Default-Gateway Using EIGRP PE-CE

```
PE2#show ip bgp vpnv4 all 0.0.0.0
BGP routing table entry for 1234:2:0.0.0.0/0, version 124
Paths: (1 available, best #1, table intranet)
  Local, imported path from 1234:1:0.0.0.0/0
    11.11.11.11 (metric 30) from 22.22.22.22 (22.22.22.22)
      Origin incomplete, metric 281856, localpref 100, valid, internal, best
      Extended Community: RT:1234:123 Cost:pre-bestpath:129:281856 0x8800:0:5
        0x8801:1:25856 0x8802:65281:256000 0x8803:65281:1500
        0x8804:1:2802450433 0x8805:9:0
      Originator: 11.11.11.11, Cluster list: 22.22.22.22, 21.21.21.21
```

PE2 prefers the DEFAULT from PE1 as the metric received in the Pre-Bestpath is better than the metric received via EIGRP

via MPLS-Core: 281856 via EIGRP: 1762048

```
PE2#show ip eigrp vrf intranet topology
P 0.0.0.0/0, 1 successors, FD is 281856, tag is 5
         via VPNv4 Sourced (281856/0)
         via 167.51.51.1 (1762048/256256), Serial2/0
```

Metric Determines which path is better

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Default-Gateway Using EIGRP PE-CE

```
PE2#show ip eigrp vrf intranet topology 0.0.0.0 0.0.0.0
IP-EIGRP topology entry for 0.0.0.0/0
  State is Passive, Query origin flag is 1, 1 Successor(s), FD is 281856
  0.0.0.0, from 0.0.0.0, Send flag is 0x0
      Composite metric is (281856/0), Route is External (VPNv4 Sourced)
      External data:
        Originating router is 167.10.0.1
        AS number of route is 1
        External protocol is BGP, external metric is 0
        Administrator tag is 5 (0x00000005)
  167.51.51.1 (Serial2/0), from 167.51.51.1, Send flag is 0x0
      Composite metric is (1762048/256256), Route is External
      External data:
        Originating router is 2.2.2.2
        AS number of route is 2
        External protocol is BGP, external metric is 0
        Administrator tag is 6 (0x00000006)
        Exterior flag is set
```

The EIGRP-VRF topology shows the metrics received via the EIGRP Path and via MP-BGP path. The Composite-metric received from the MPLS-VPN core (281856) is better than the one received from the EIGRP-neighbor (1762048)

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Default-Gateway Using EIGRP PE-CE

```
PE2#show ip bgp vpnv4 all 0.0.0.0
BGP routing table entry for 1234:2:0.0.0.0/0, version 105
Paths: (2 available, best #2, table intranet)
  Local, imported path from 1234:1:0.0.0.0/0
    11.11.11.11 (metric 30) from 22.22.22.22 (22.22.22.22)
      Origin incomplete, metric 281856, localpref 100, valid, internal
      Extended Community: RT:1234:123
        Cost:pre-bestpath:129:2000000 (default-2145483647)
  Local
    167.51.51.1 (via intranet) from 0.0.0.0 (12.12.12.12)
      Origin incomplete, metric 1762048, localpref 100, weight 32768, valid, sourced, best
      Extended Community: RT:1234:123
        Cost:pre-bestpath:129:1762048 (default-2145721599)
```

We can use the Pre-Bestpath option to overwrite metric that will be used in the best path calculation.

The Pre-Bestpath metric from 11.11.11.11 is compared with from 167.51.51.1

2000000 vs. 1762048 Lower Metric wins.

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Default-Gateway Using EIGRP PE-CE

Configuration Example

```
router bgp 65000
 !
 address-family vpnv4
 neighbor 21.21.21.21 activate
 neighbor 21.21.21.21 send-community extended
 neighbor 21.21.21.21 route-map COST out
 exit-address-family
!
route-map COST permit 1
 set extcommunity cost pre-bestpath 129 2000000
!
```

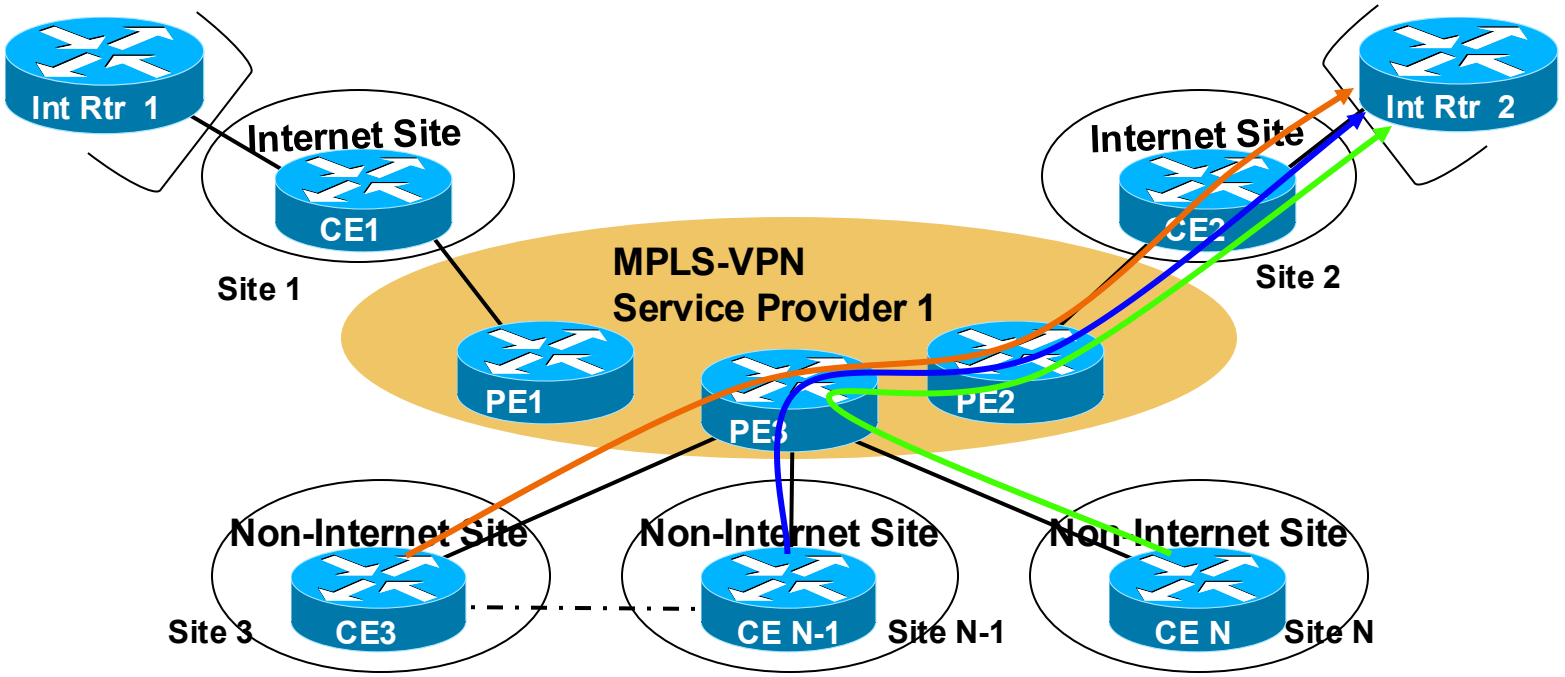
PE3 also now prefers it via PE2

Customer at NON-Internet Site may prefer for their Internet traffic to go via PE1


```
PE3#show ip bgp vpnv4 all 0.0.0.0
BGP routing table entry for 1234:3:0.0.0.0/0, version 88
Paths: (2 available, best #2, table intranet)
  Local, imported path from 1234:1:0.0.0.0/0
    11.11.11.11 (metric 30) from 23.23.23.23 (23.23.23.23)
      Origin incomplete, metric 281856, localpref 100, valid, internal
      Extended Community: RT:1234:123
        Cost:pre-bestpath:129:2000000 (default-2145483647)
  Local, imported path from 1234:2:0.0.0.0/0
    12.12.12.12 (metric 30) from 23.23.23.23 (23.23.23.23)
      Origin incomplete, metric 1762048, localpref 100, valid, internal, best
      Extended Community: RT:1234:123
        Cost:pre-bestpath:129:1762048 (default-2145721599)
```


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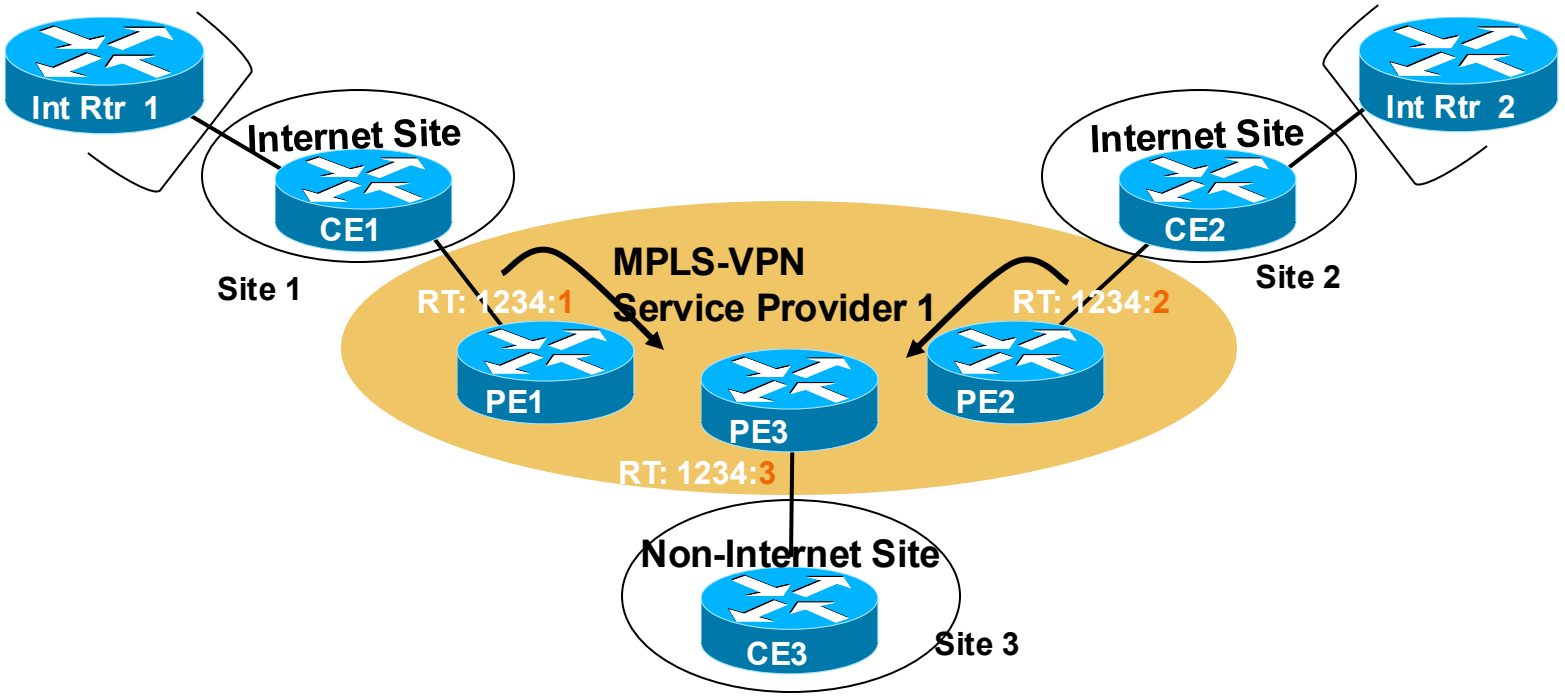
Default-Gateway Using EIGRP PE-CE



- As seen, Non-Internet-Site may start preferring the Site 2 for Internet Connection via PE2
- All other Non-Internet-Sites will also prefer via PE2
- This may not be Optimal as All Other Non-Internet Sites may prefer Deterministically via Site 2 using PE2

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Using RT for the Default Gateway



- Assign an Unique RT value at each PE router
- Non-Internet Connected site PE router can use an in-bound policy using a pre-bestpath or weight to change the best path that is received.
- PE3, in the above example uses: RT with 1234.1 with preferred pre-bestpath than one with the 1234:2

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Using RT for the Default Gateway

```
PE3#show ip bgp vpnv4 all 0.0.0.0
BGP routing table entry for 1234:3:0.0.0.0/0, version 103
Paths: (2 available, best #2, table intranet)
  Local, imported path from 1234:2:0.0.0.0/0
    12.12.12.12 (metric 30) from 23.23.23.23 (23.23.23.23)
      Origin incomplete, metric 1762048, localpref 100, valid, internal
      Extended Community: RT:1234:2 Cost:pre-bestpath:1:2
        Cost:pre-bestpath:129:1762048 (default-2145721599)
  Local, imported path from 1234:1:0.0.0.0/0
    11.11.11.11 (metric 30) from 23.23.23.23 (23.23.23.23)
      Origin incomplete, metric 281856, localpref 100, valid, internal, best
      Extended Community: RT:1234:1 Cost:pre-bestpath:1:1
        Cost:pre-bestpath:129:2000000 (default-2145483647)
```

Even though Cost: pre-bestpath: 129:2000000 is worse than 129:1762048
PE3 picks route from the 11.11.11.11 than from the 12.12.12.12

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Using RT for the Default Gateway

Configuration on the PE3 Router

```
router bgp 65000
 no bgp default ipv4-unicast
 bgp log-neighbor-changes
 neighbor 23.23.23.23 remote-as 65000
 neighbor 23.23.23.23 update-source Loopback0
 !
 address-family vpnv4
 neighbor 23.23.23.23 activate
 neighbor 23.23.23.23 send-community extended
 neighbor 23.23.23.23 route-map choose-internet-pop in
 exit-address-family
 !
 address-family ipv4 vrf intranet
 redistribute eigrp 1
 no synchronization
 exit-address-family
!
ip extcommunity-list 1 permit rt 1234:1
ip extcommunity-list 2 permit rt 1234:2
!
route-map choose-internet-pop permit 10
 match extcommunity 1
 set extcommunity cost pre-bestpath 1 1
!
route-map choose-internet-pop permit 20
 match extcommunity 2
 set extcommunity cost pre-bestpath 1 2
!
```

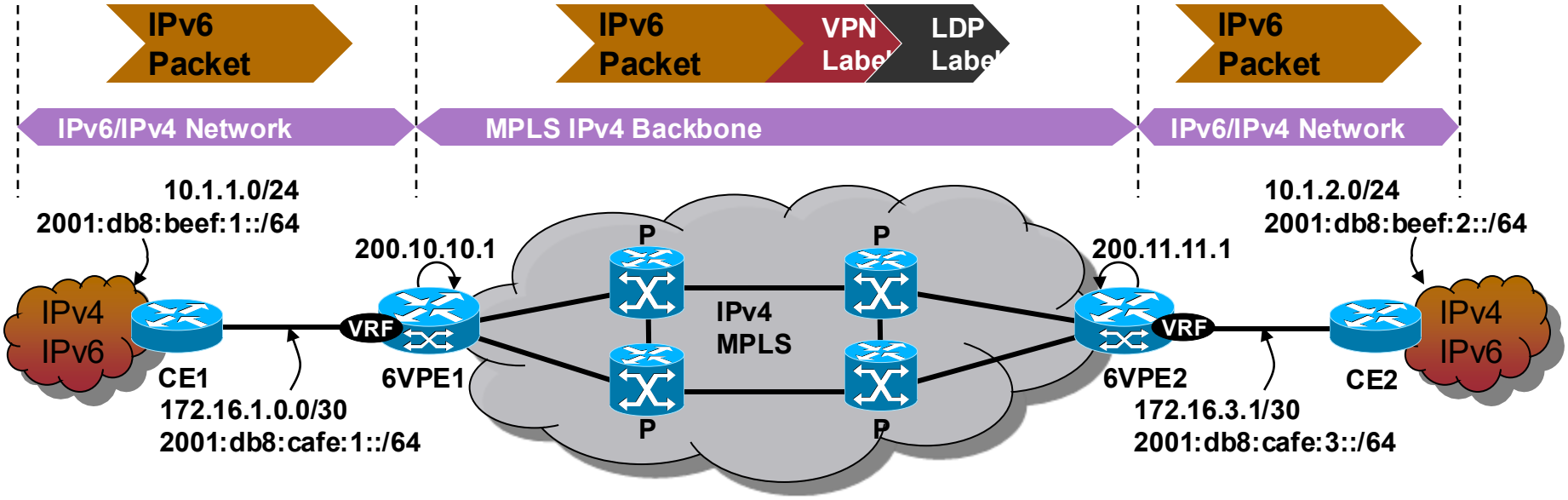

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6vPE

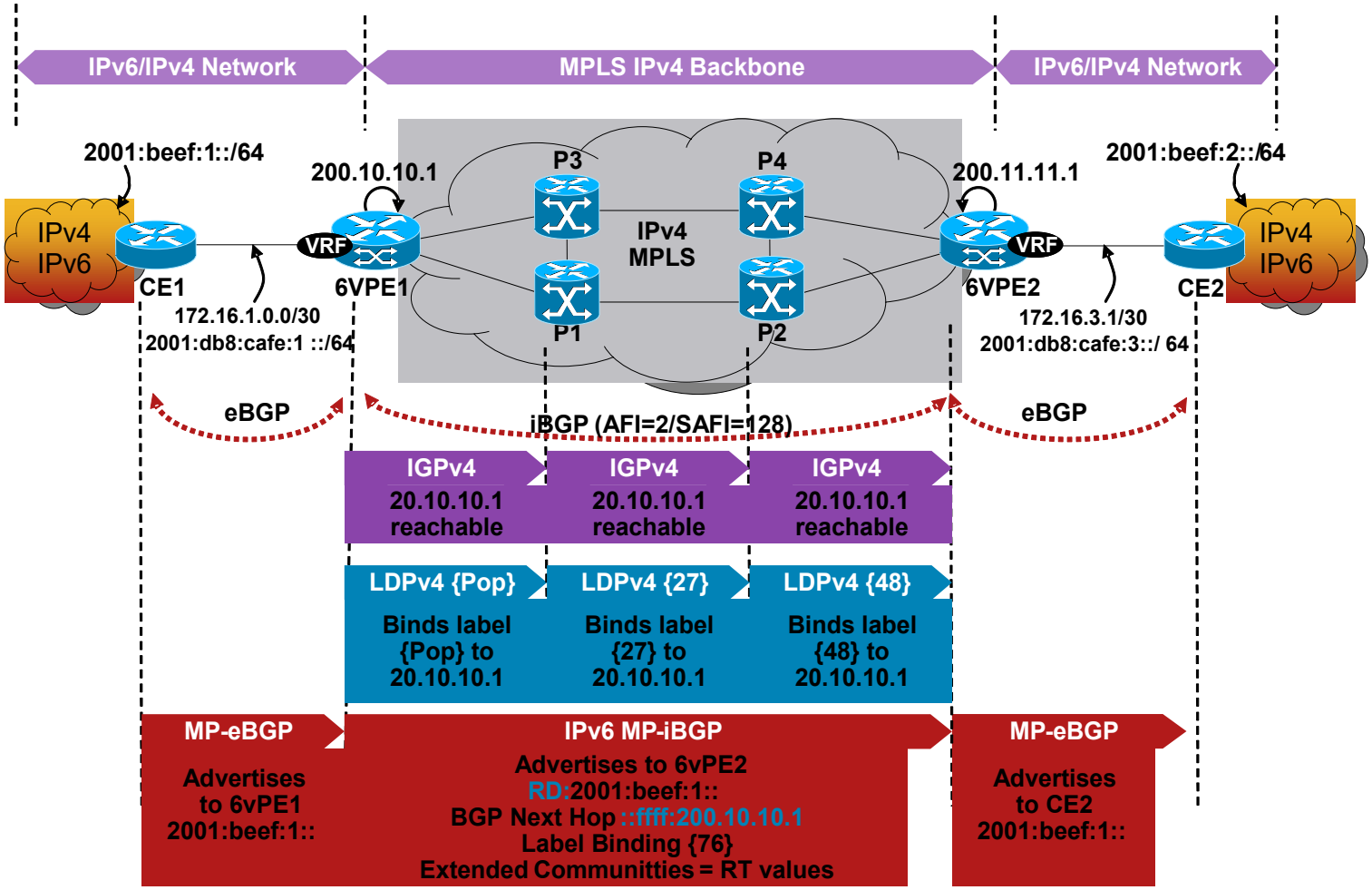
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IPv6 VPN 6VPE (RFC 4659)

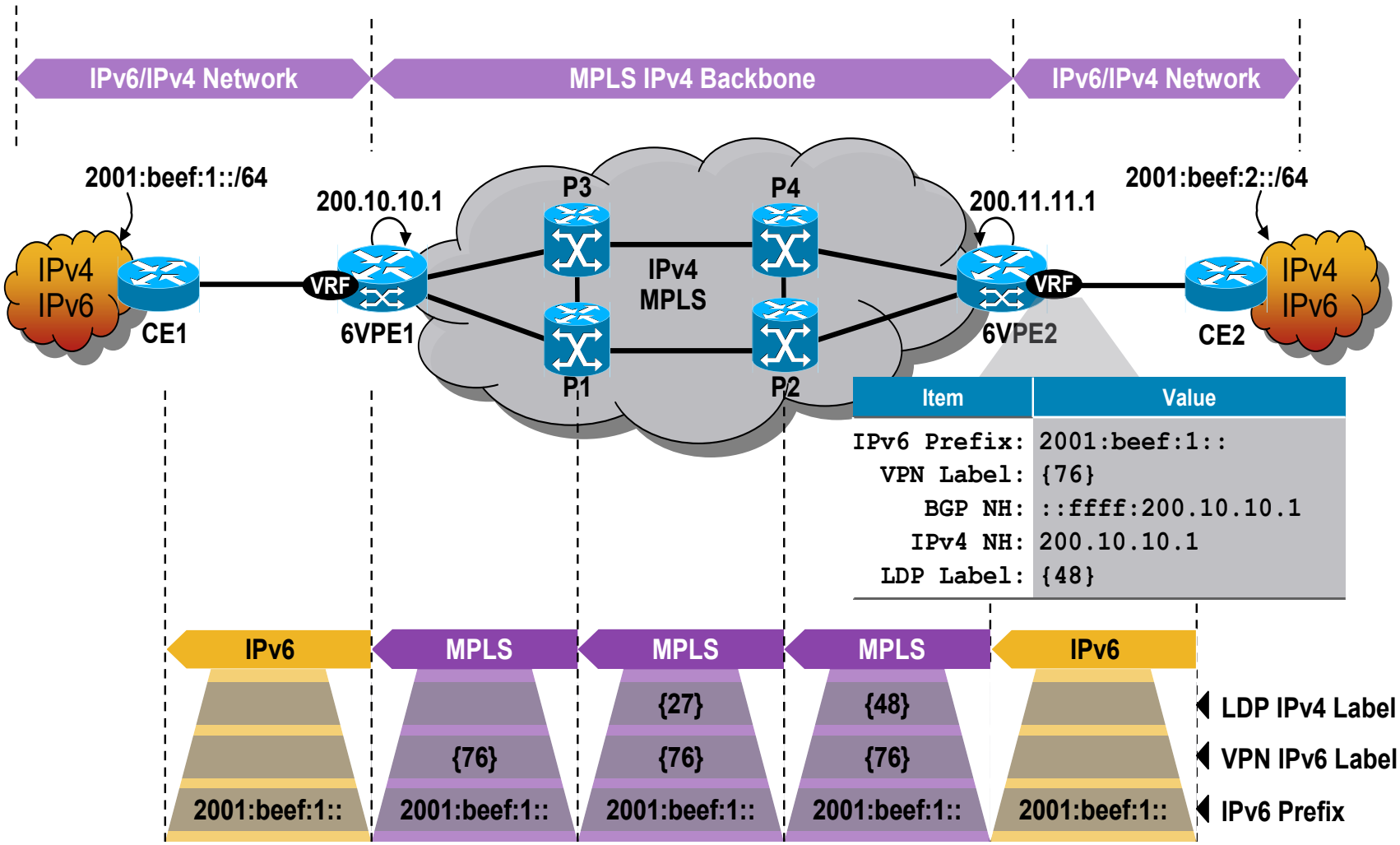


- 6VPE uses existing IPv4 MPLS infrastructure to provide IPv6 VPN
  - Core uses IPv4 control plane (LDPv4, TEv4, IGPv4)
- PEs must support dual stack IPv4+IPv6
- Offers same architectural features as MPLS-VPN for IPv4
  - RTs, VRFs, RDs are appended to IPv6 to form VPNv6 address
  - MP-BGP distributed both VPN address families
  - BGP NH uses IPv4 to IPv6 mapped address format ::ffff:A.B.C.D
- VRF can contain both VPNv4 and VPNv6 routes

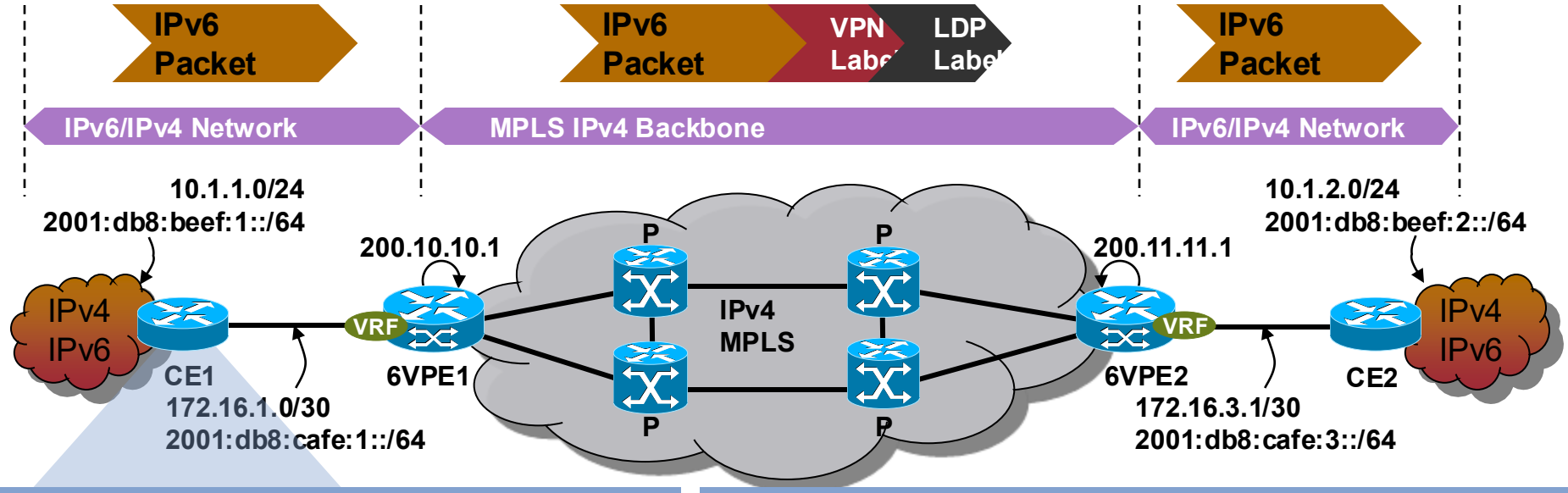
6vPE Control Plane



6vPE Data Plane



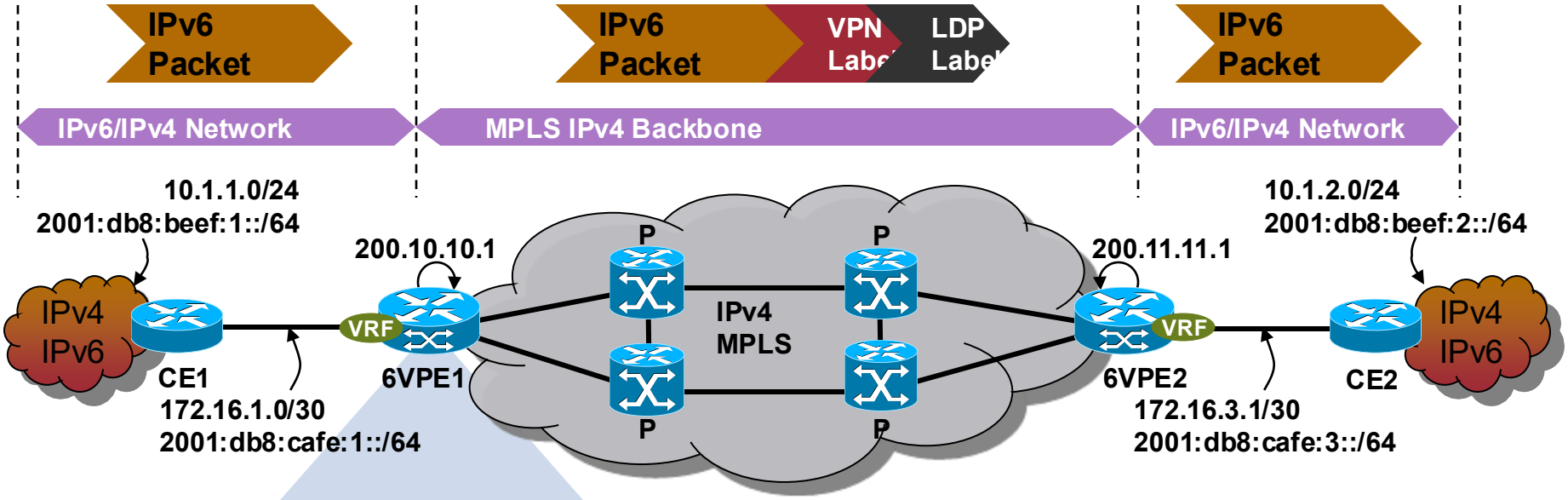
CE1 Configuration




```
ipv6 unicast-routing
ipv6 cef
!
interface Ethernet0/0
 description Link to PE1
 ip address 172.16.1.1 255.255.255.0
 ipv6 address 2001:db8:cafe:1::1/64
!
interface Ethernet1/0
 description to GREEN LAN
 ip address 10.1.1.1 255.255.255.0
 ipv6 address 2001:db8:beef:1::1/64
 ipv6 rip GREEN enable
```

```
router bgp 500
 neighbor 2001:db8:cafe:1::2 remote-as 100
 neighbor 172.16.1.2 remote-as 100
 !
 address-family ipv4
 redistribute eigrp 100
 neighbor 172.16.1.2 activate ← 6VPE1
 exit-address-family
 !
 address-family ipv6
 neighbor 2001:db8:cafe:1::2 activate ← 6VPE1
 redistribute rip GREEN
 exit-address-family
```

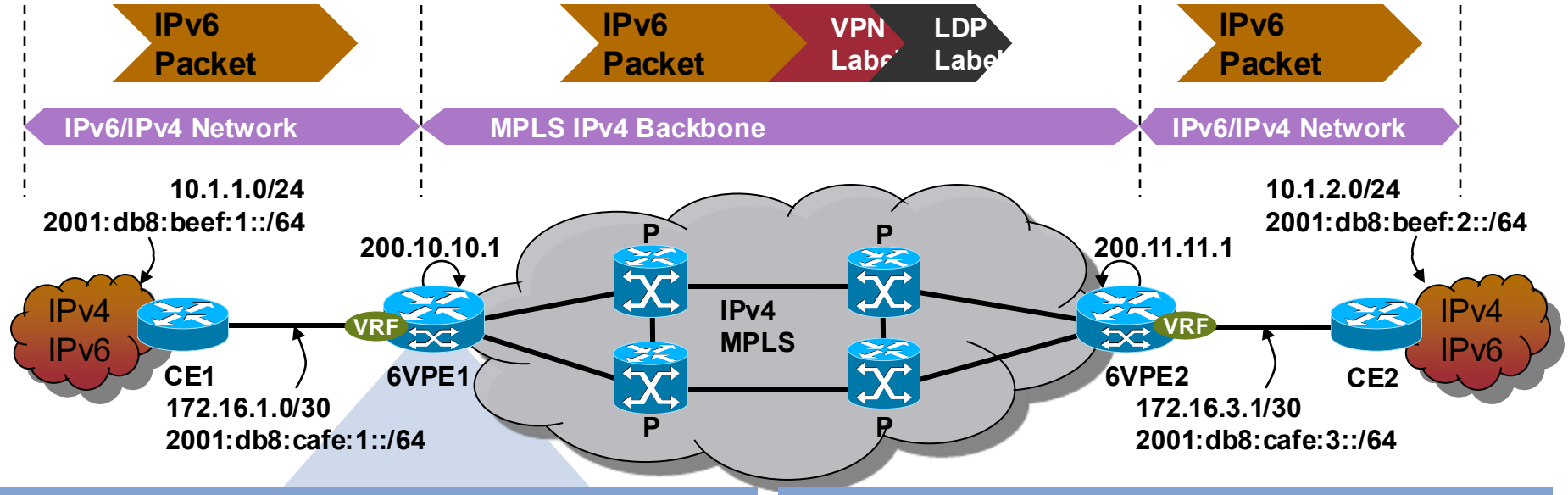
New Multi-AF VRF Configuration




```
vrf definition GREEN
 rd 200:1
 ! Common RT policies go here
 address-family ipv4
  route-target export 200:1
  route-target import 200:1
 exit-address-family
 !
 address-family ipv6
  route-target export 200:1
  route-target import 200:1
 exit-address-family
```

- New VRF AF definition
  - Allows address-families
  - Each with unique or common policies
- `vrf upgrade-cli multi-af-mode {common-policies | non-common-policies} [vrf <name>]`
  - This command can update existing VRF definitions

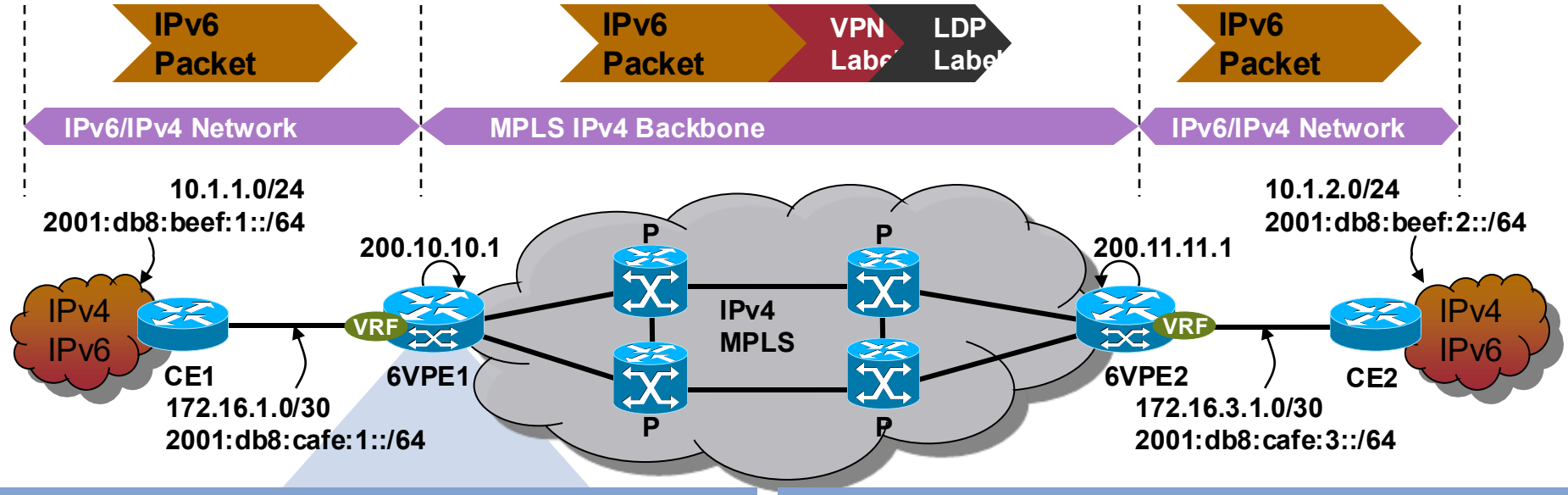
6VPE1 General Configuration




```
ipv6 unicast-routing
ipv6 cef
!
interface Loopback0
 ip address 200.10.10.1  255.255.255.255
!
interface Ethernet0/0
 Description Link to CE1
 vrf forwarding GREEN
 ip address 172.16.1.2 255.255.255.0
 ipv6 address 2001:db8:cafe:1::2/64
```

```
!
interface Ethernet2/0
 description Link to Core Network
 ip address 192.168.1.1 255.255.255.252
 mpls ip
!
router ospf 1
 log-adjacency-changes
 redistribute connected subnets
 passive-interface Loopback0
 network 192.168.1.0 0.0.0.255 area 0
```

6VPE1 BGP Configuration




```
router bgp 100
 neighbor 200.11.11.1 remote-as 100
 neighbor 200.11.11.1 update-source lo0
 !
 address-family ipv4 ← Internet Routes
 neighbor 200.11.11.1 activate
 no auto-summary
 no synchronization
 exit-address-family
 !
 address-family vpnv4 ← To 6VPE2
 neighbor 200.11.11.1 activate
 neighbor 200.11.11.1 send-community ext
 exit-address-family
```

```
 address-family vpnv6 ← To 6VPE2
 neighbor 200.11.11.1 activate
 neighbor 200.11.11.1 send-community ext
 exit-address-family
!
 address-family ipv4 vrf GREEN ← To CE1
 redistribute connected
 neighbor 172.16.1.1 remote-as 500
 neighbor 172.16.1.1 activate
 exit-address-family
 !
 address-family ipv6 vrf GREEN ← To CE1
 neighbor 2001:db8:cafe:1::1 remote-as 500
 neighbor 2001:db8:cafe:1::1 activate
 exit-address-family
```


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